

spaCRT: a computationally efficient and statistically accurate differential expression test for single-cell CRISPR screens

Biostatistics seminar at OSU
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Co-authors



Jyotishka Ray Choudhury
(Georgia Tech)



Gene
(Wharton)

A model for genetic origins of human diseases

A model for genetic origins of human diseases

Normal

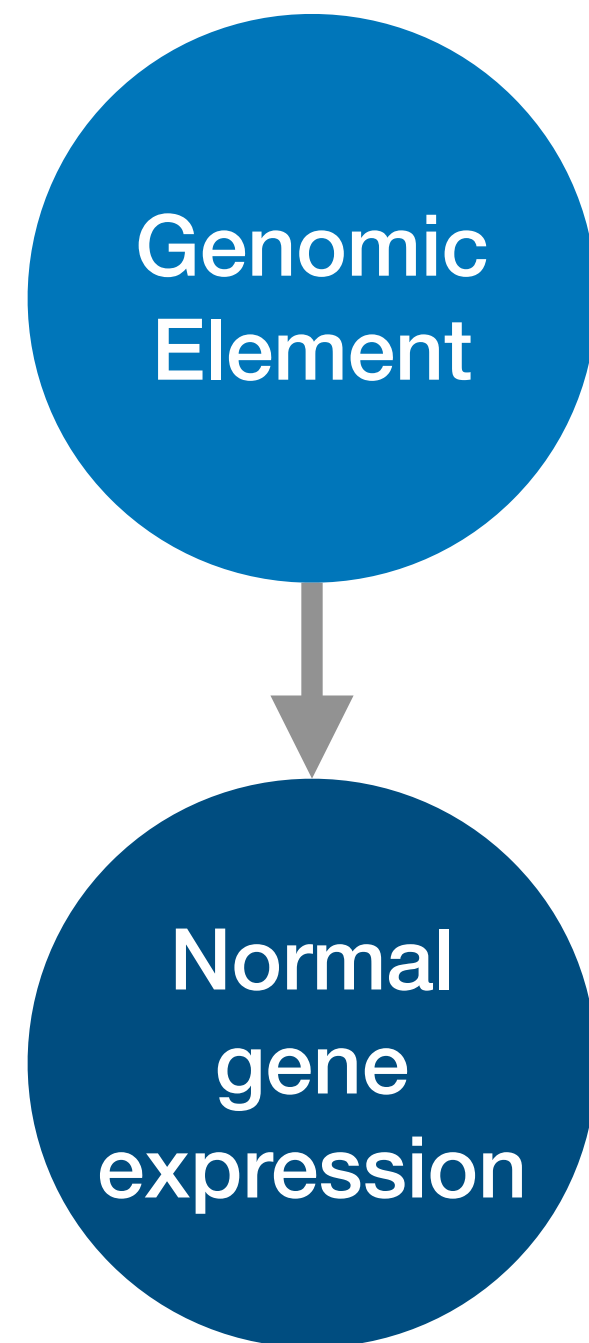
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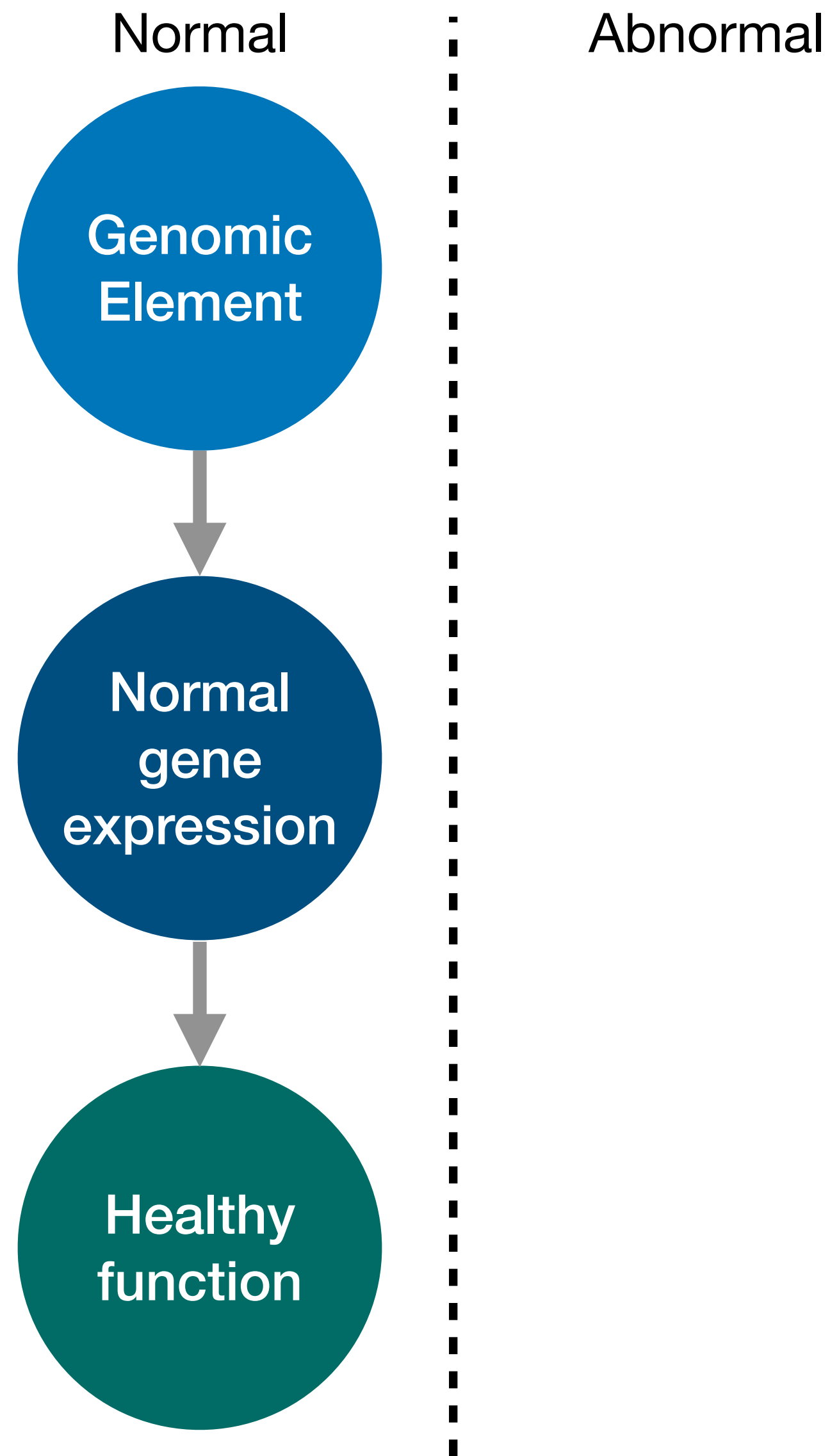


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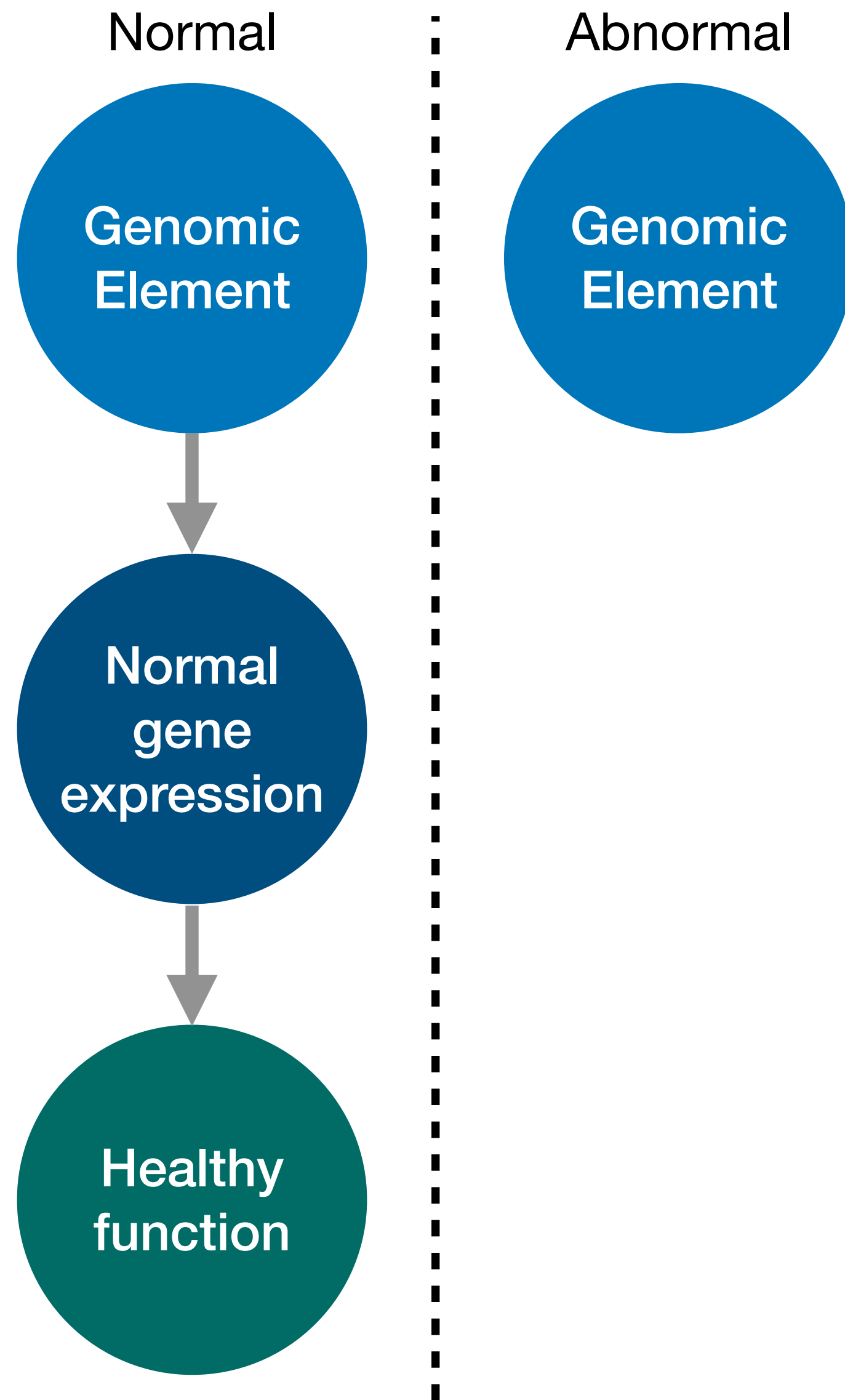
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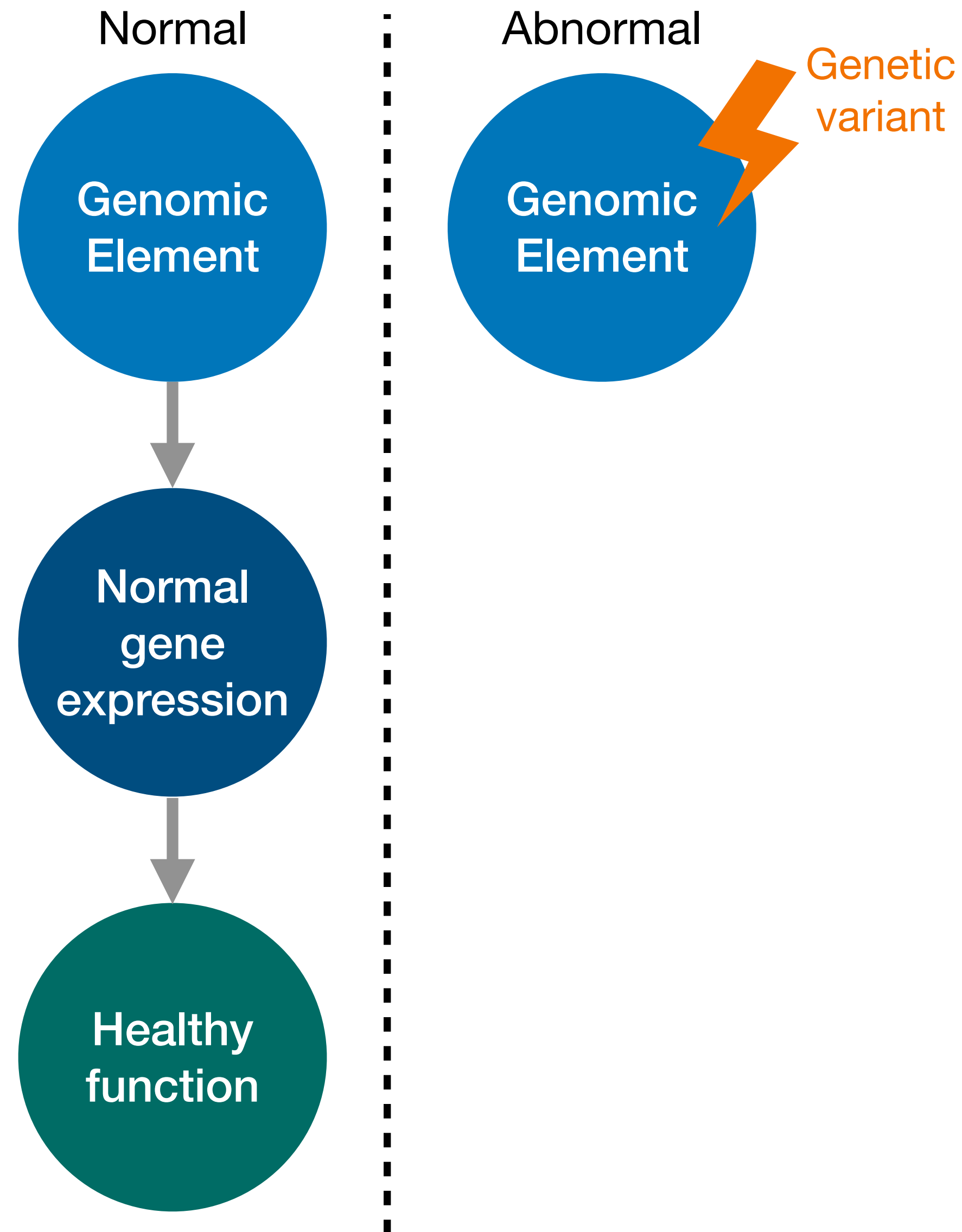
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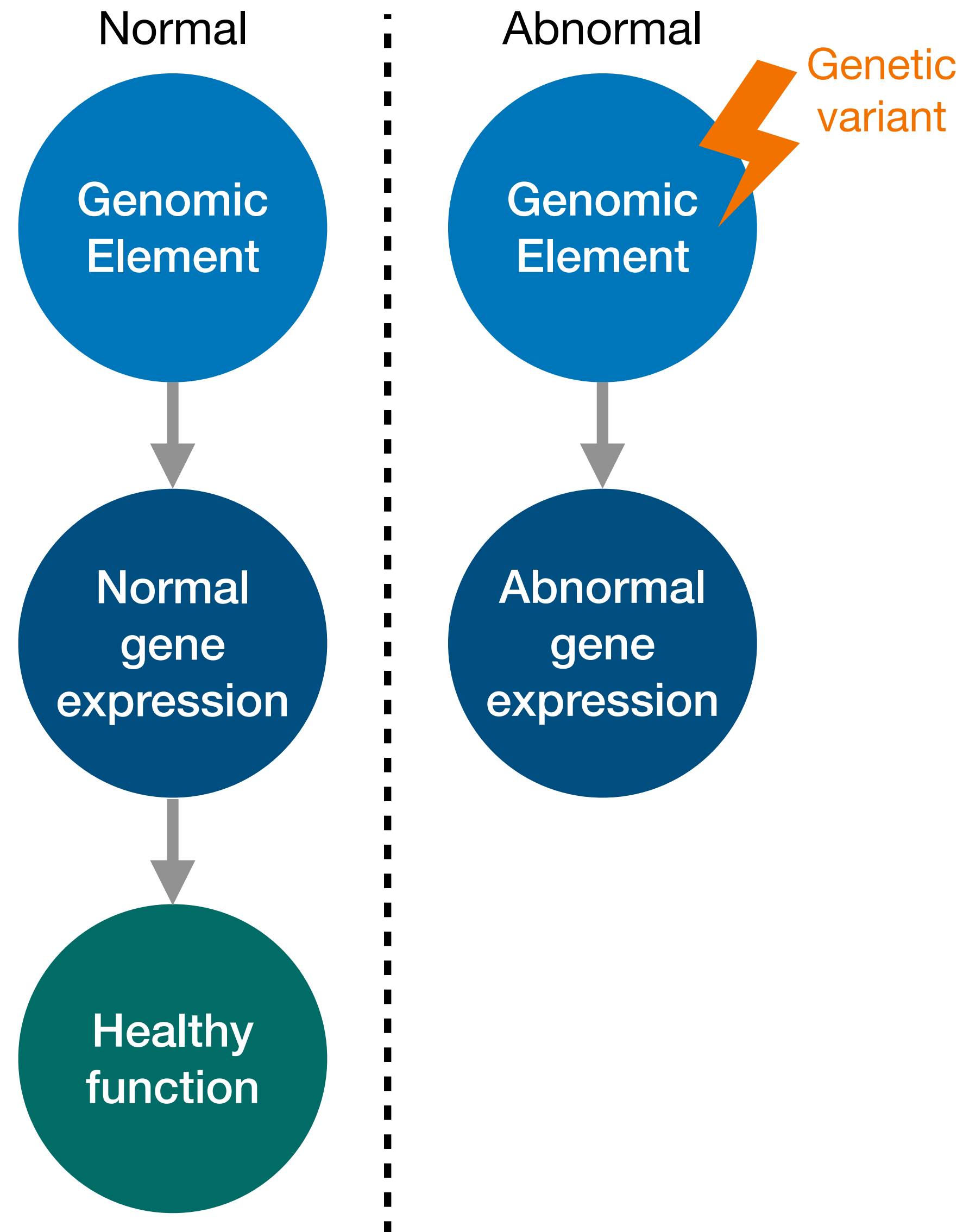
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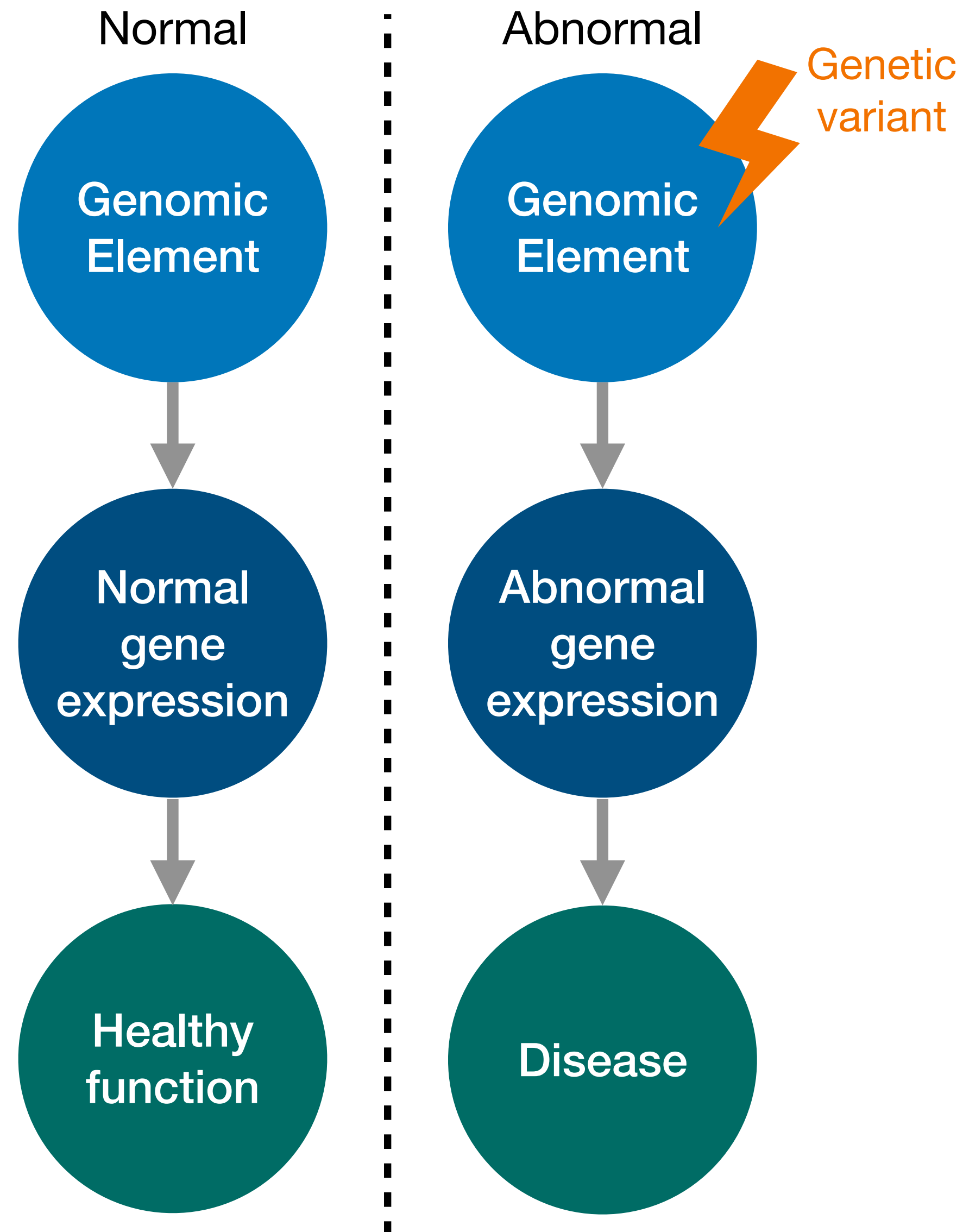
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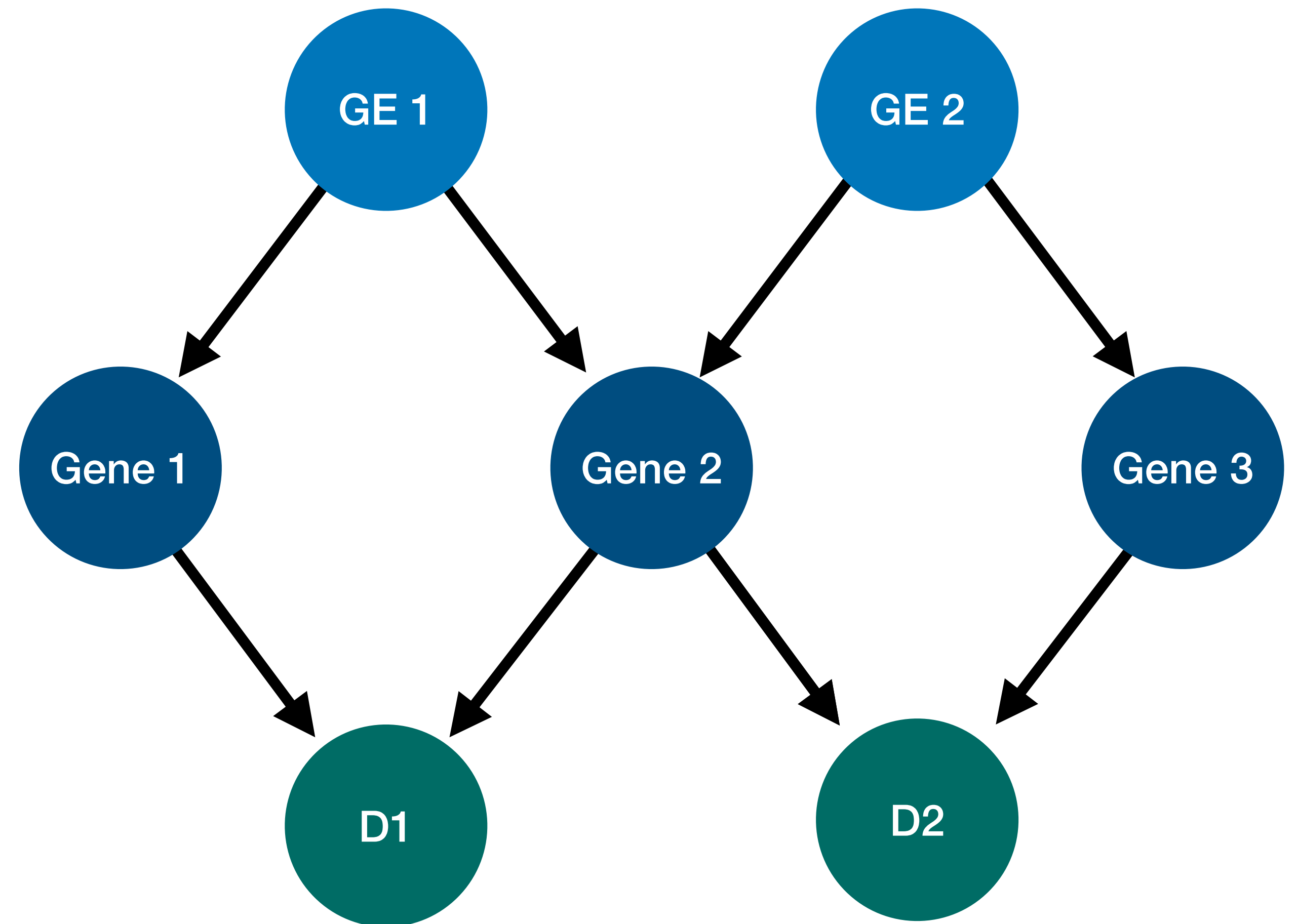
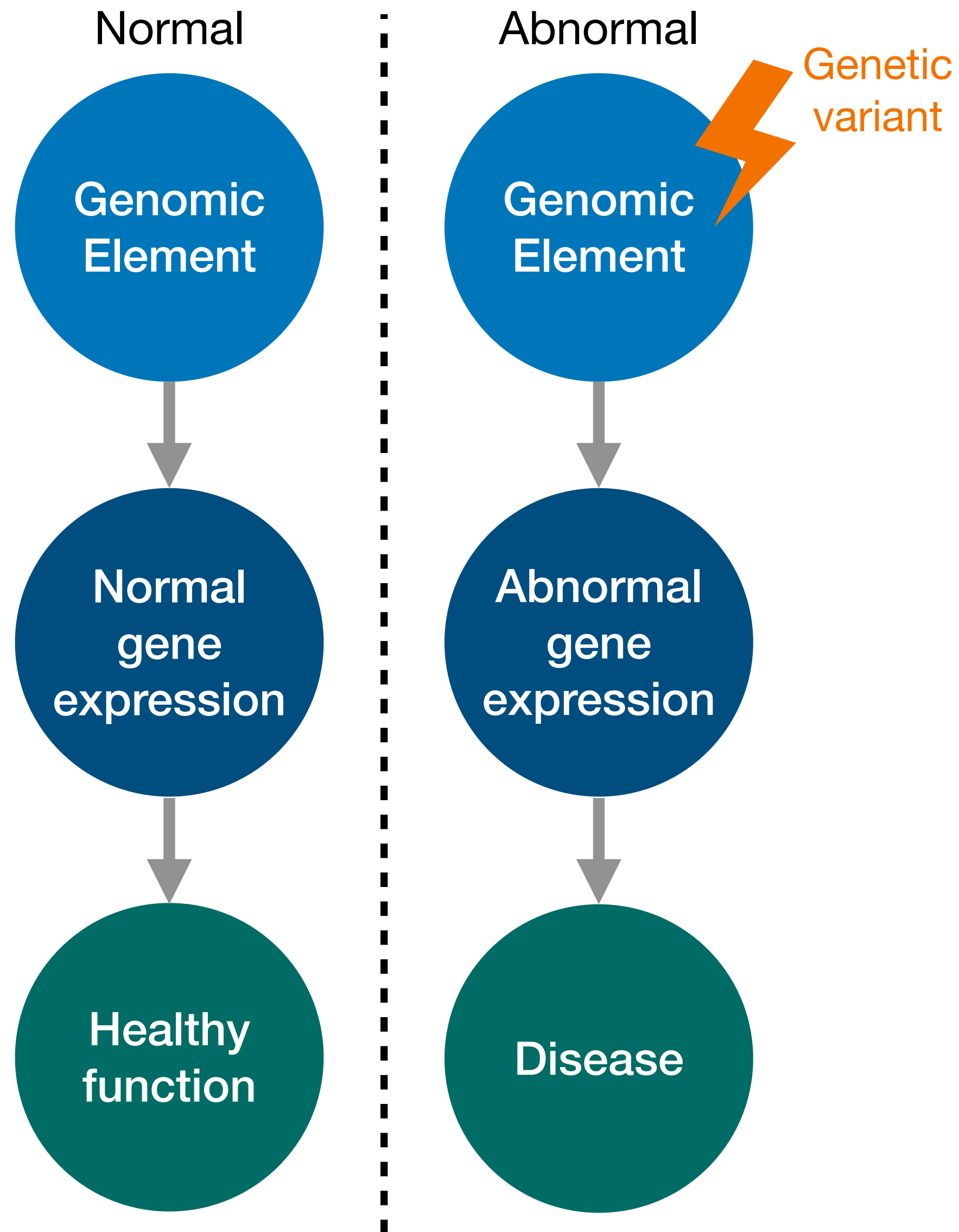
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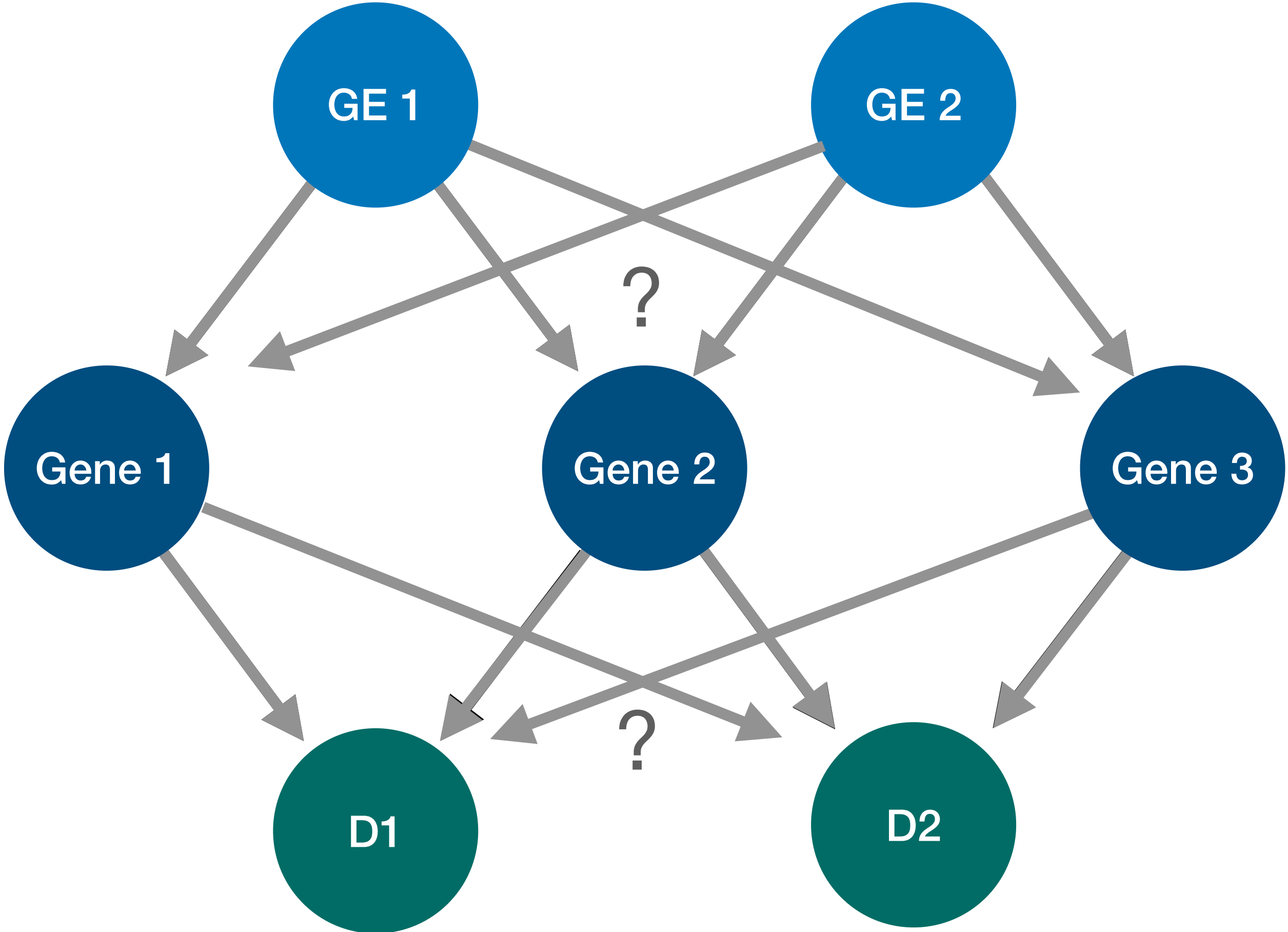
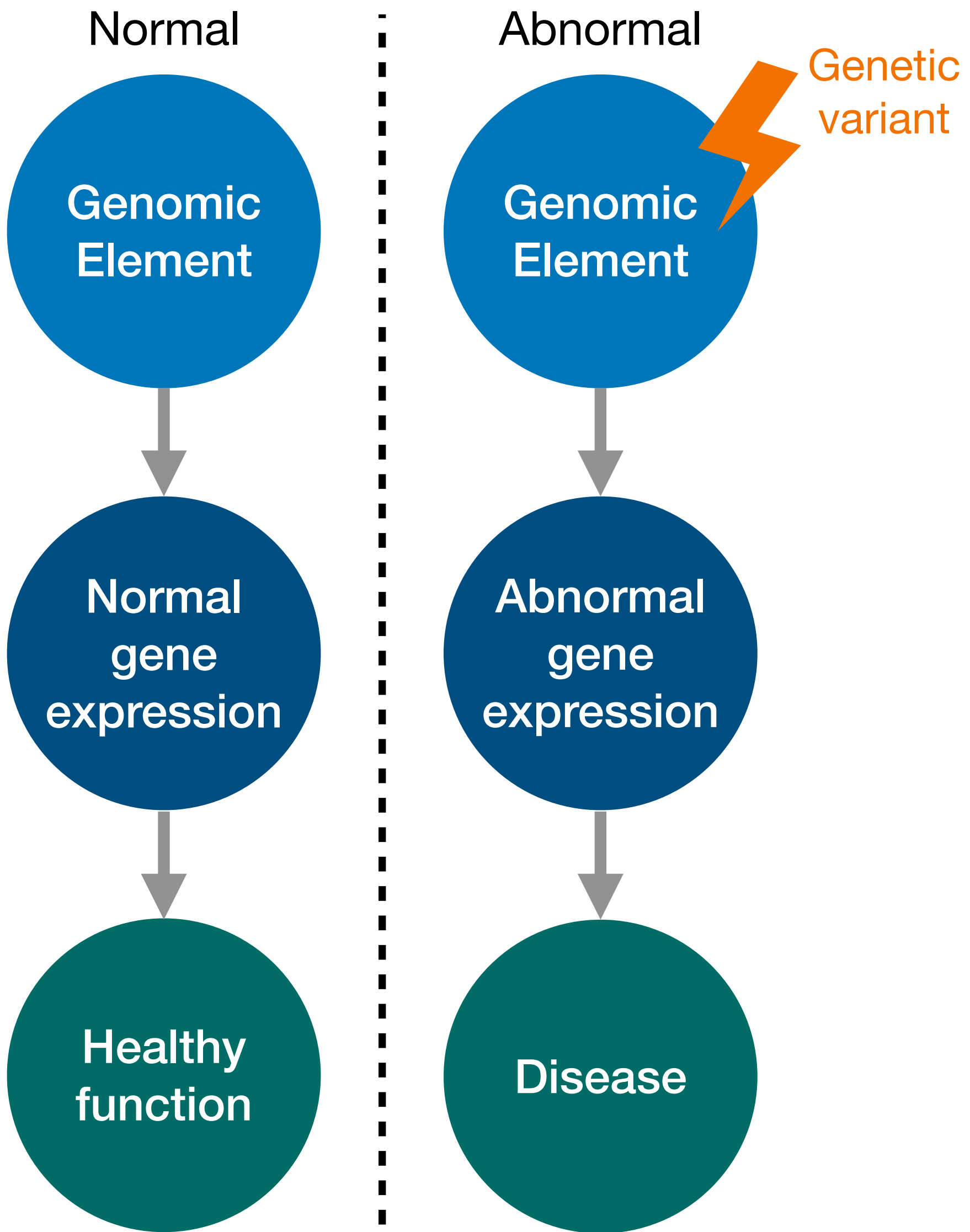
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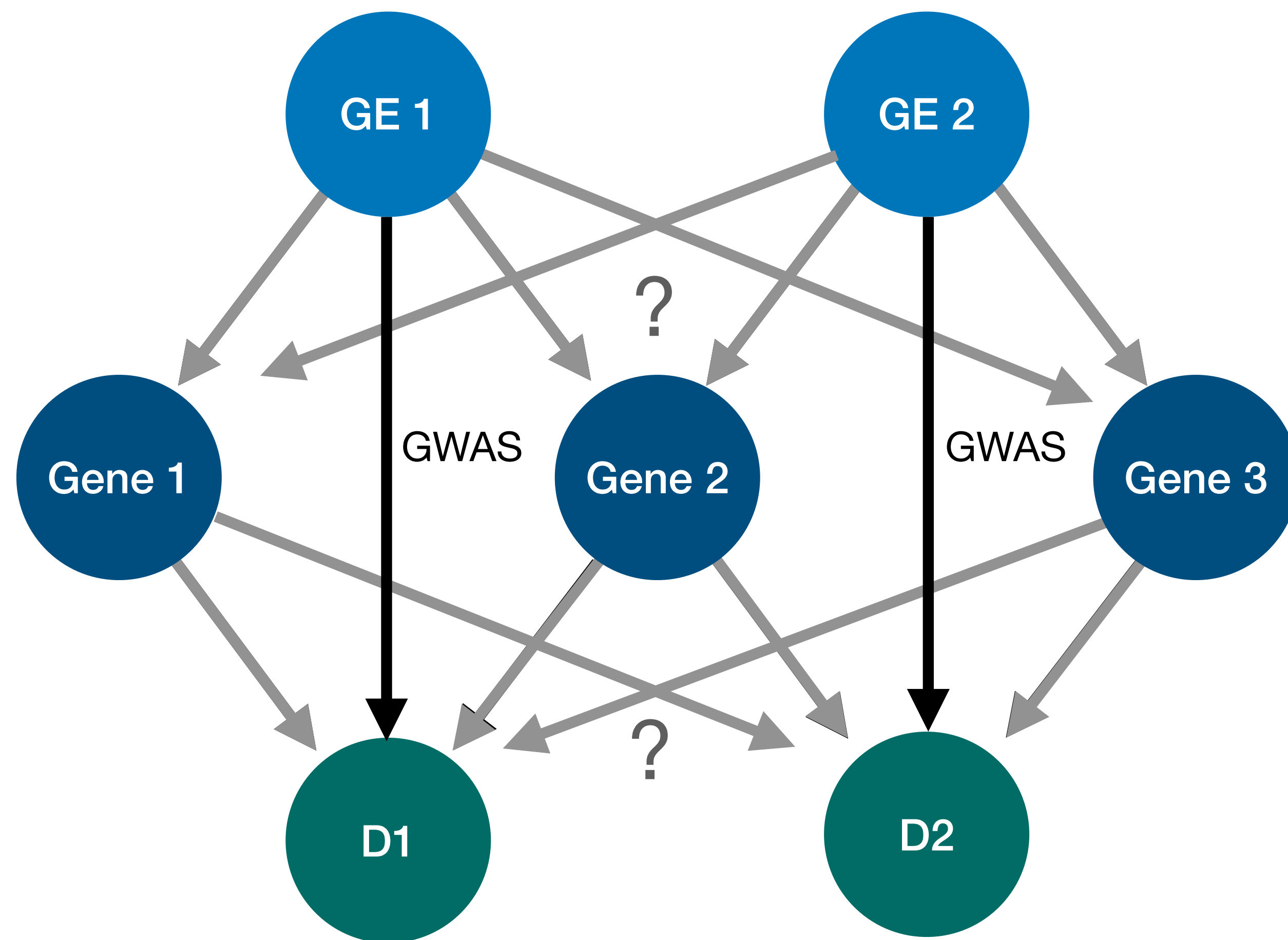
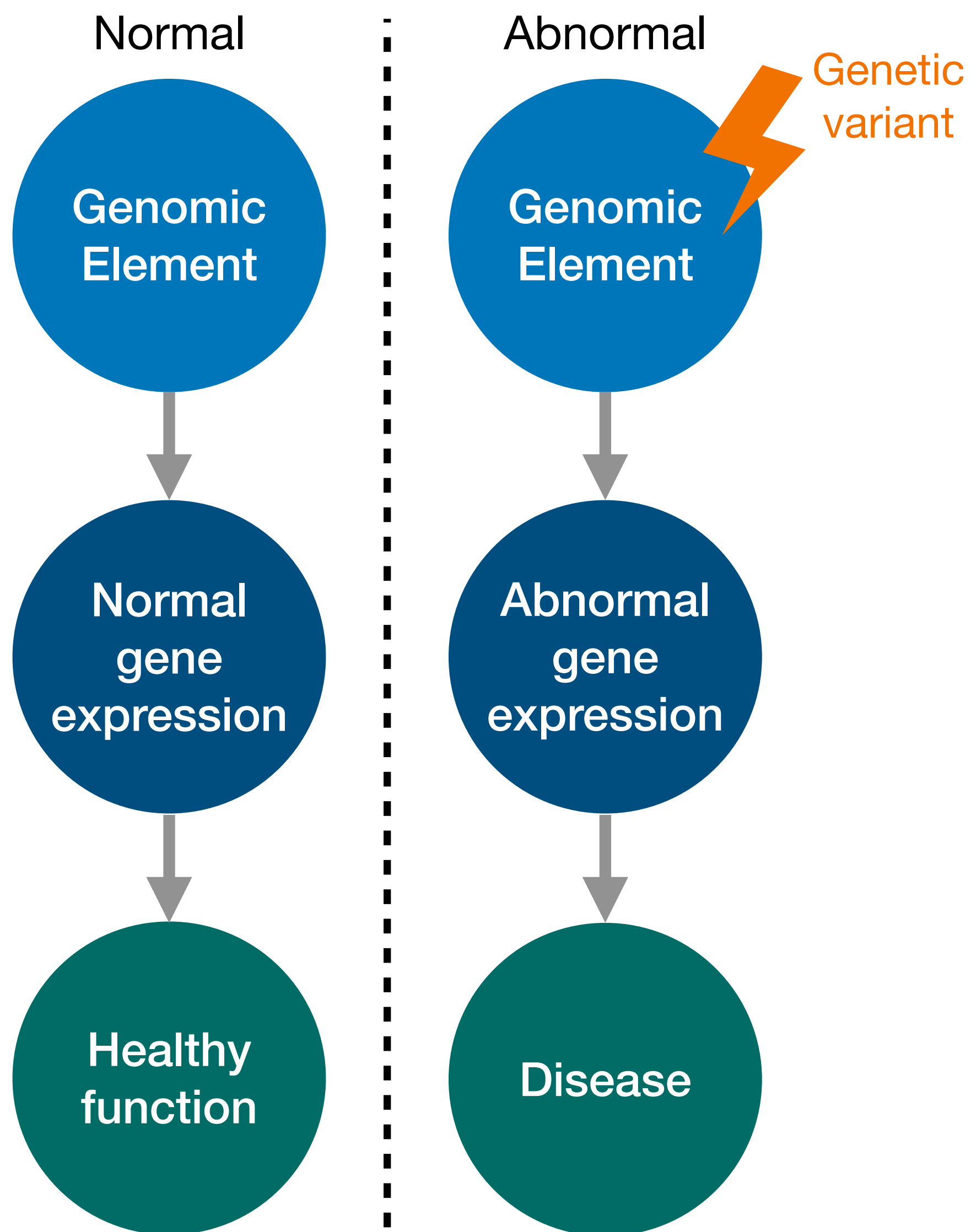
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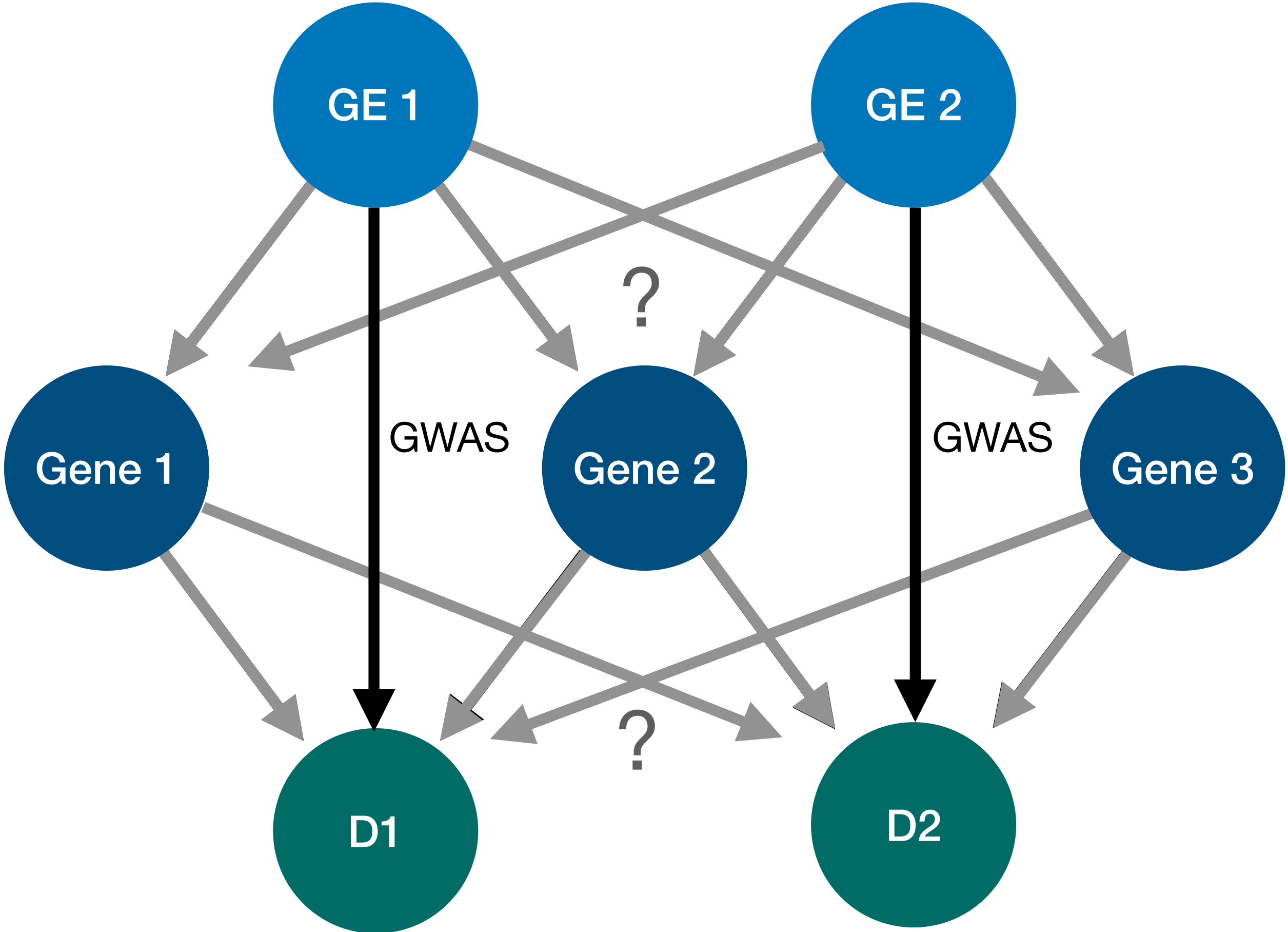
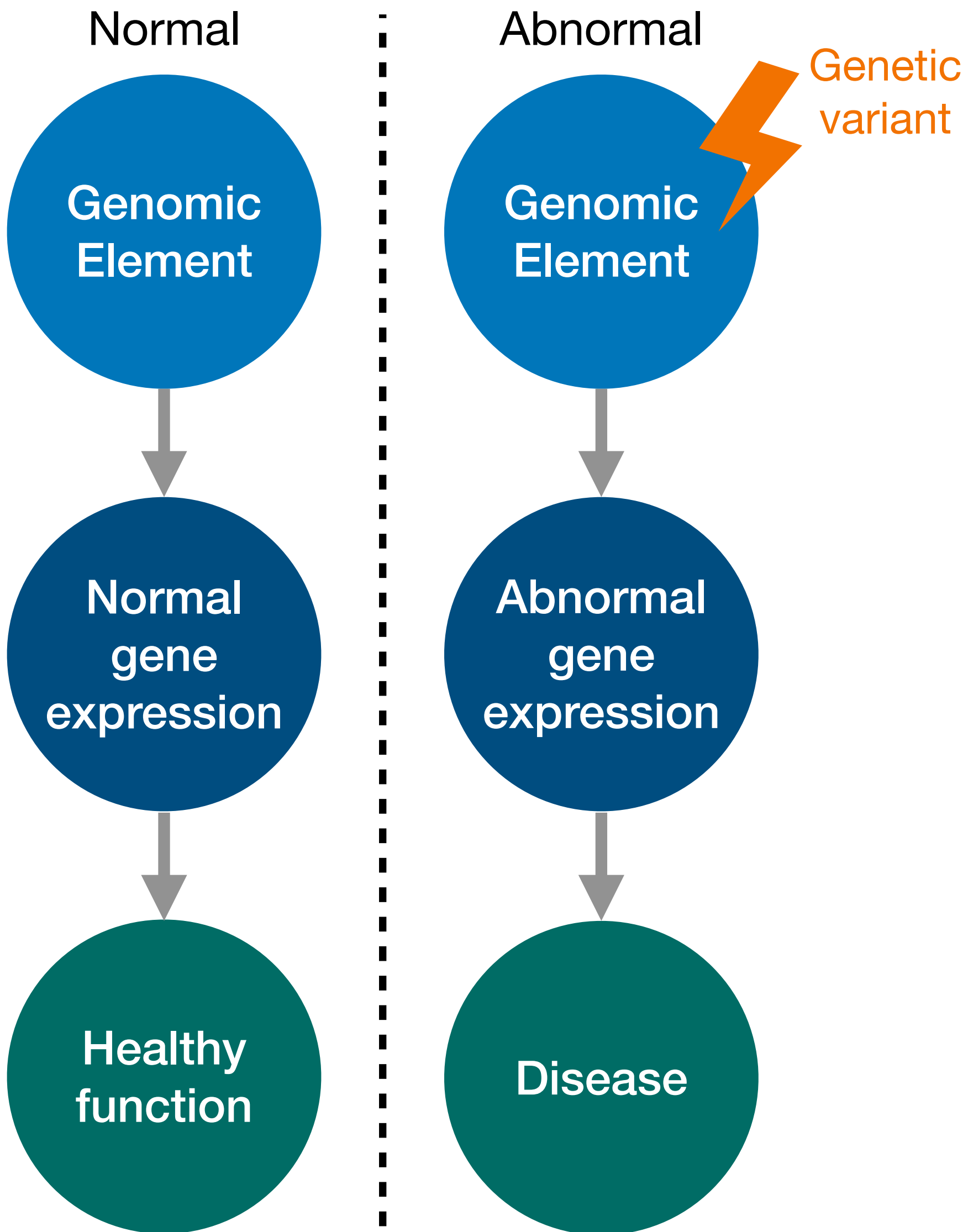
Uncovering genetic origins of human diseases



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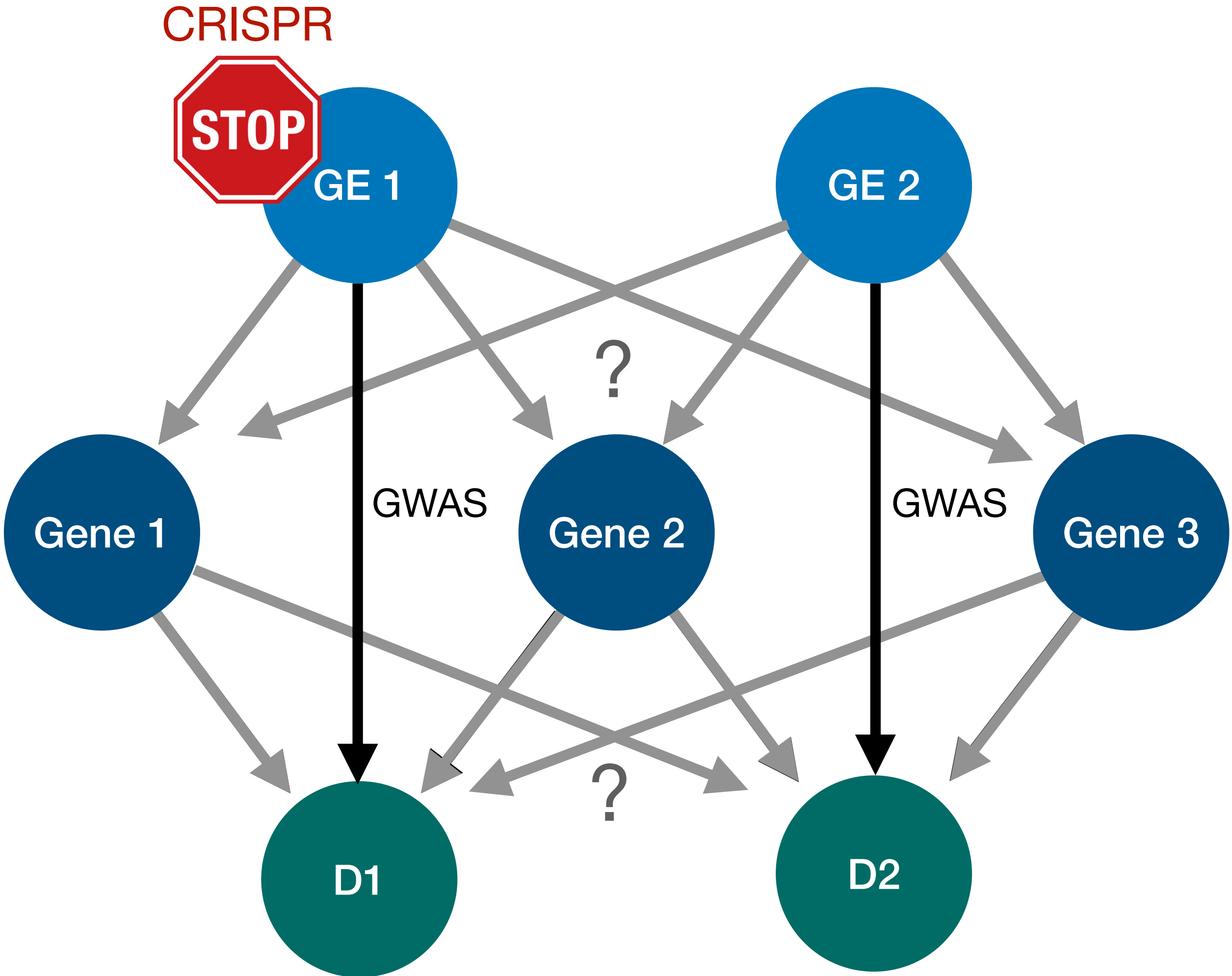
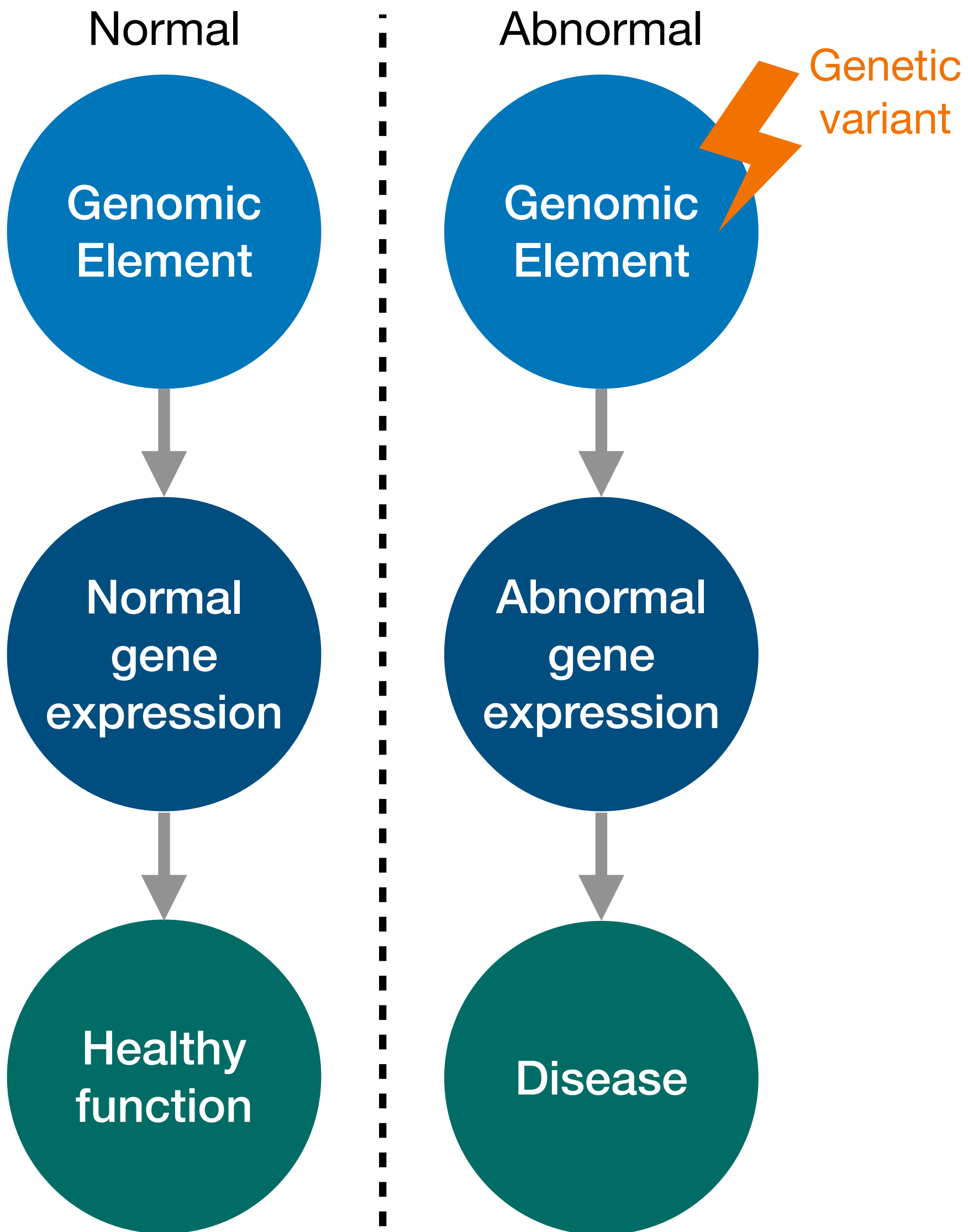


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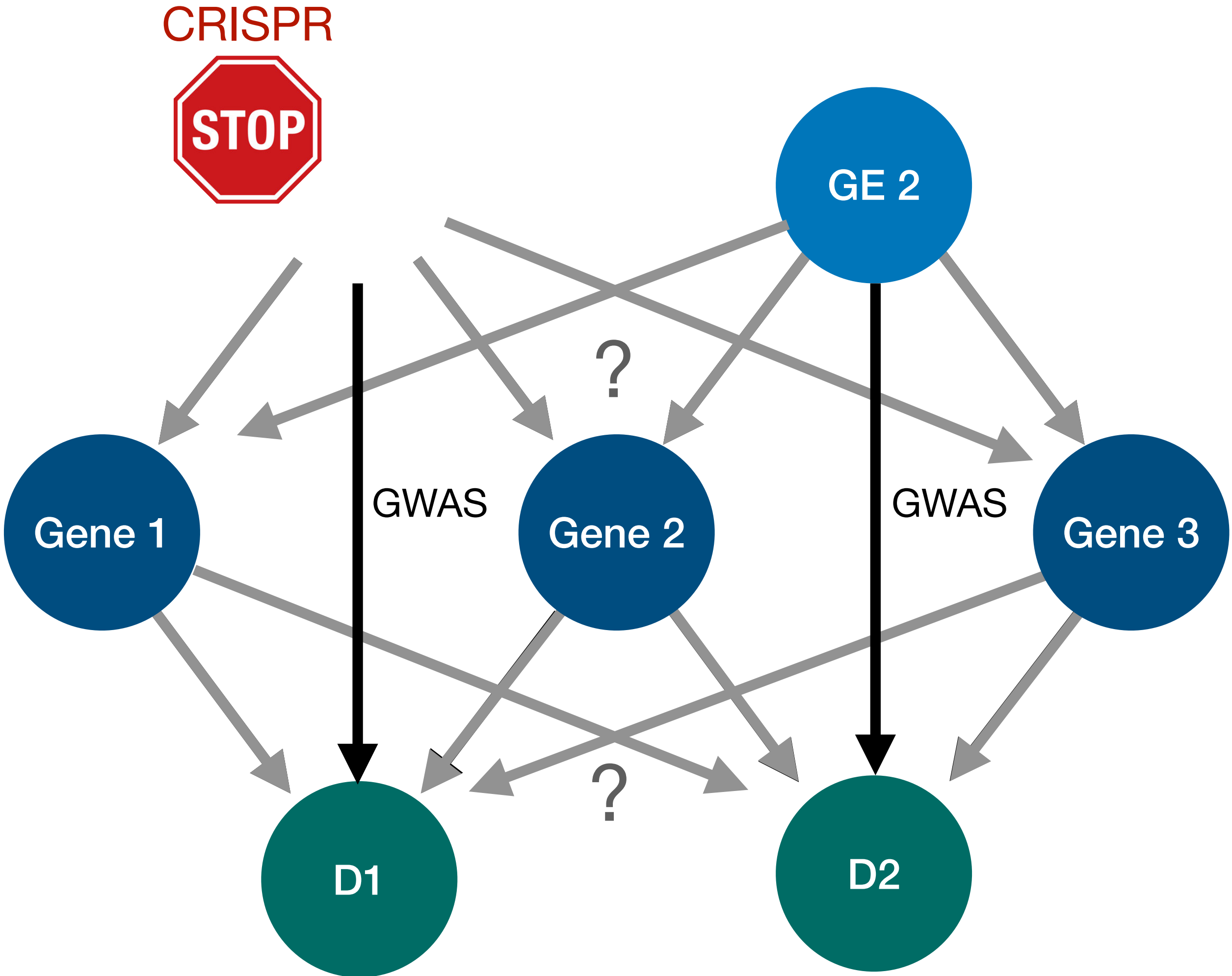
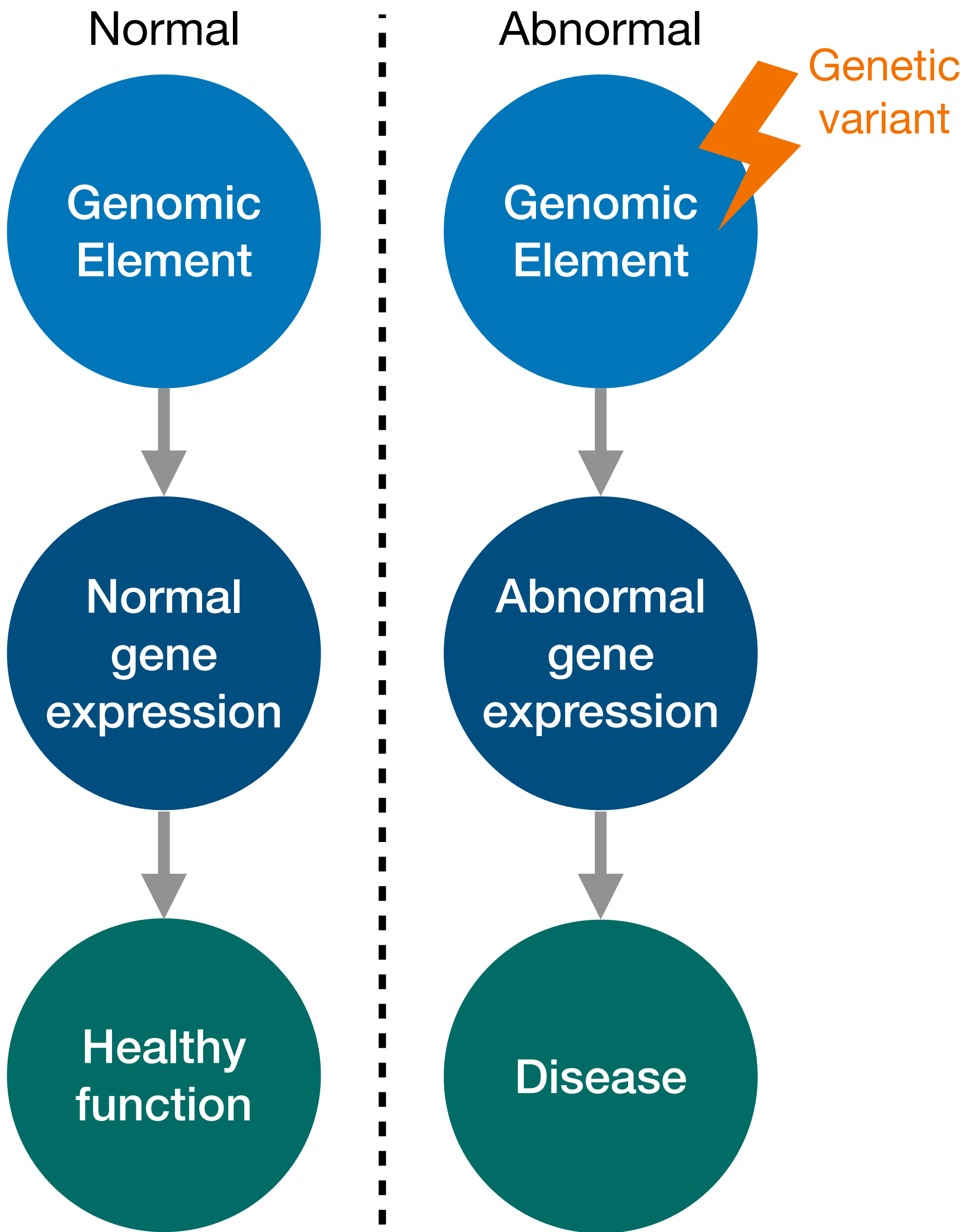
Single-cell CRISPR screens

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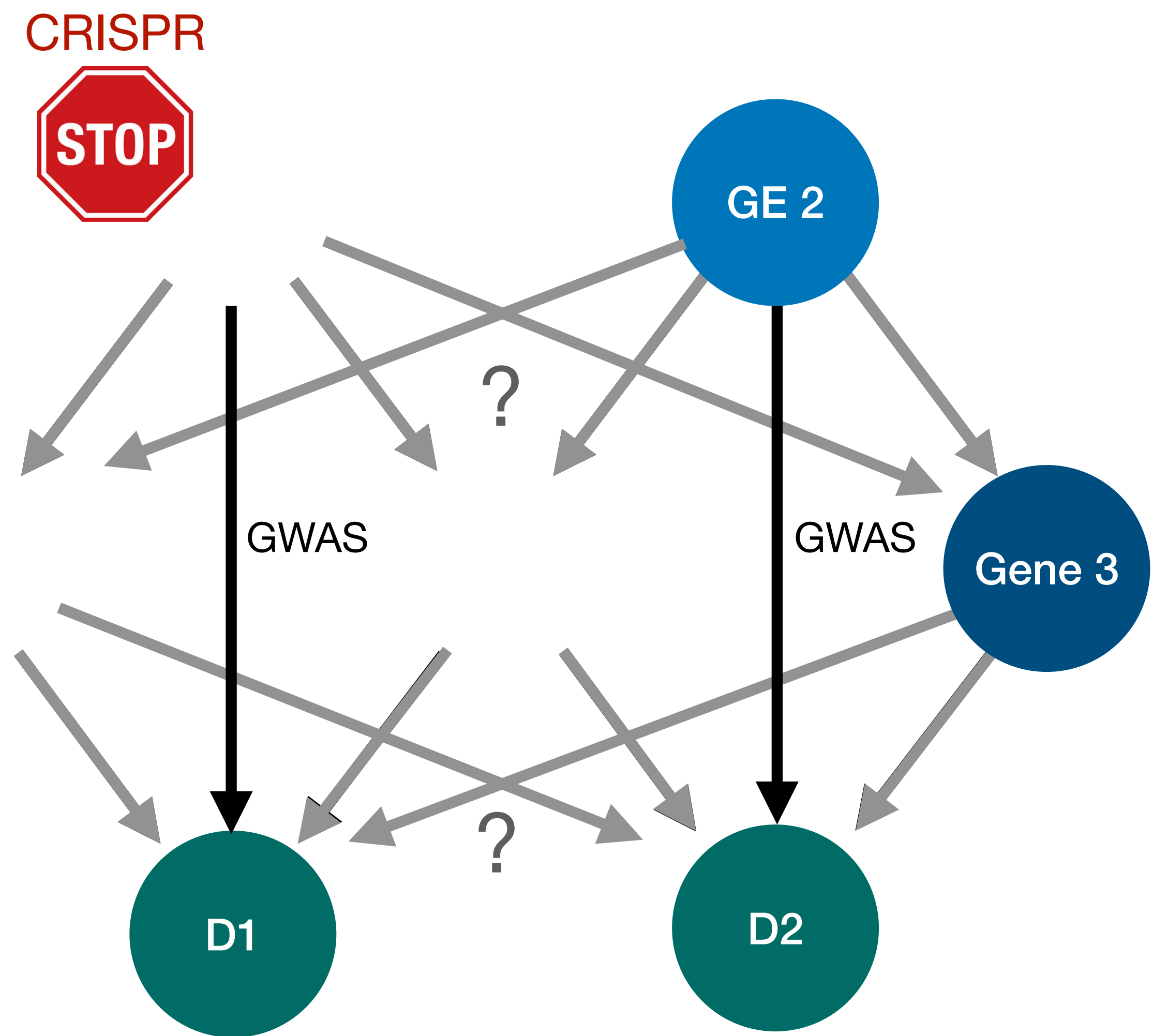
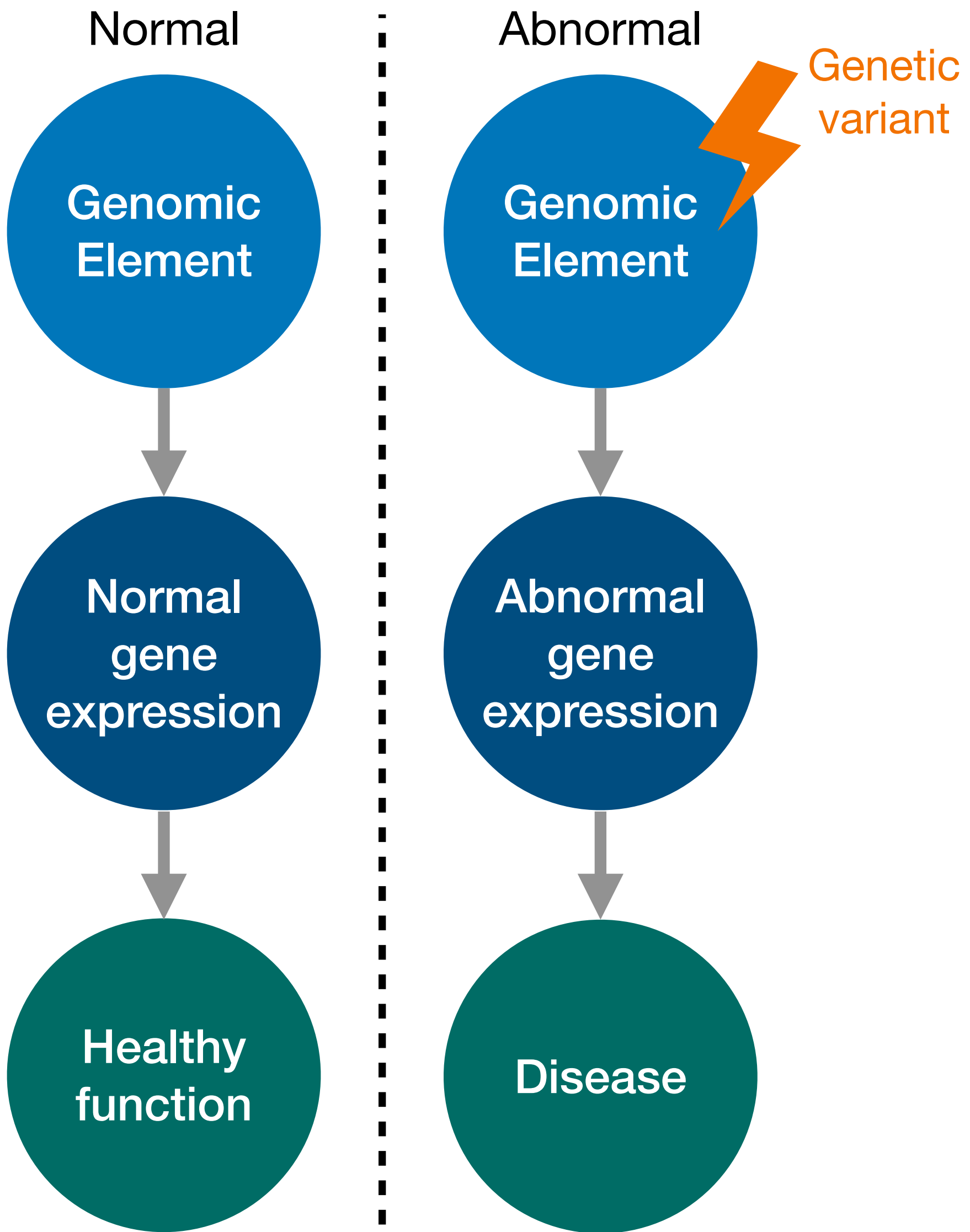
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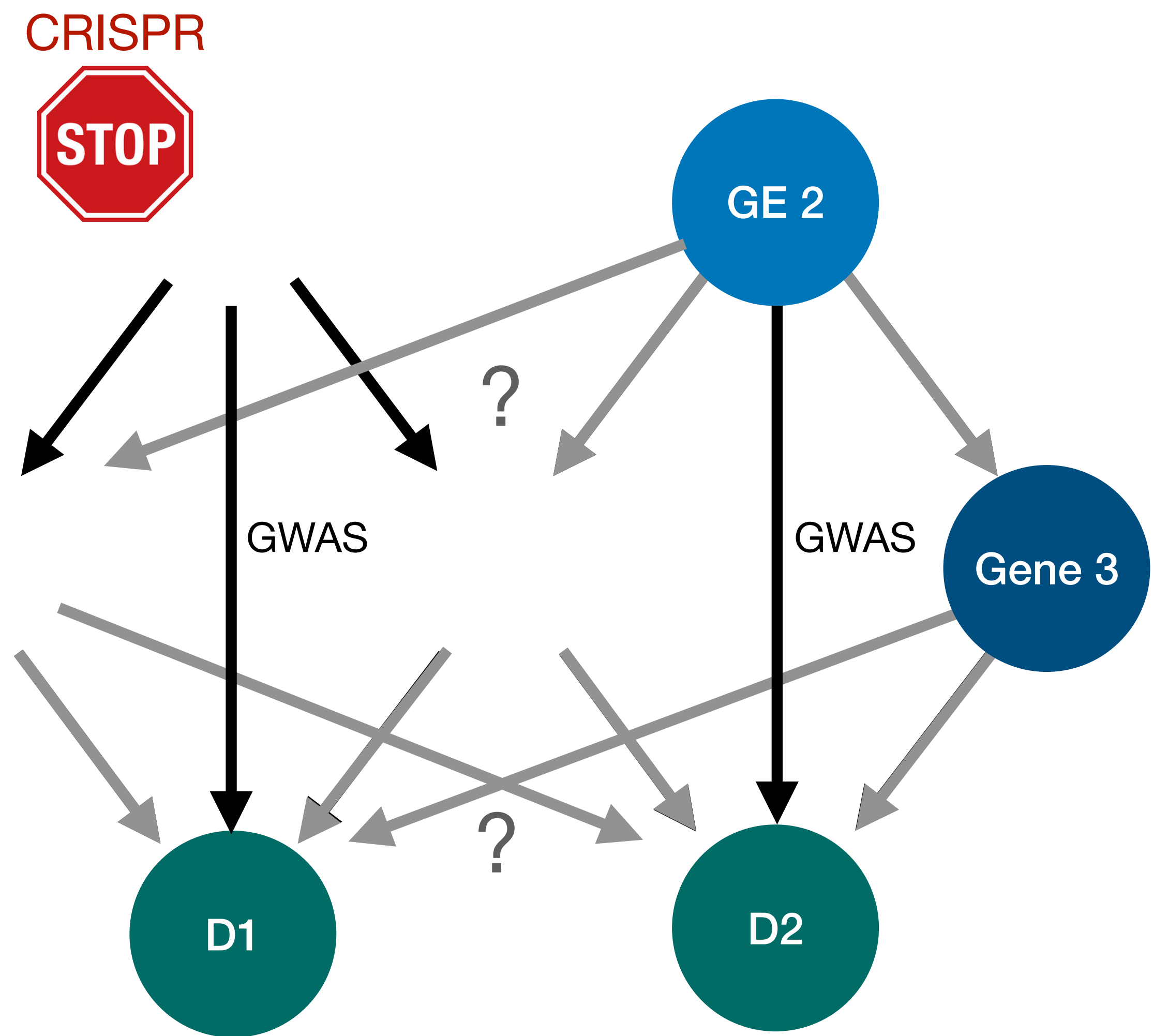
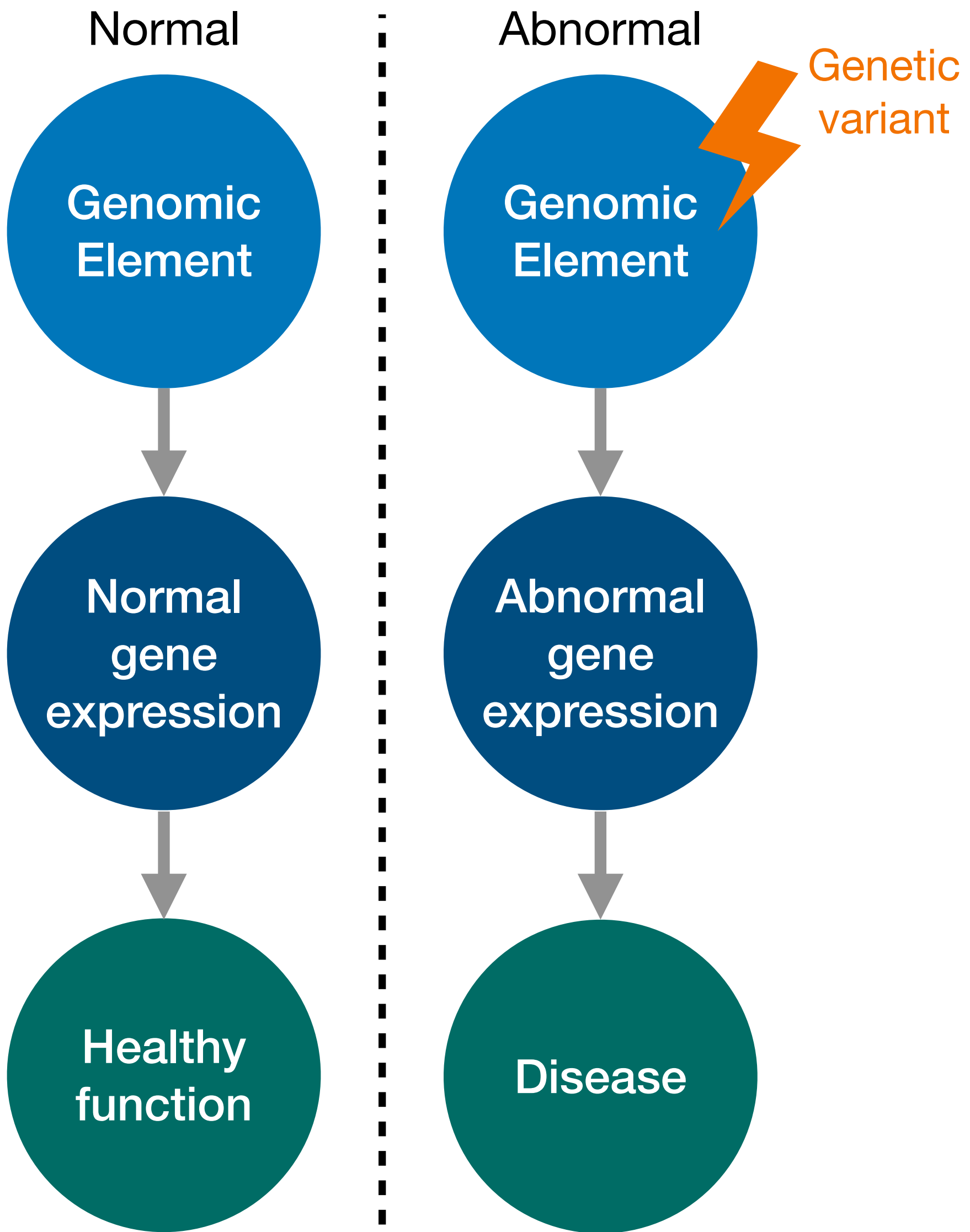
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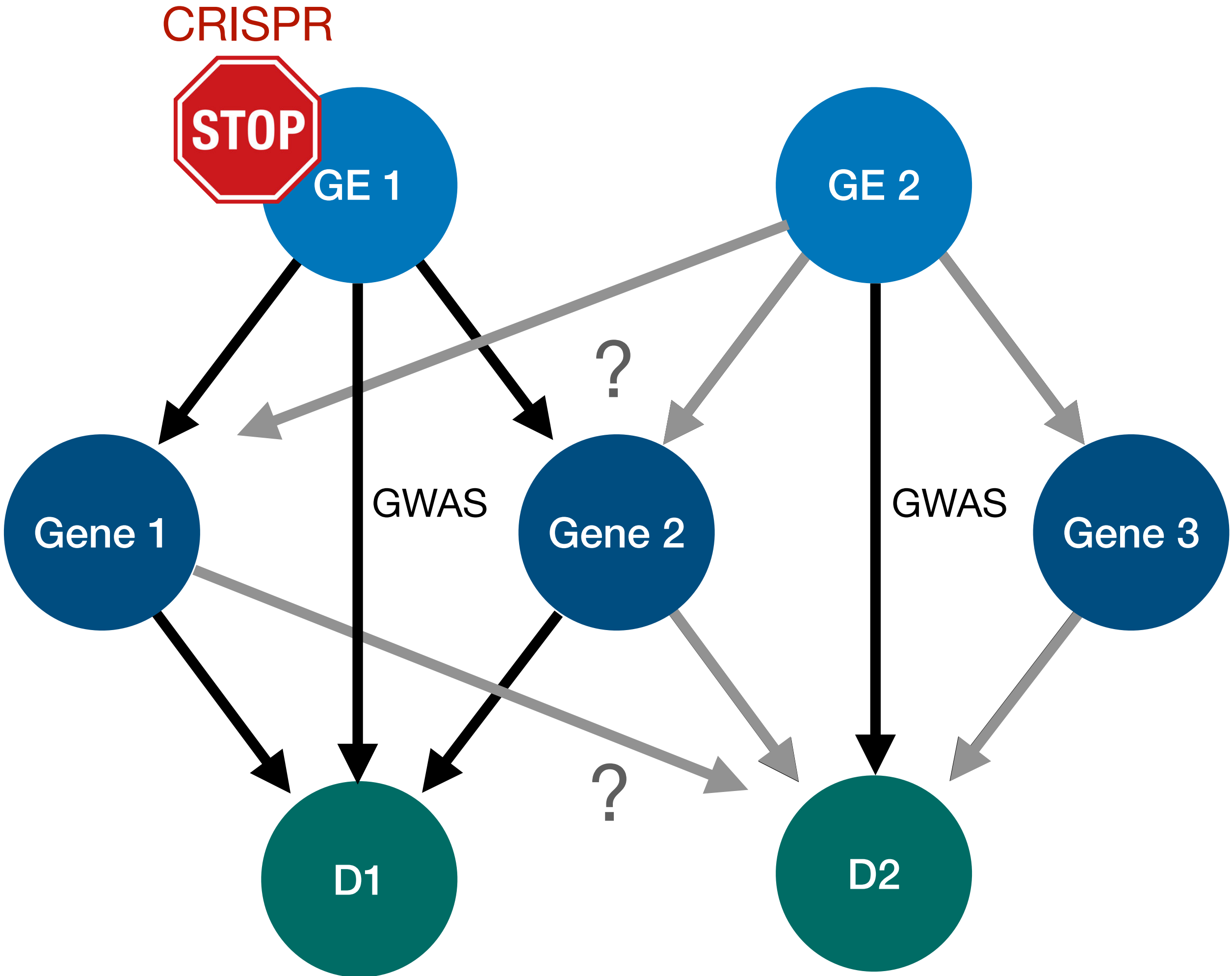
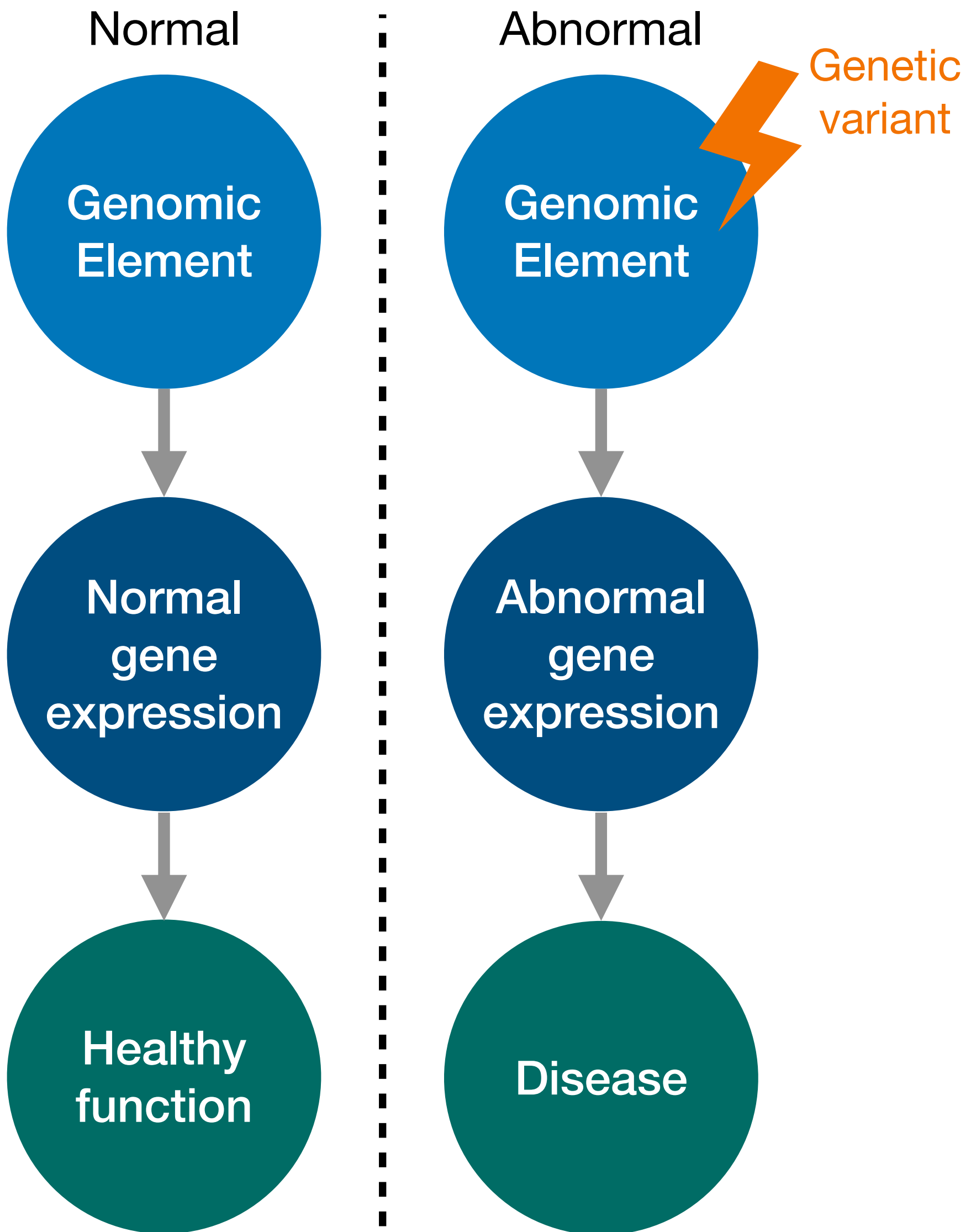
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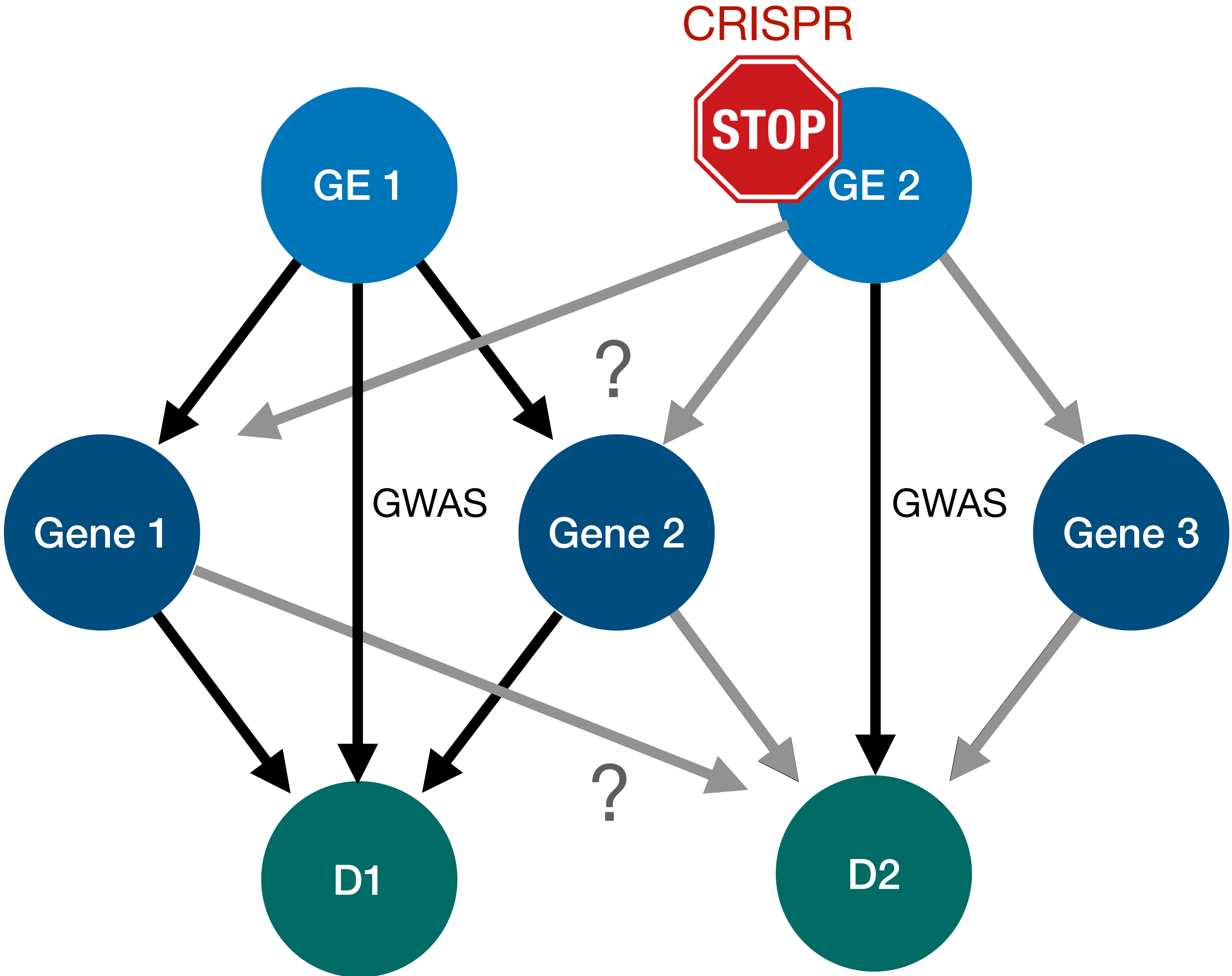
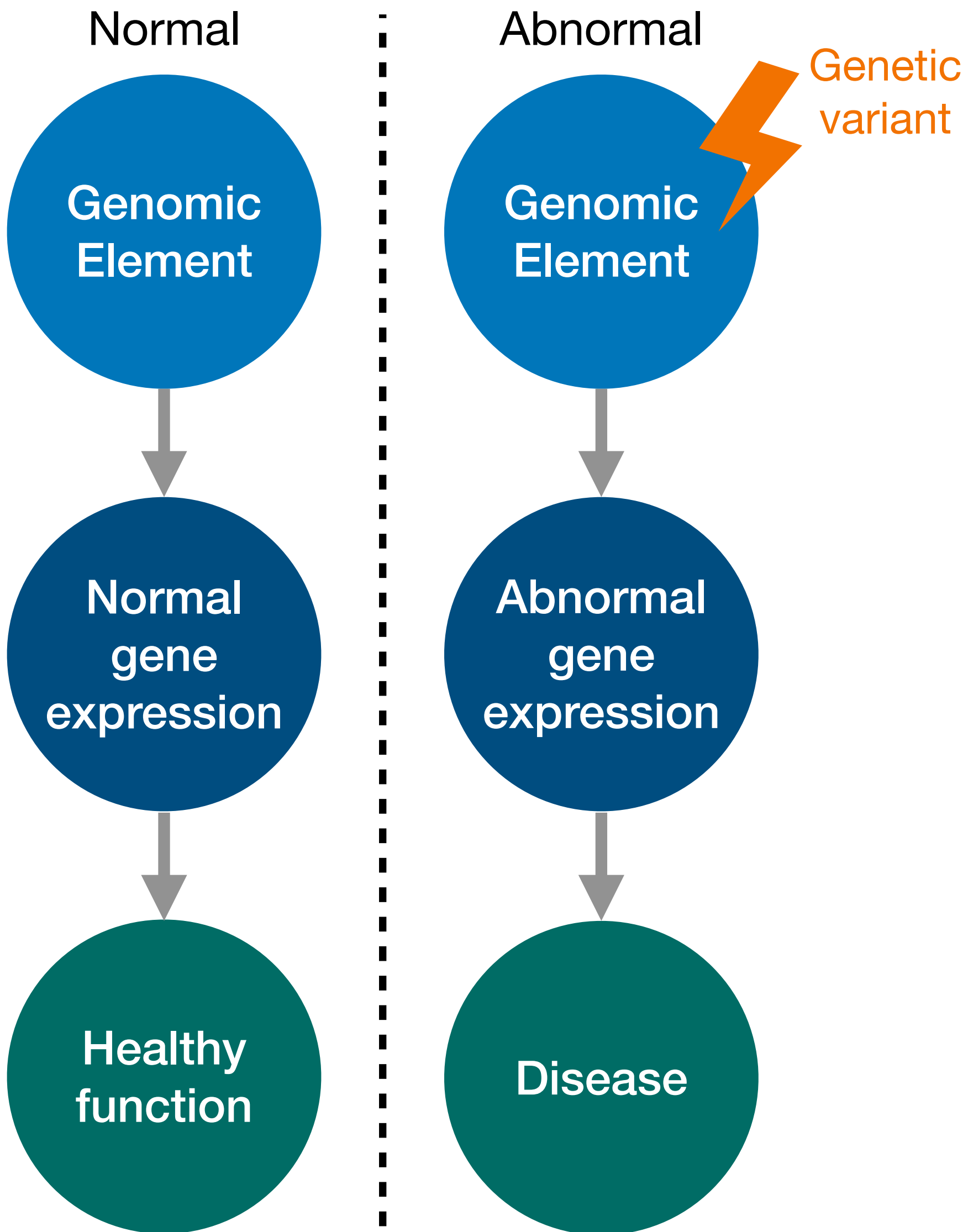
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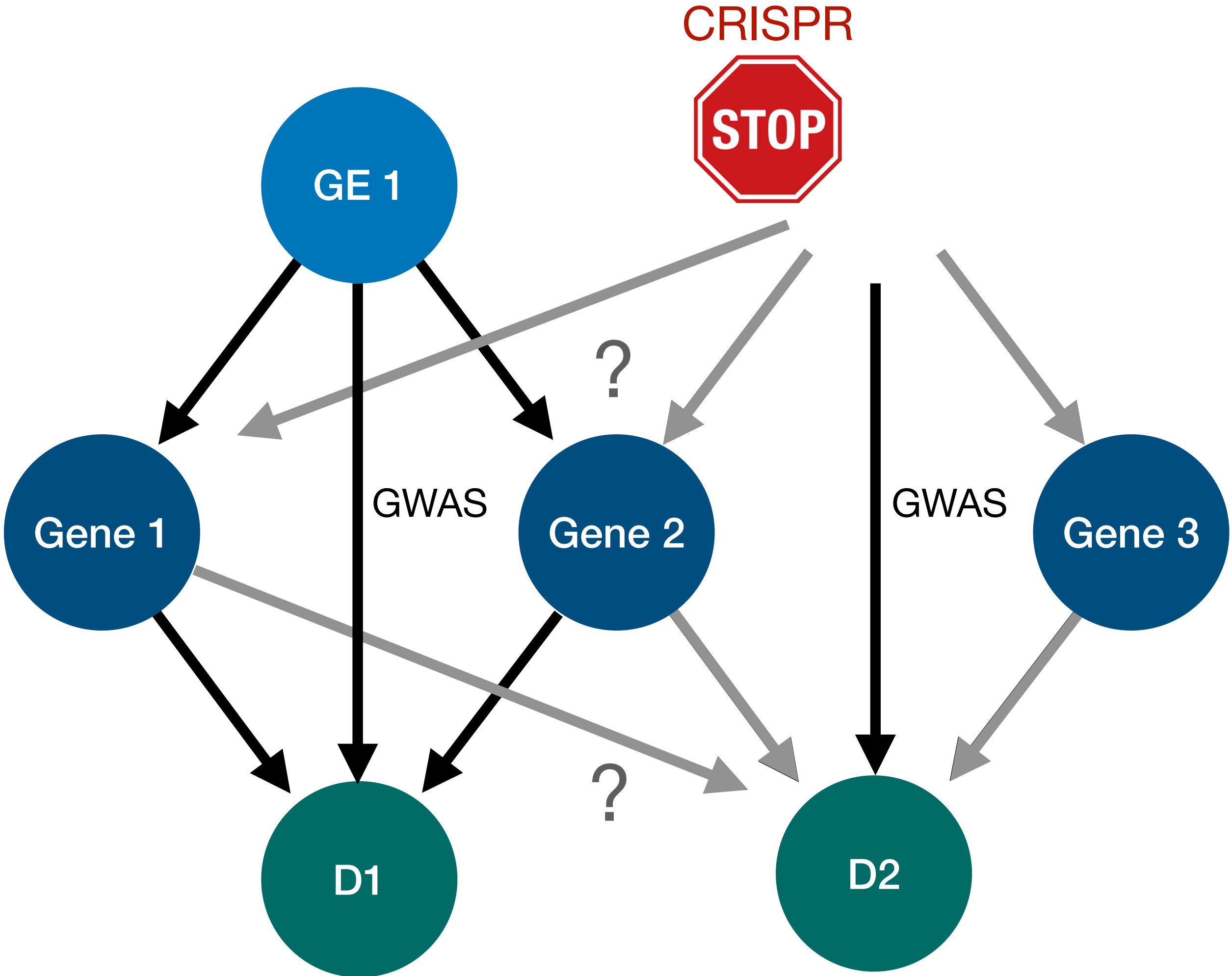
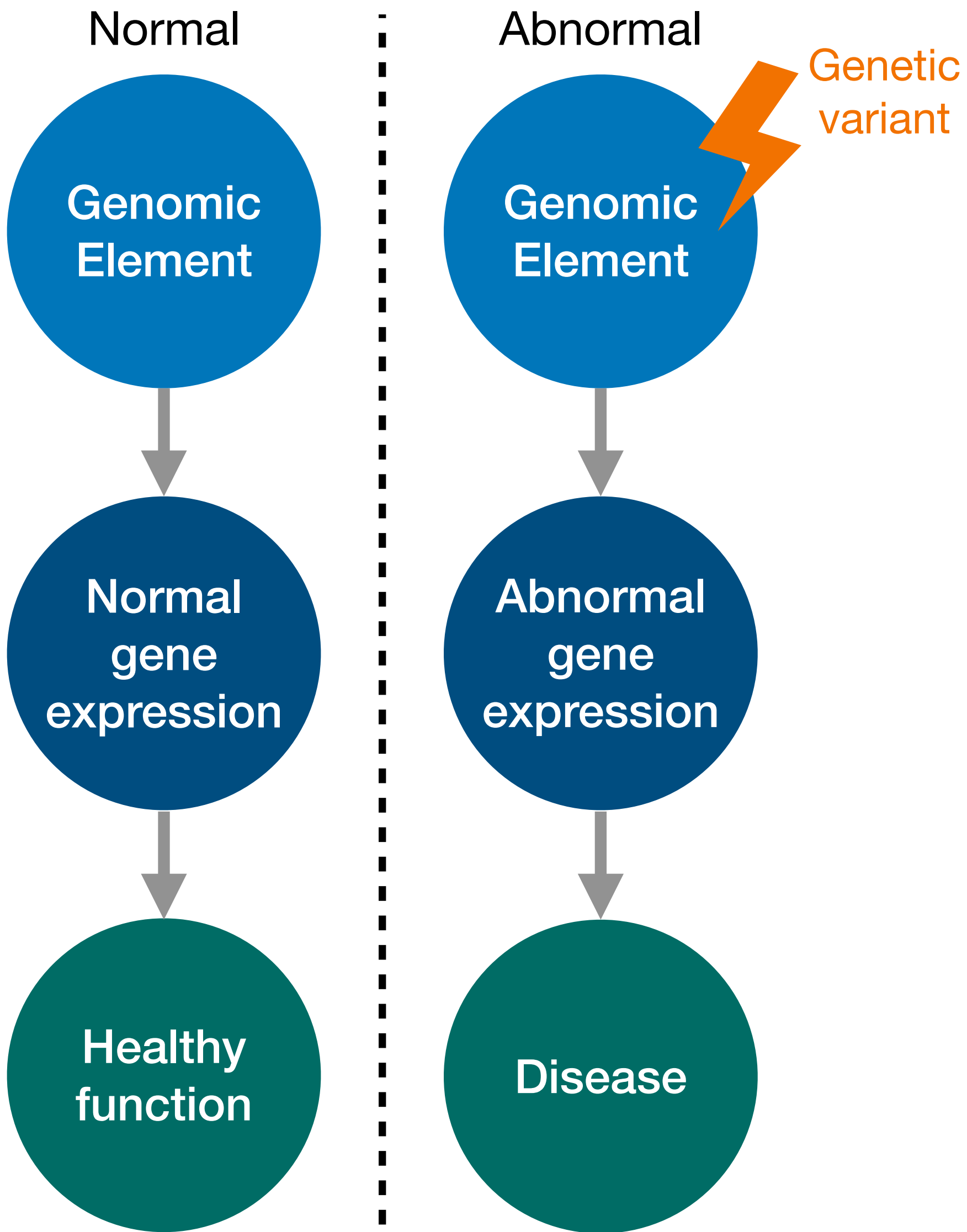
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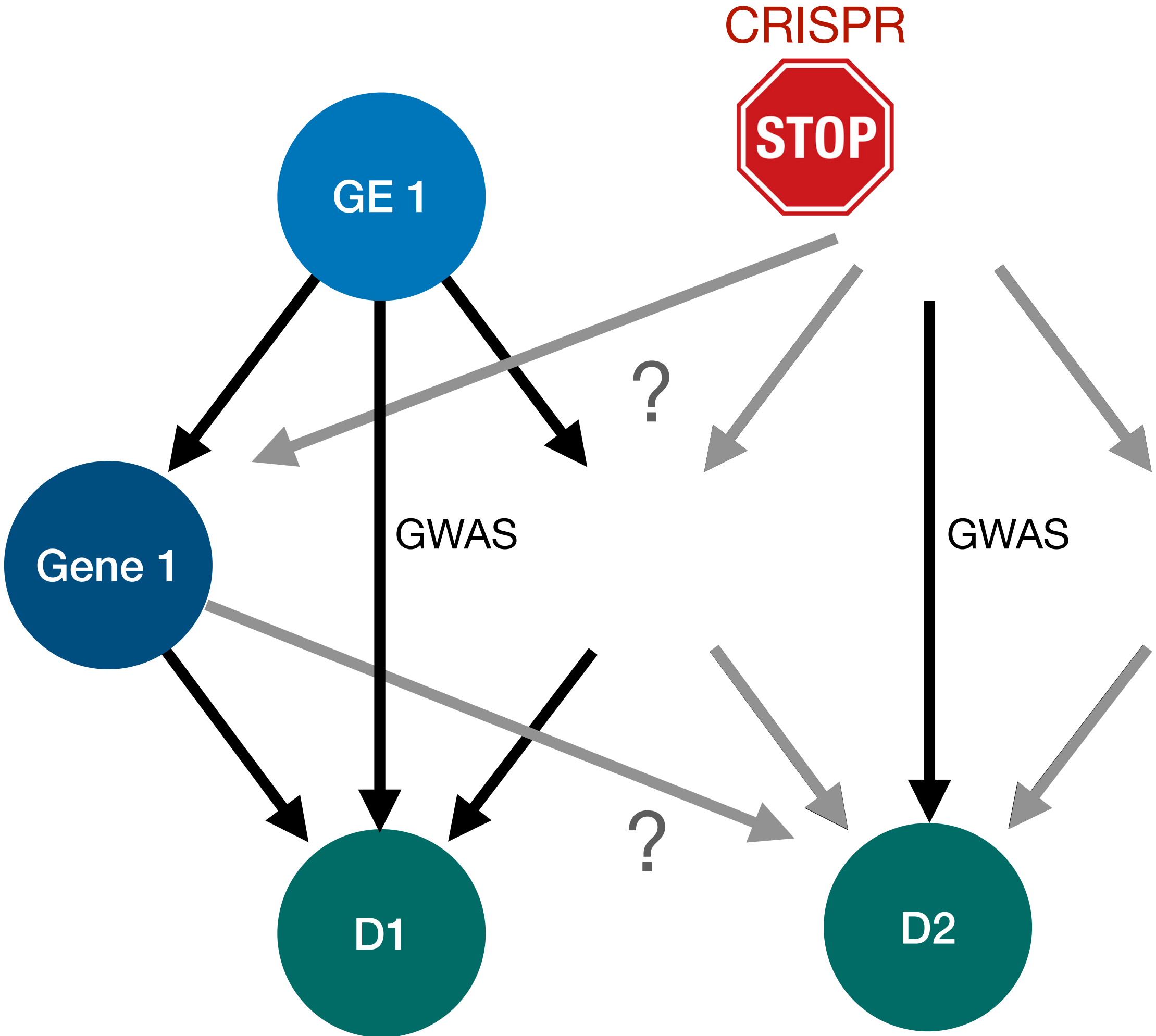
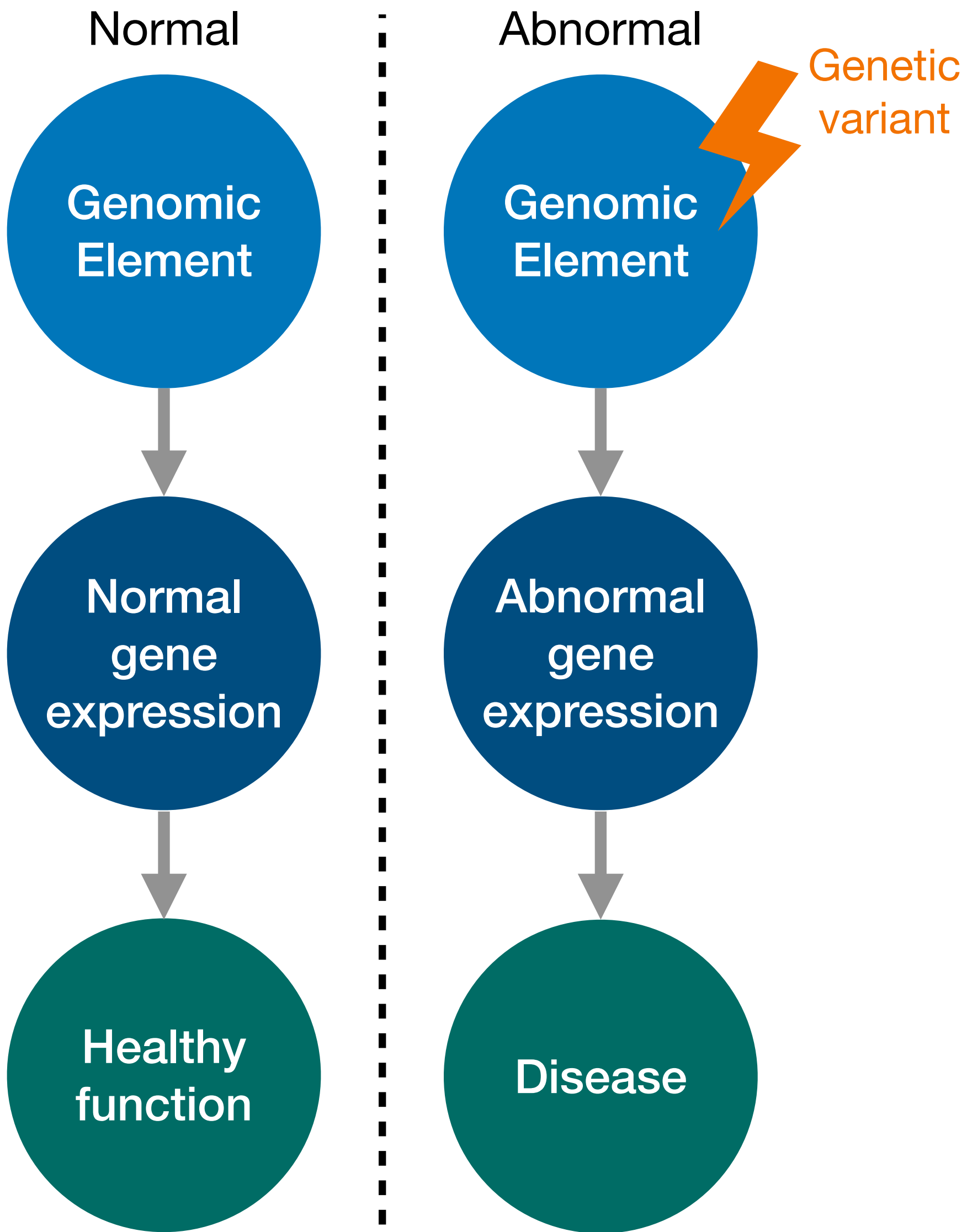
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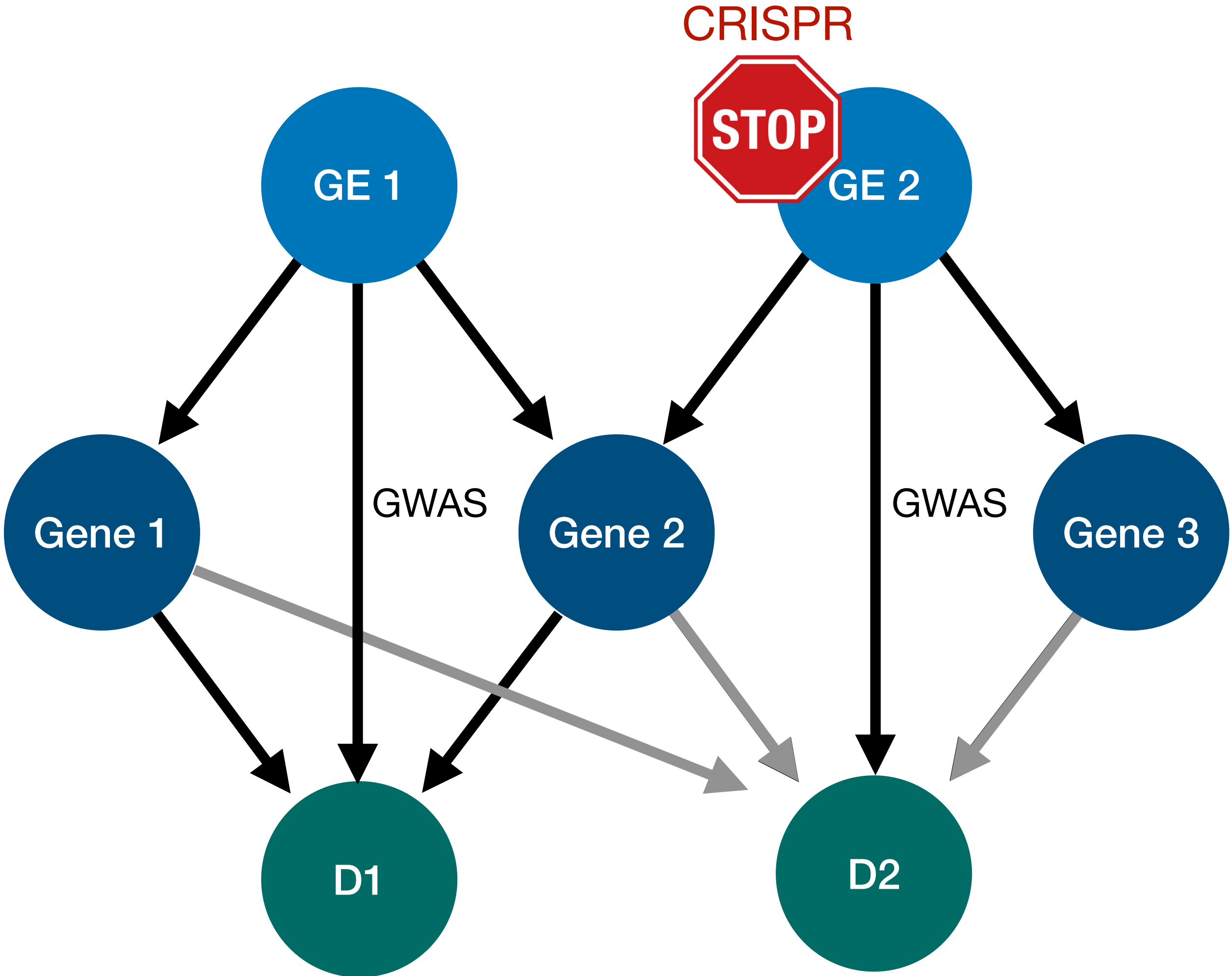
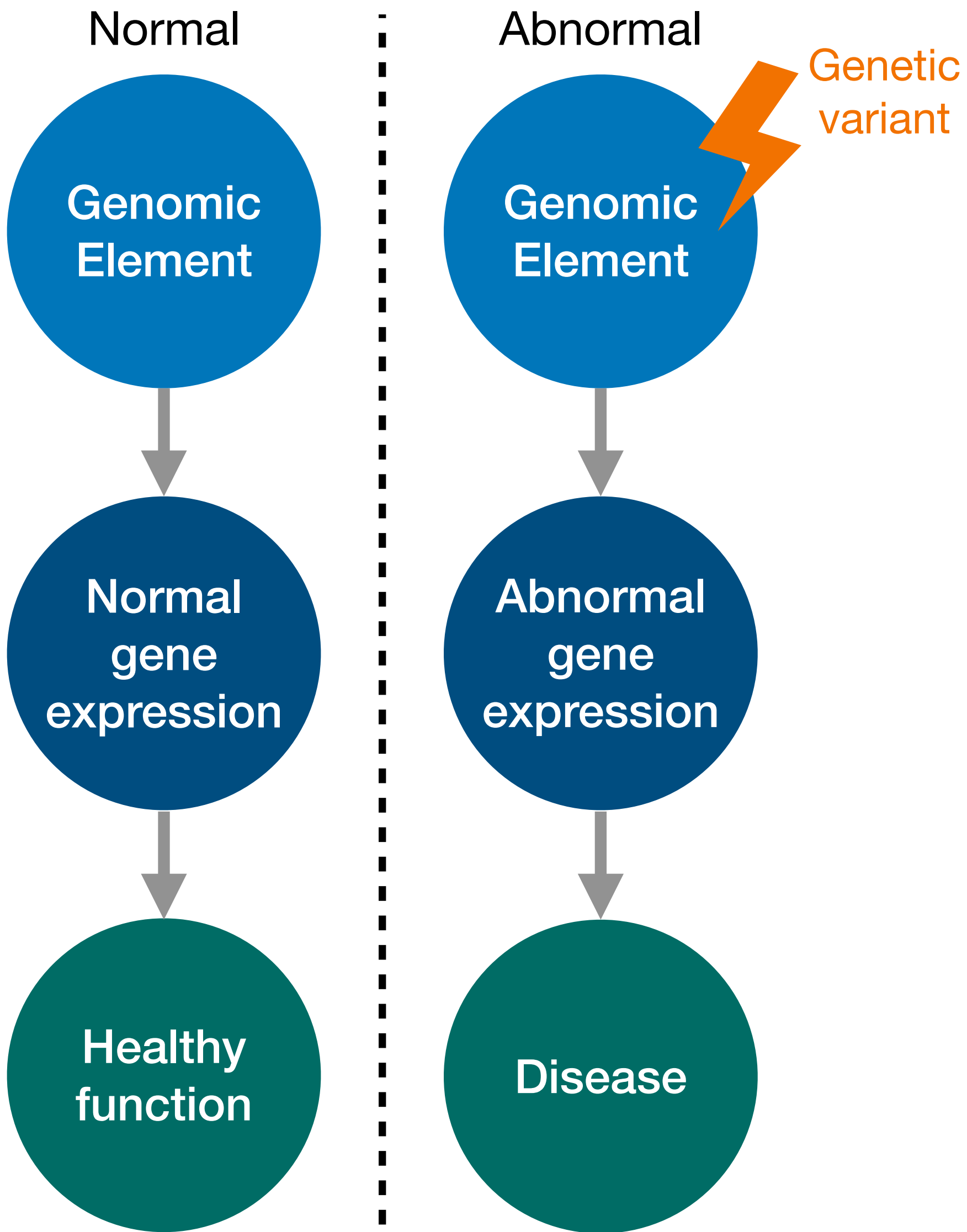
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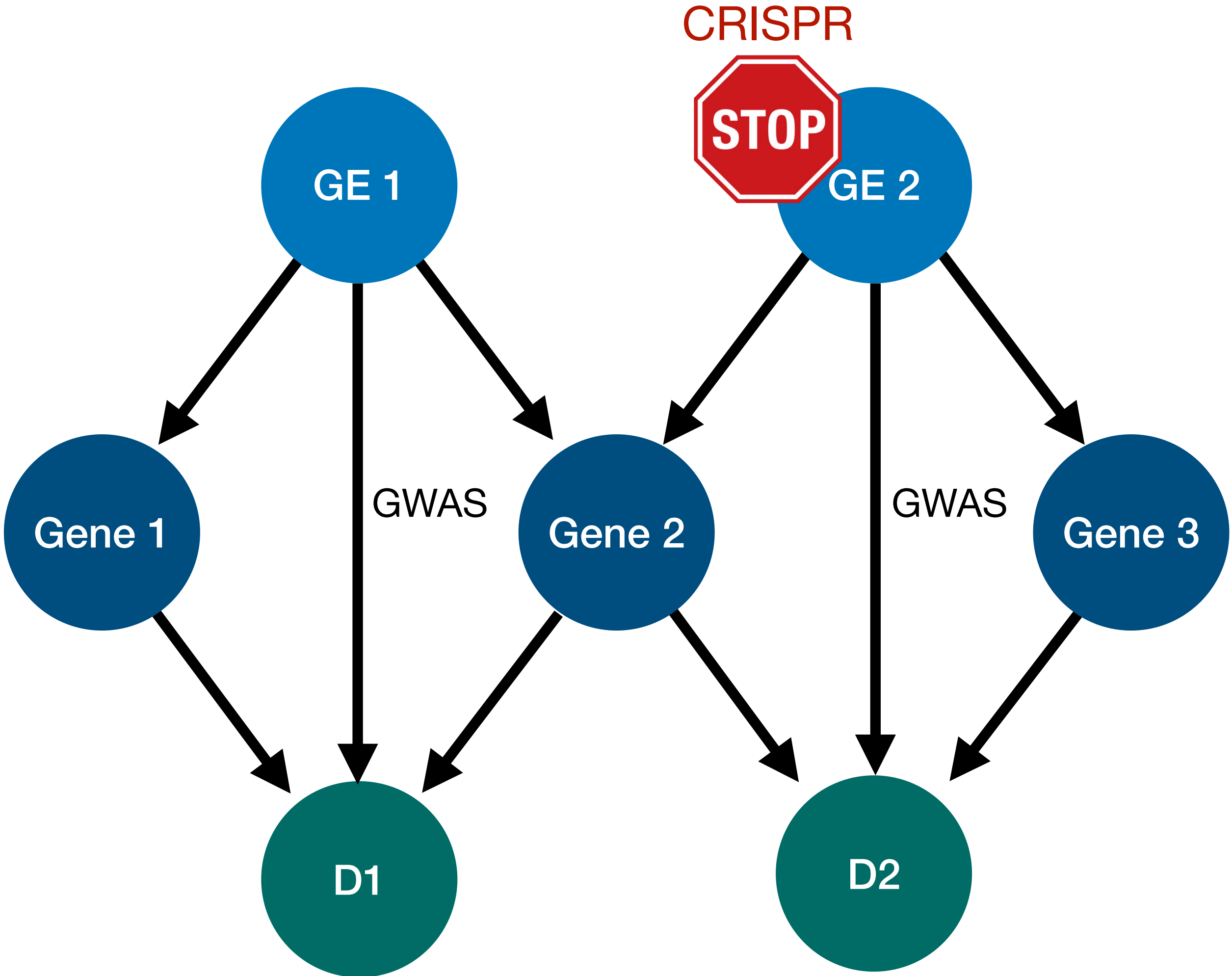
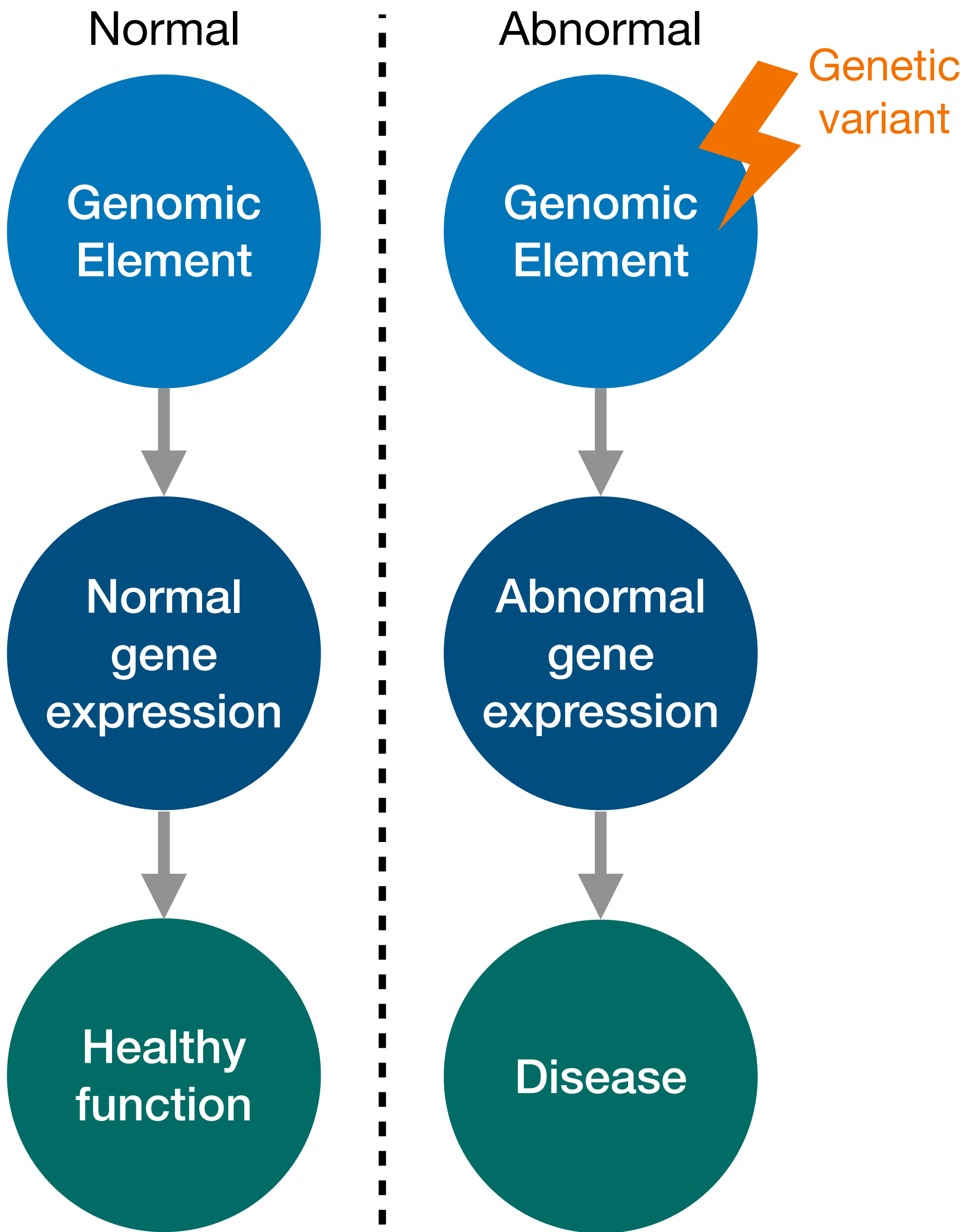
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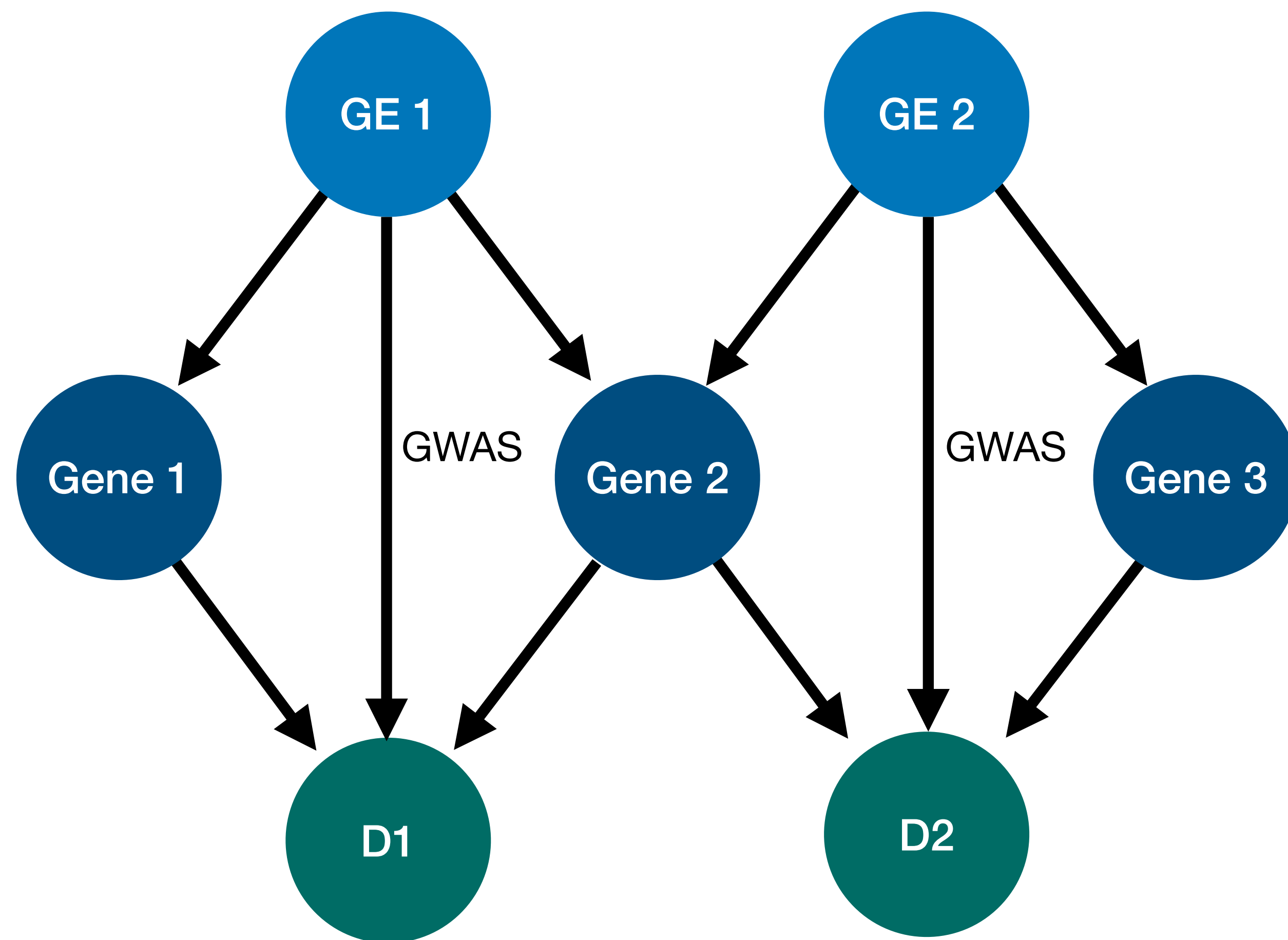
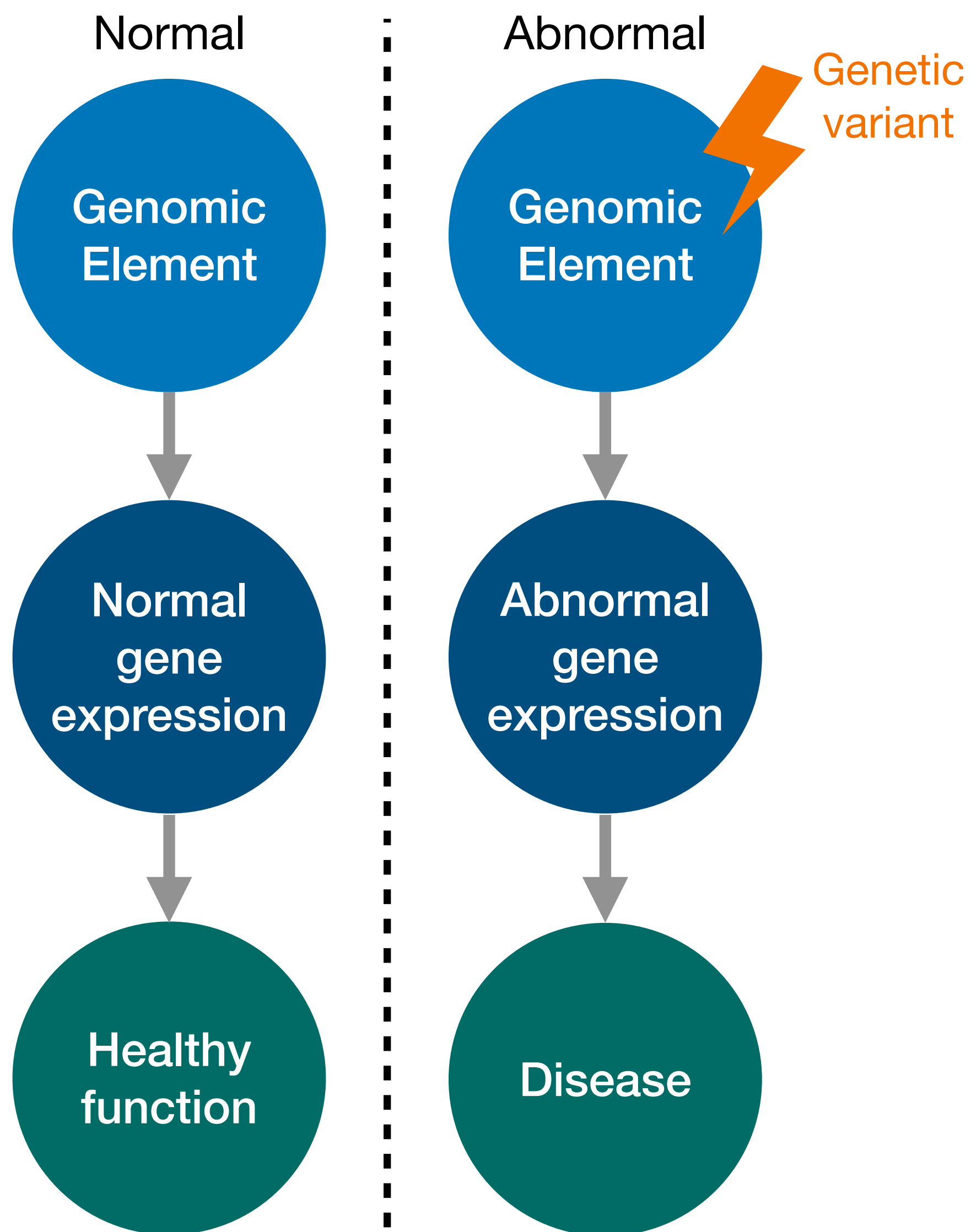
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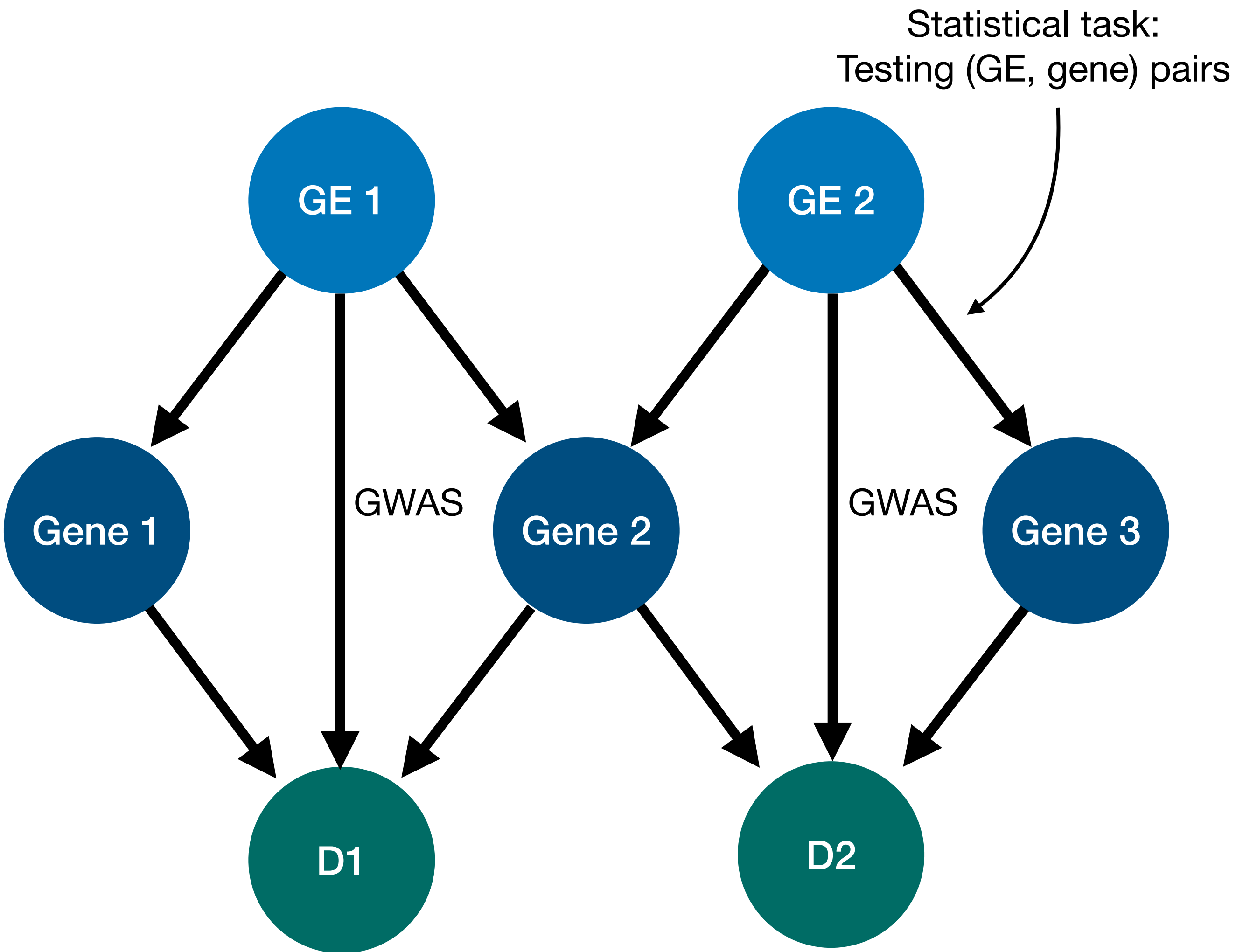
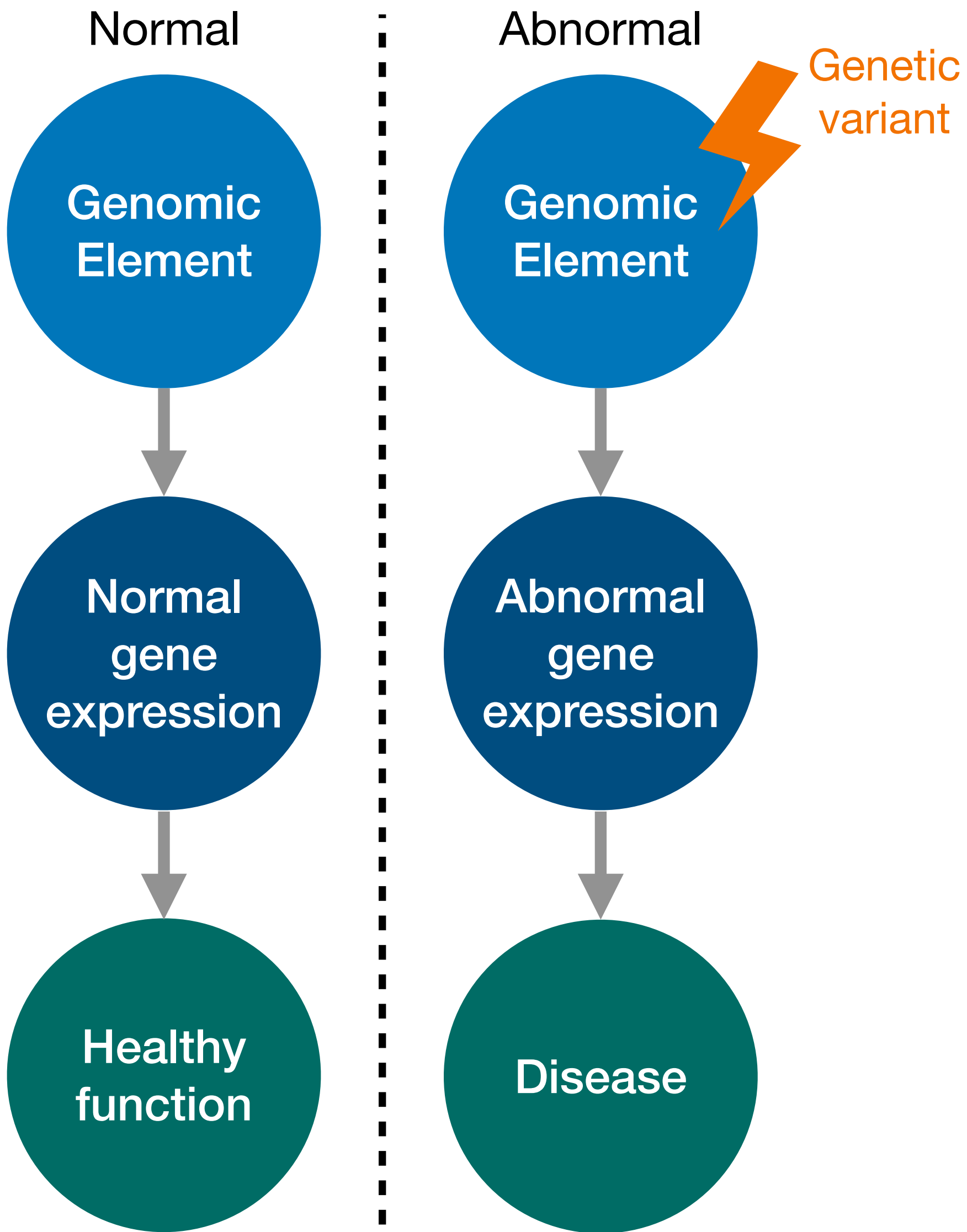
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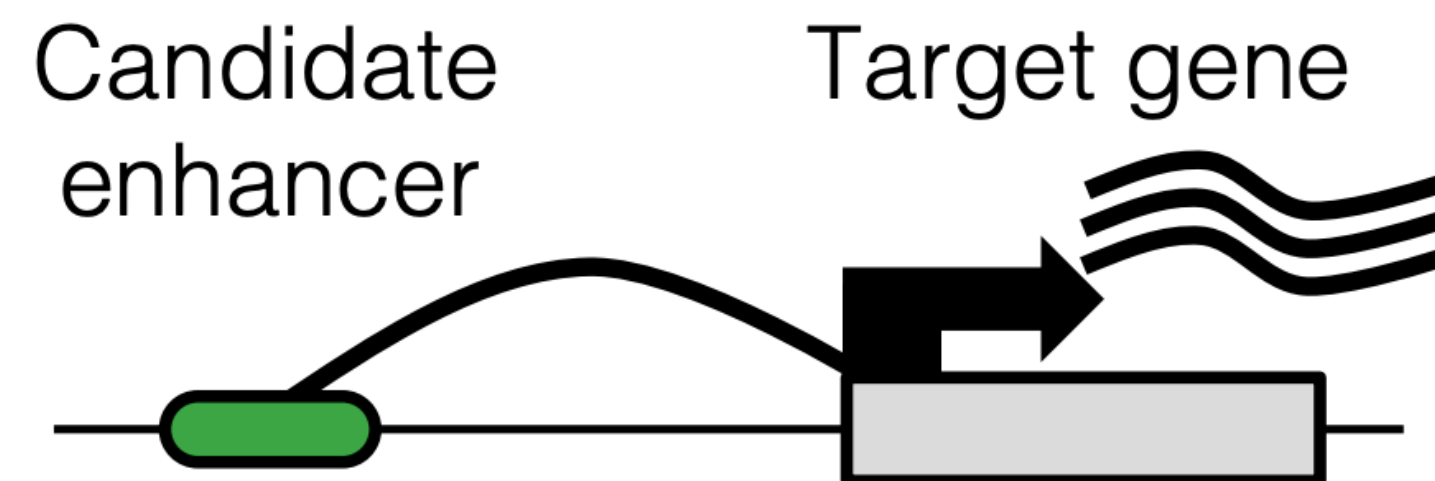
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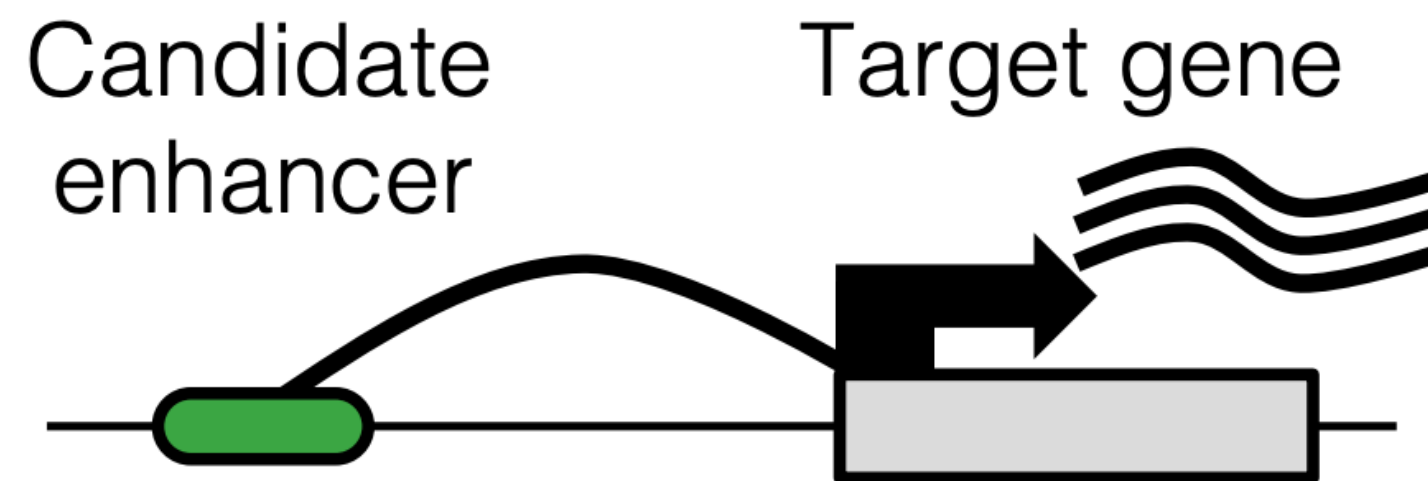
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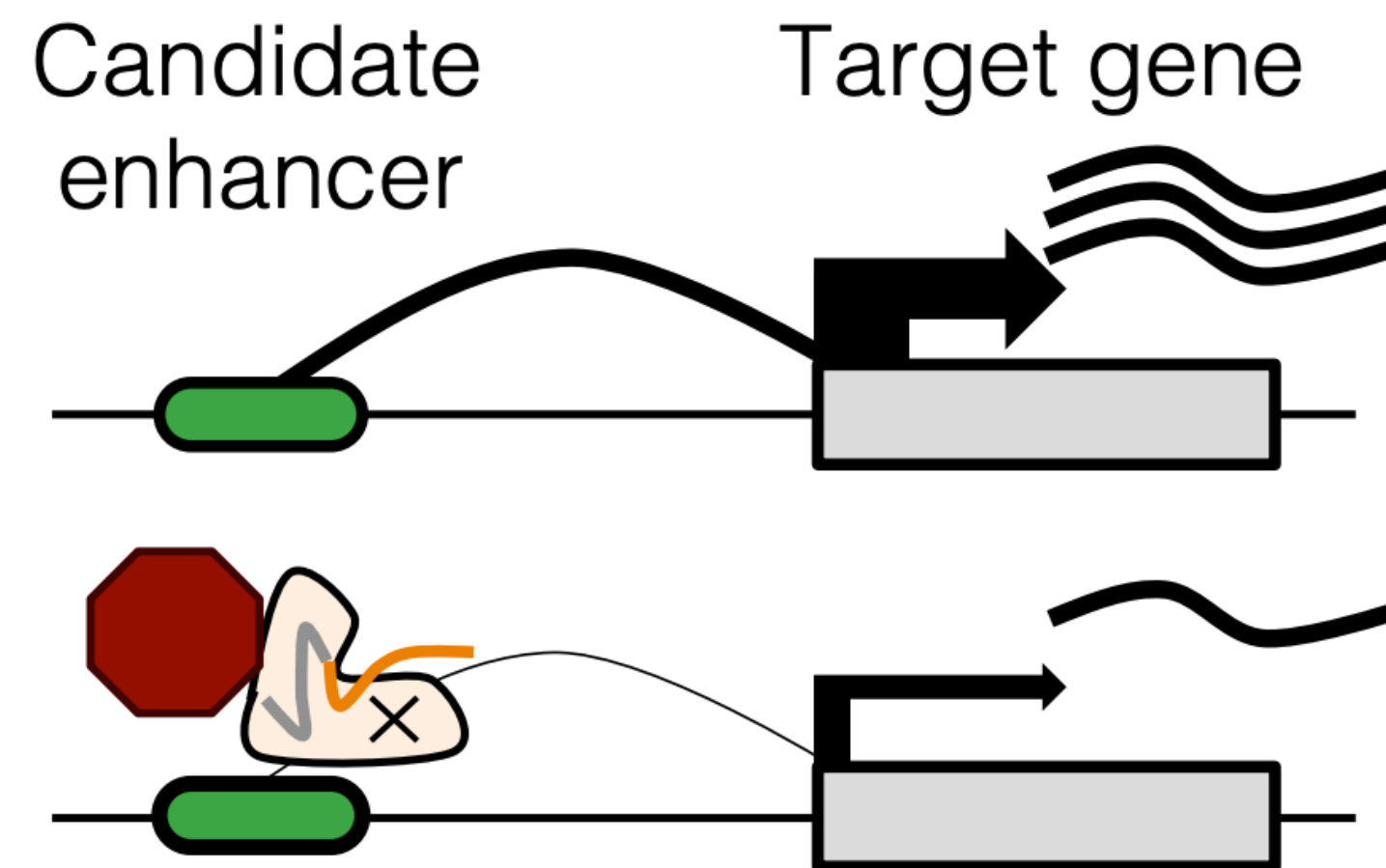
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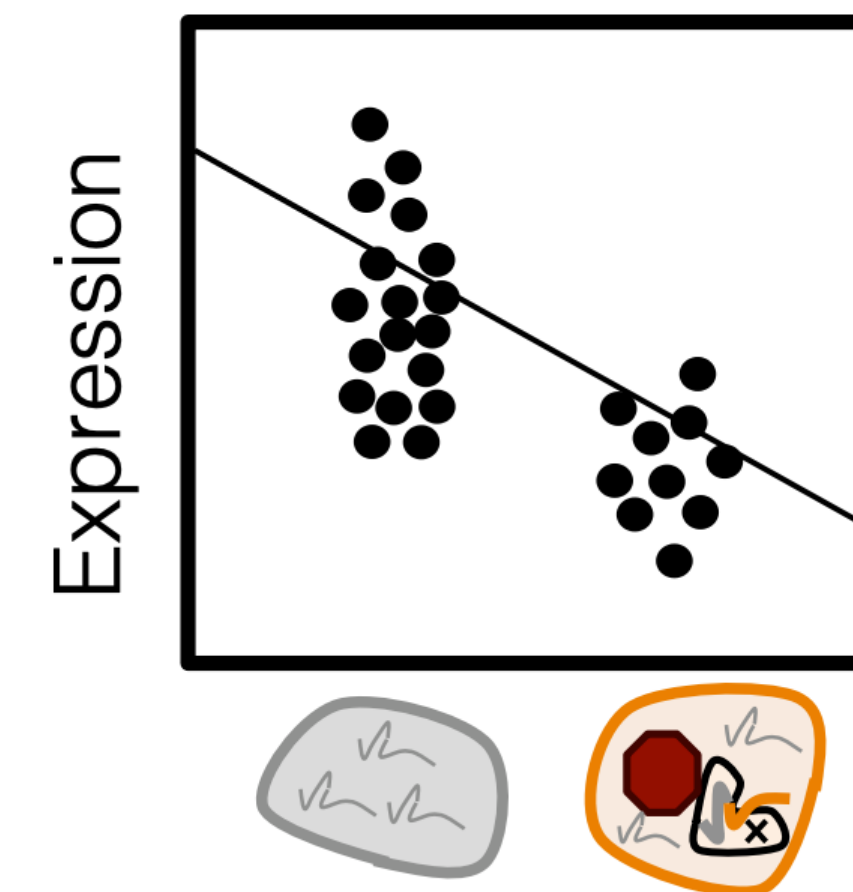
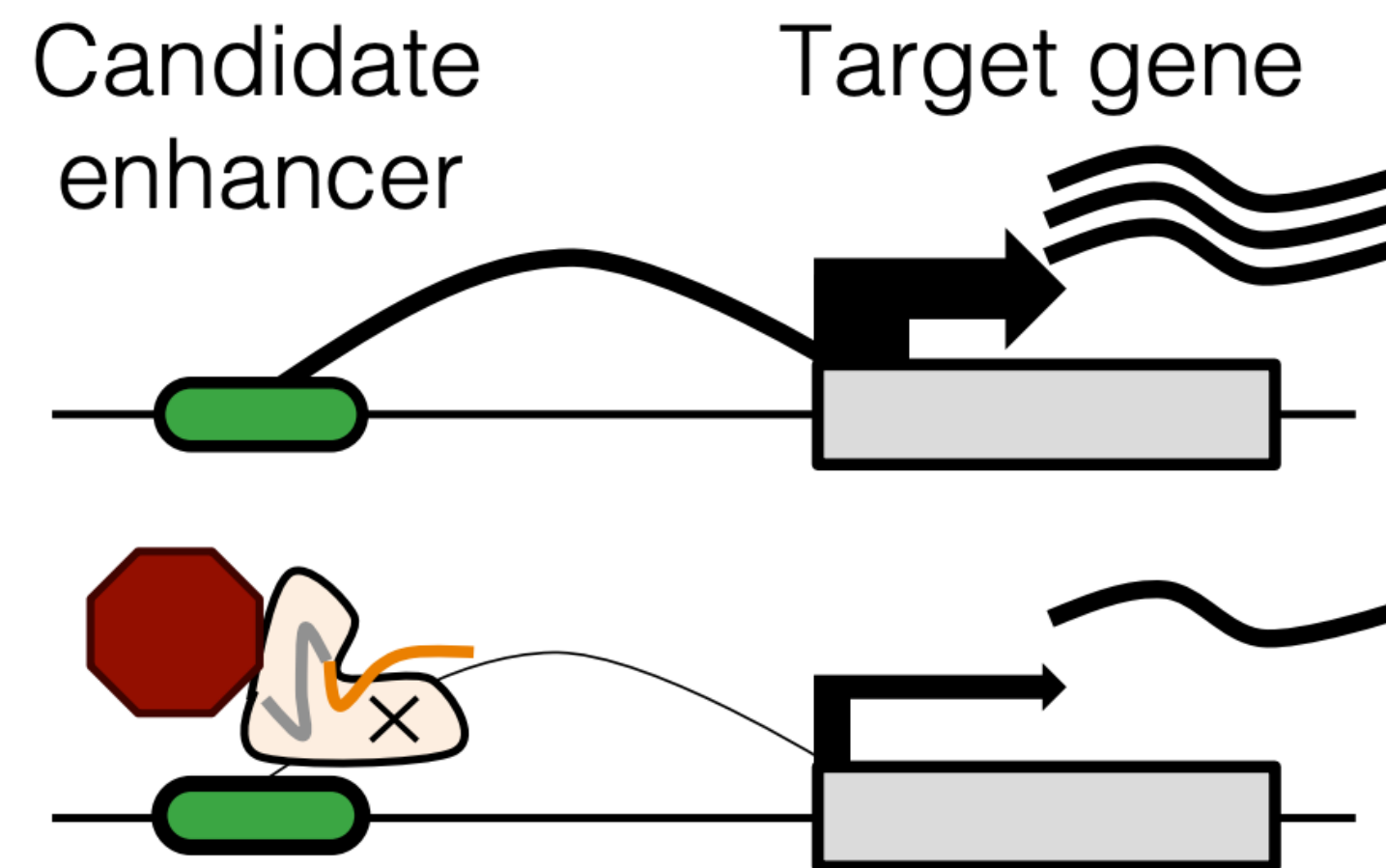
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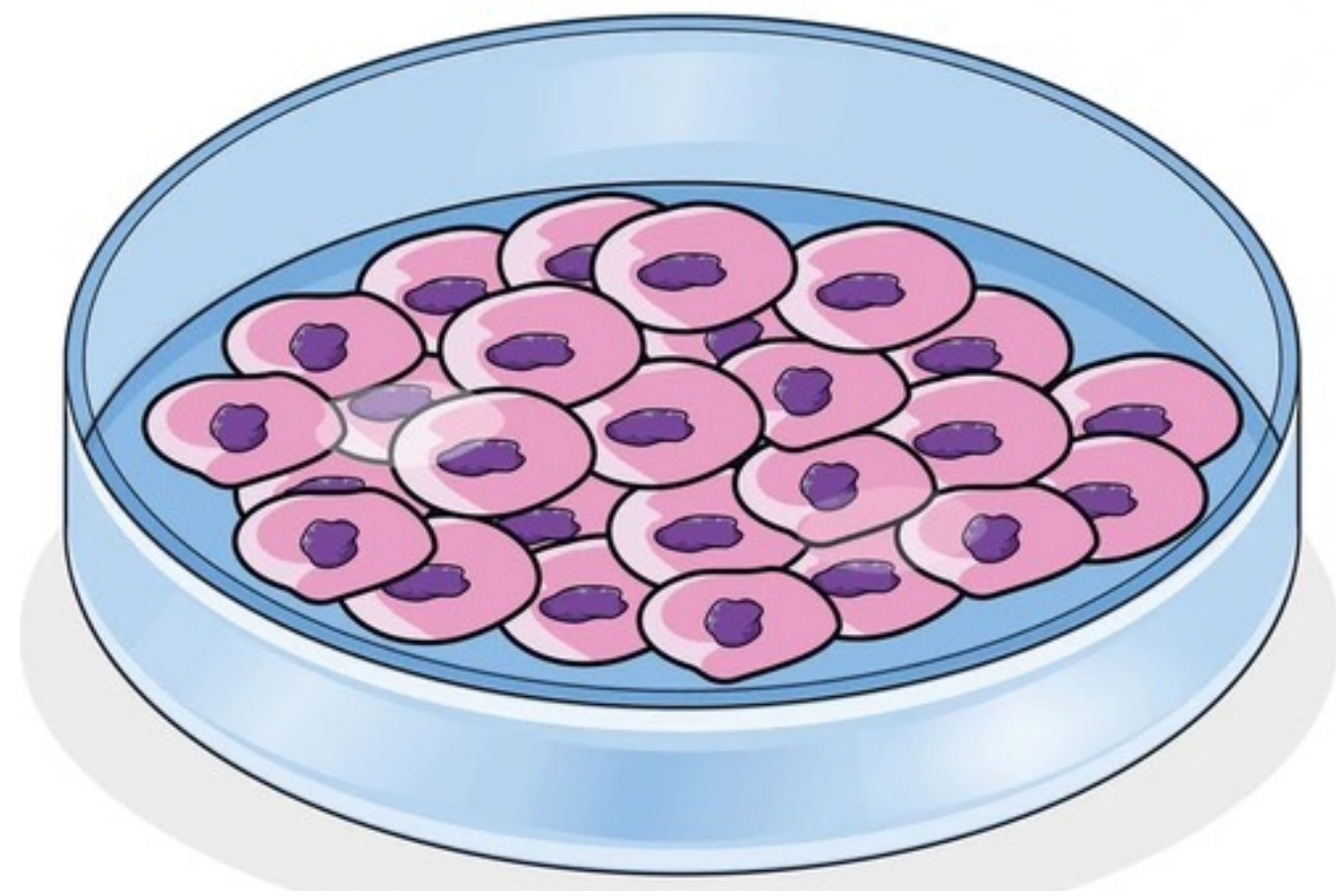
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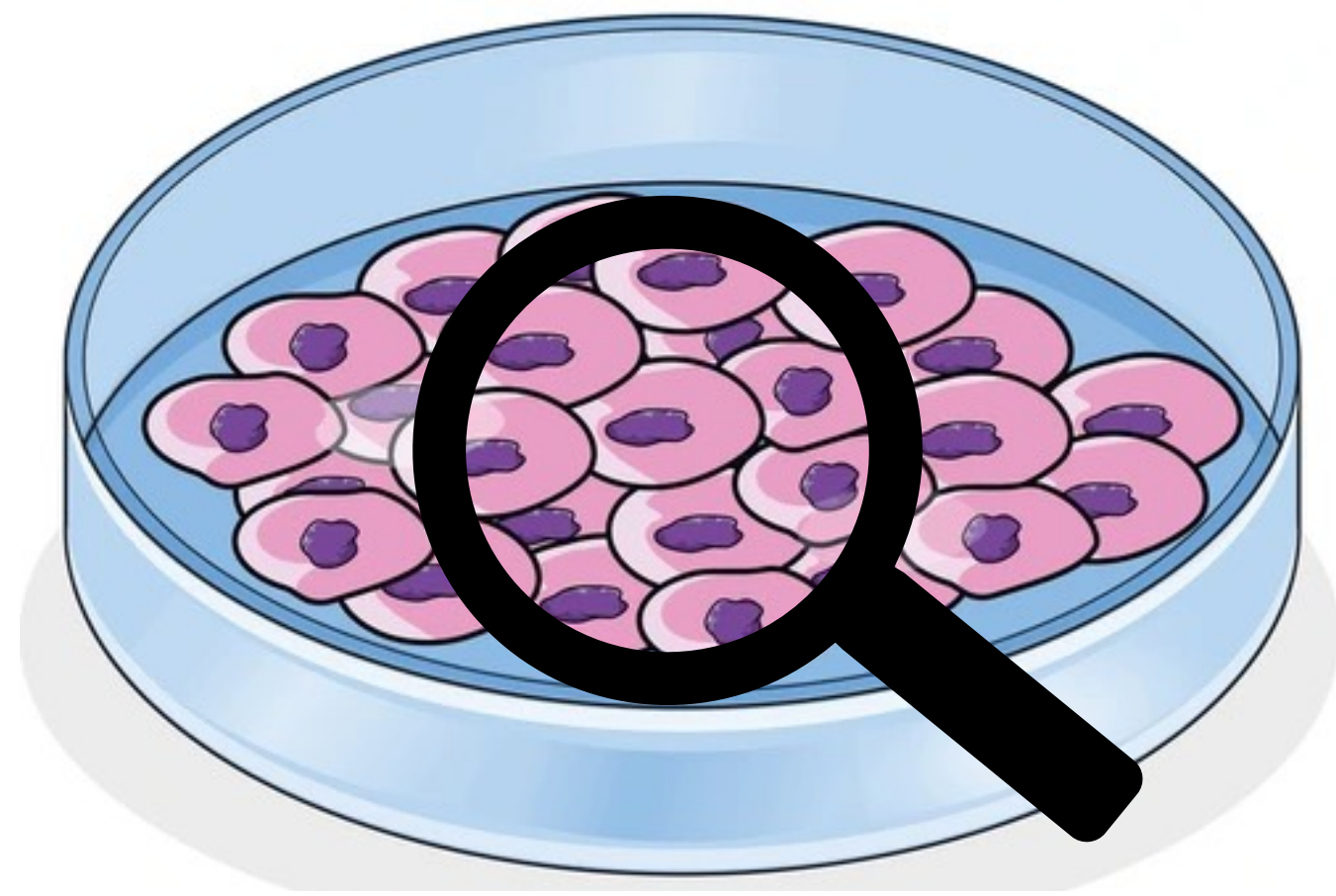
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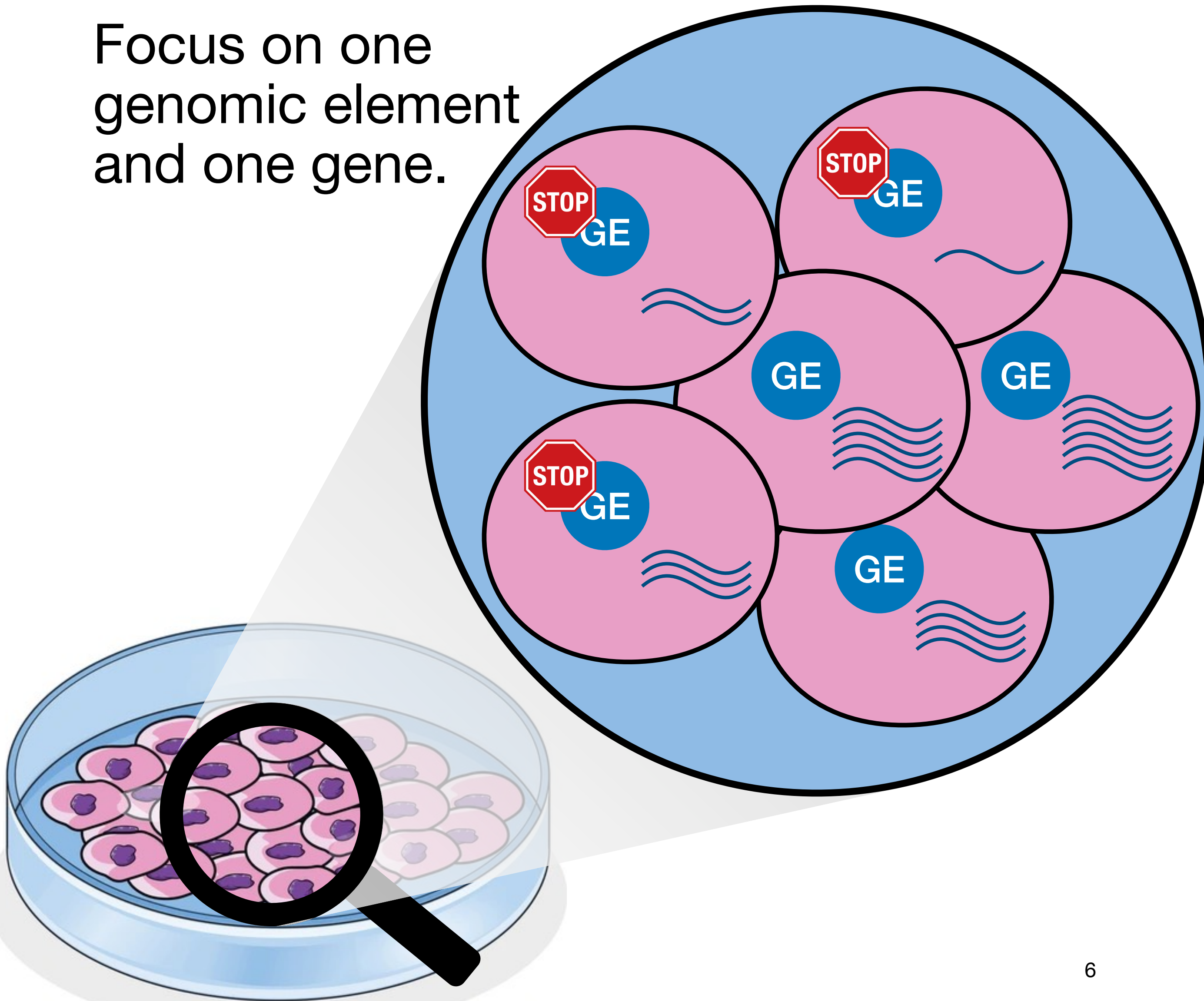
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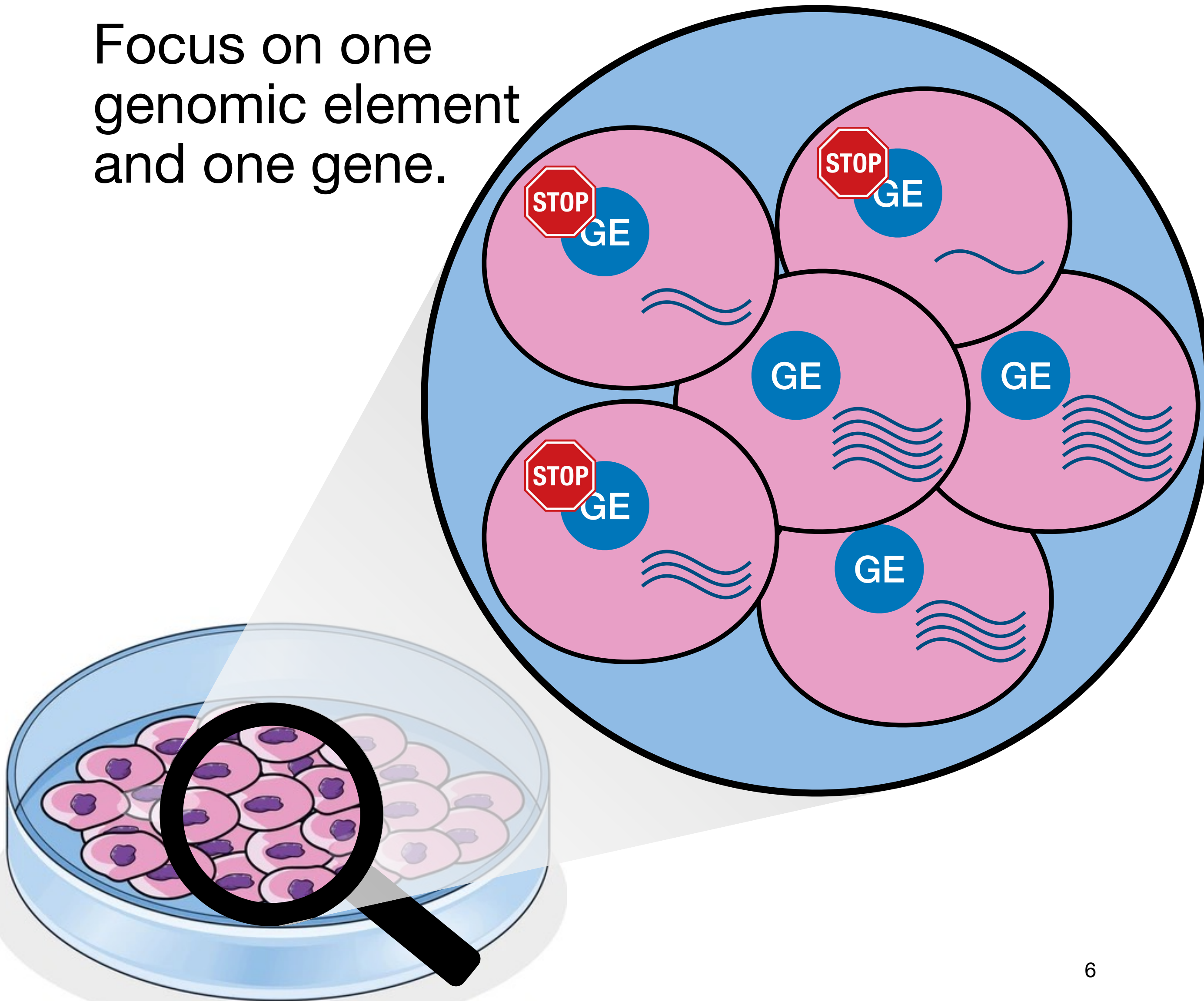
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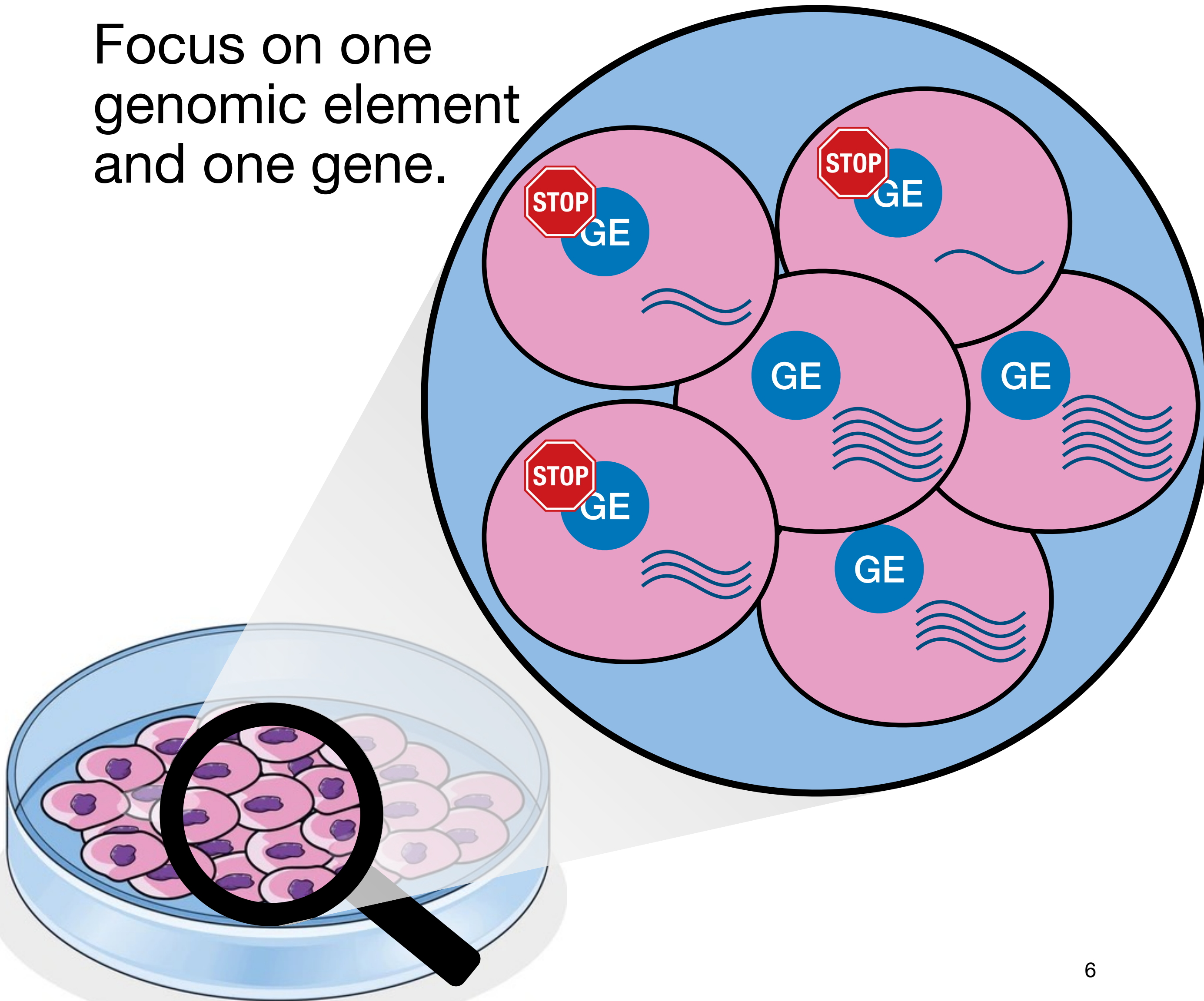


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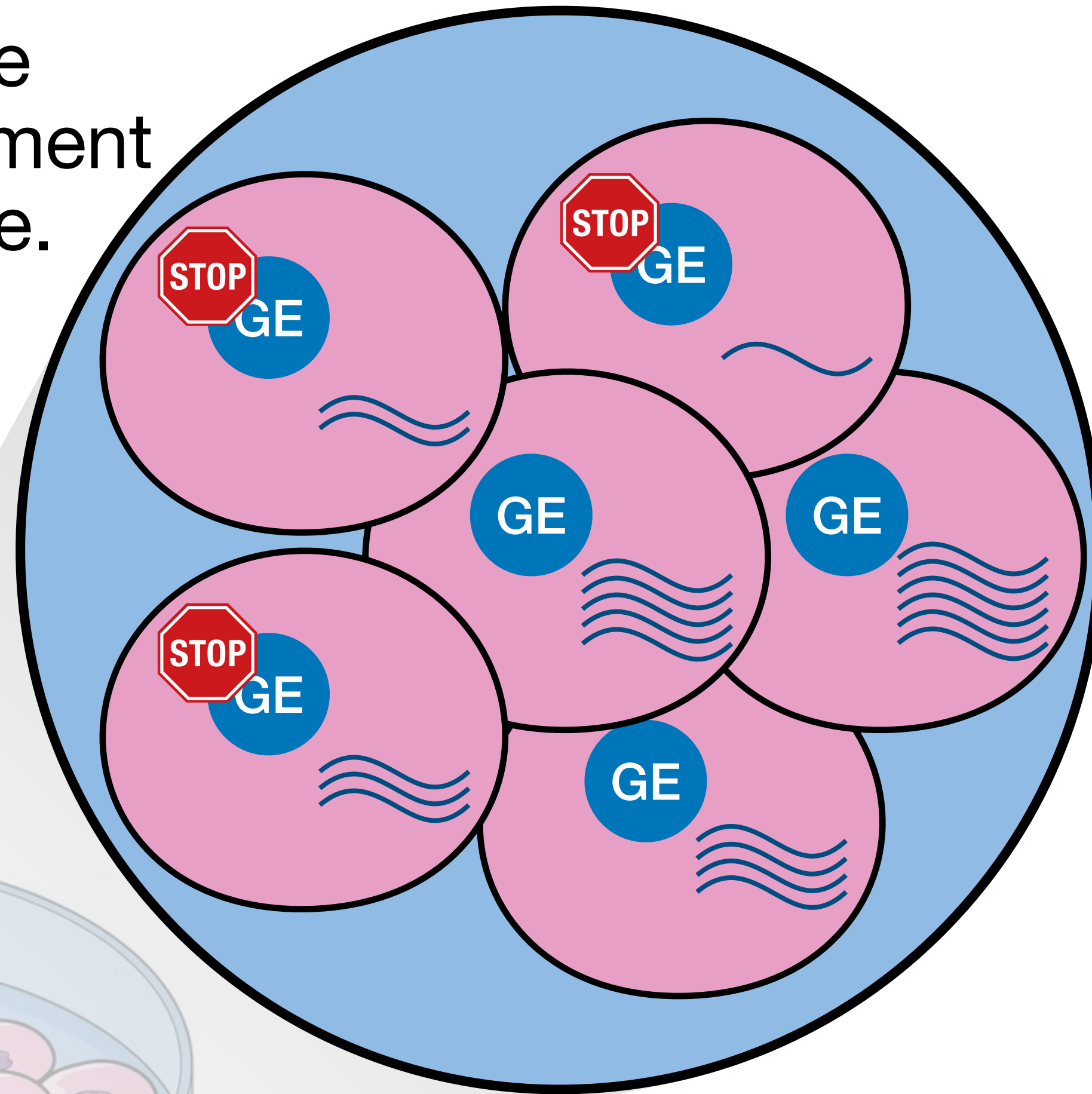
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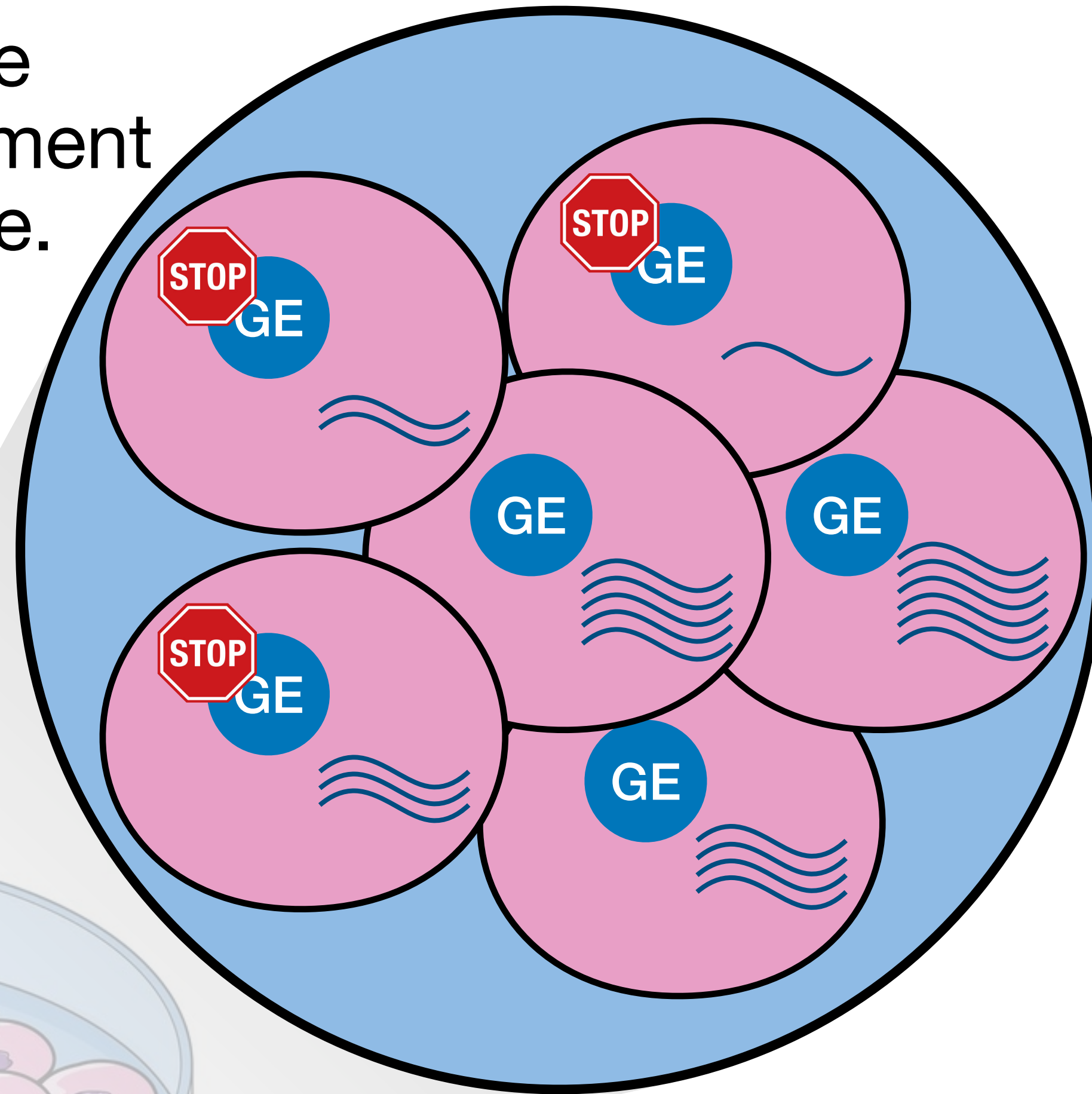


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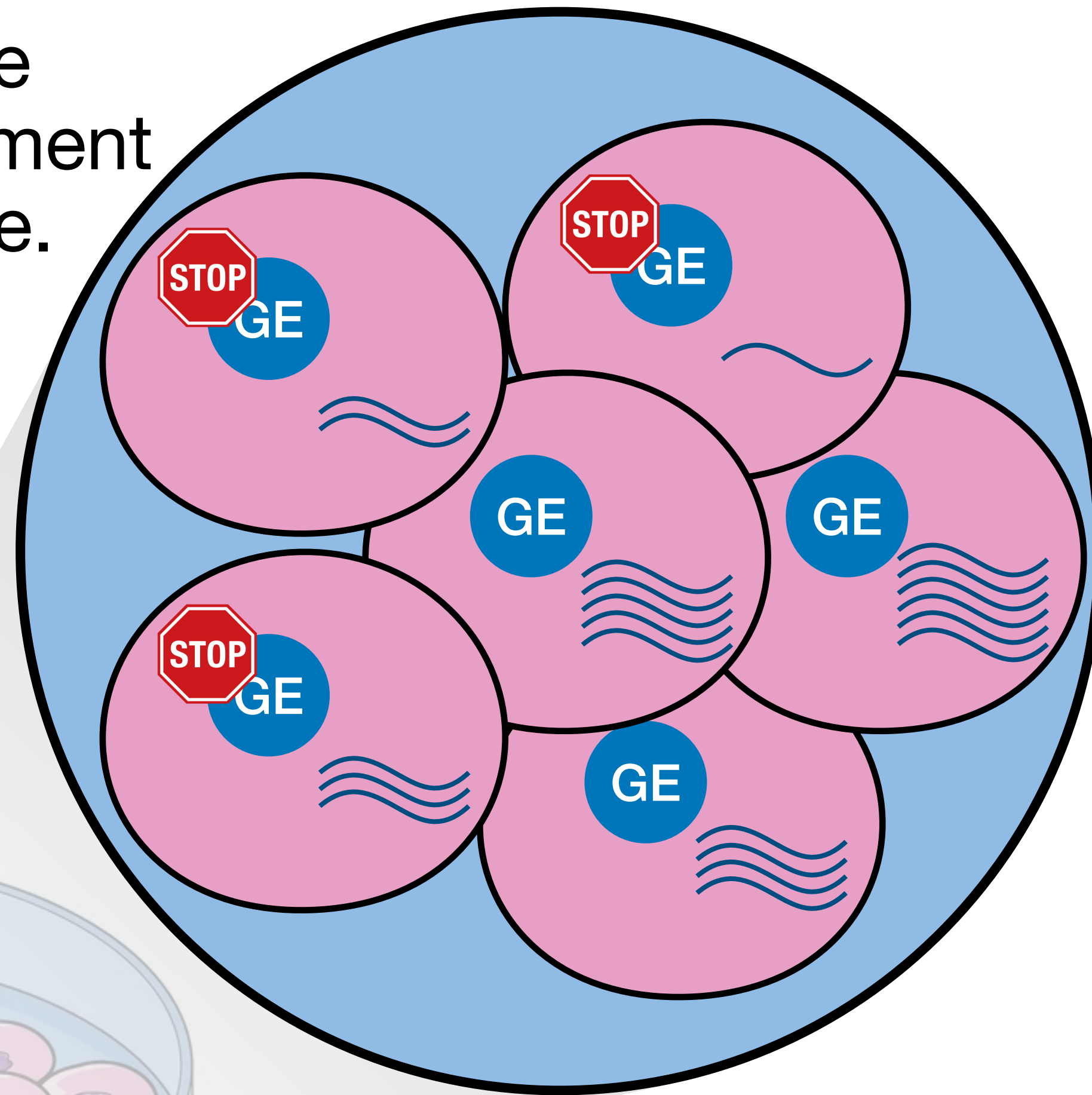


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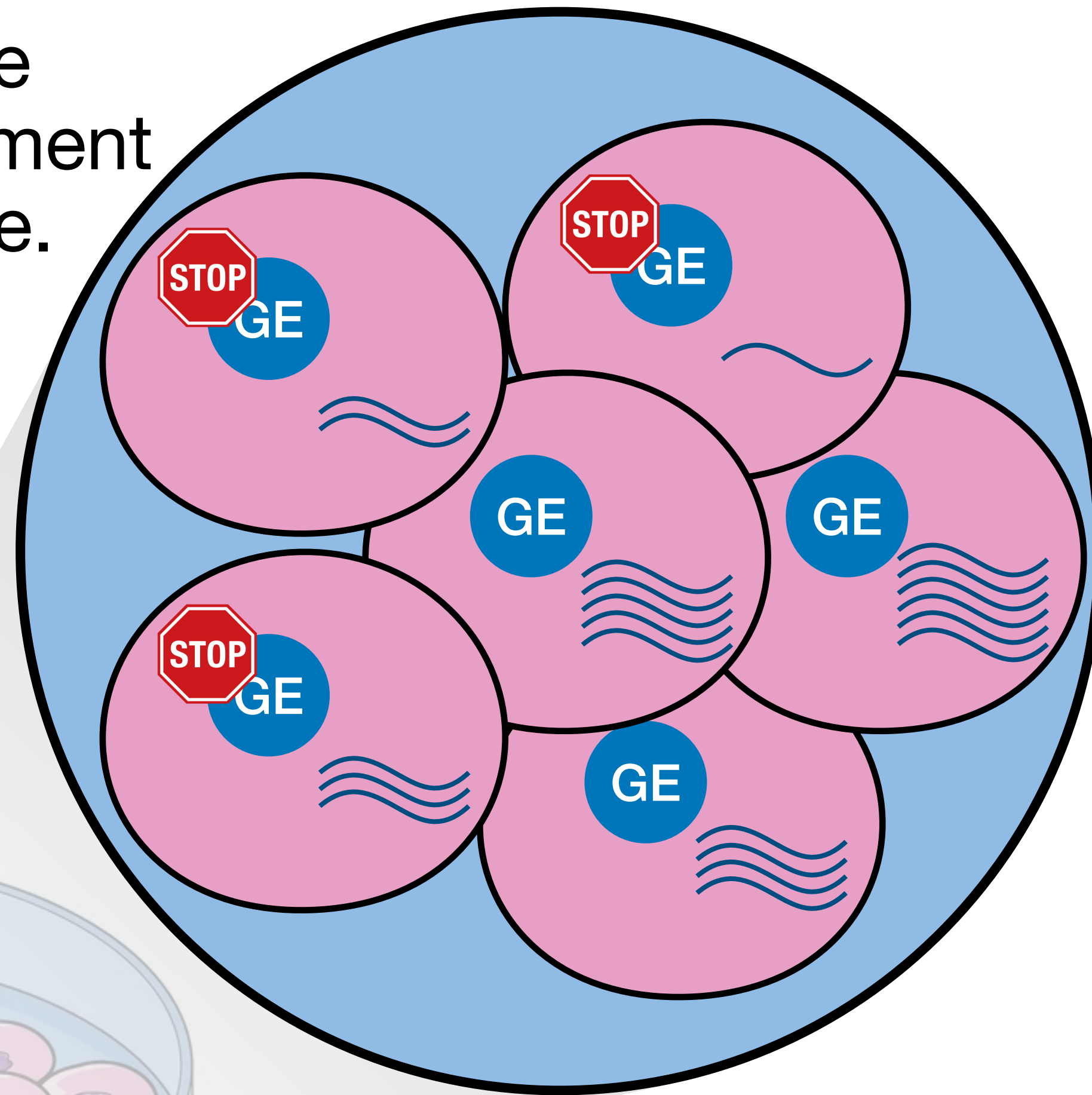
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In Gasperini (2019) data, we have $n \approx 200,000$ cells, $\approx 13,000$ genes, and $\approx 6,000$ genomic elements.

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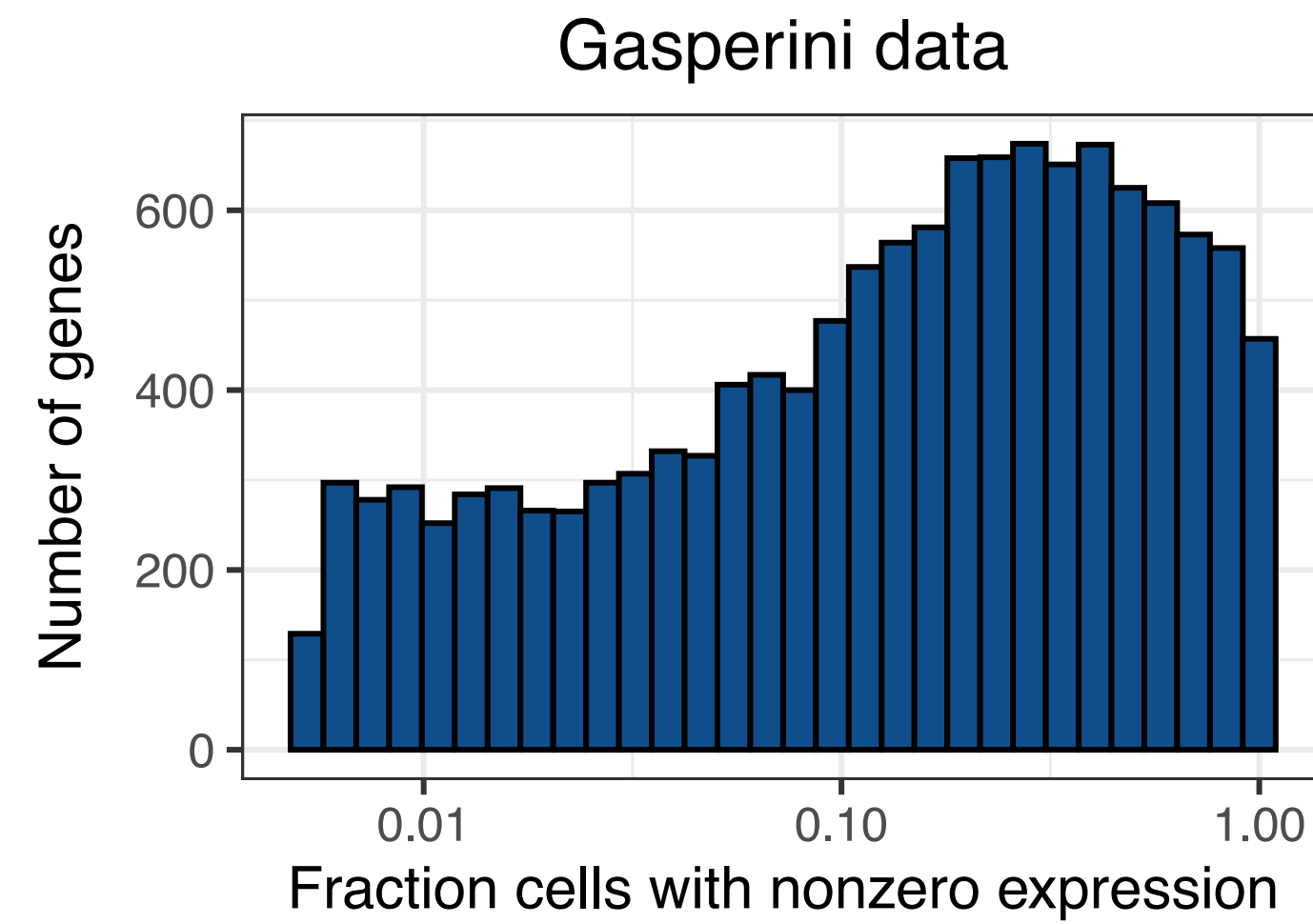
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The goal is to test

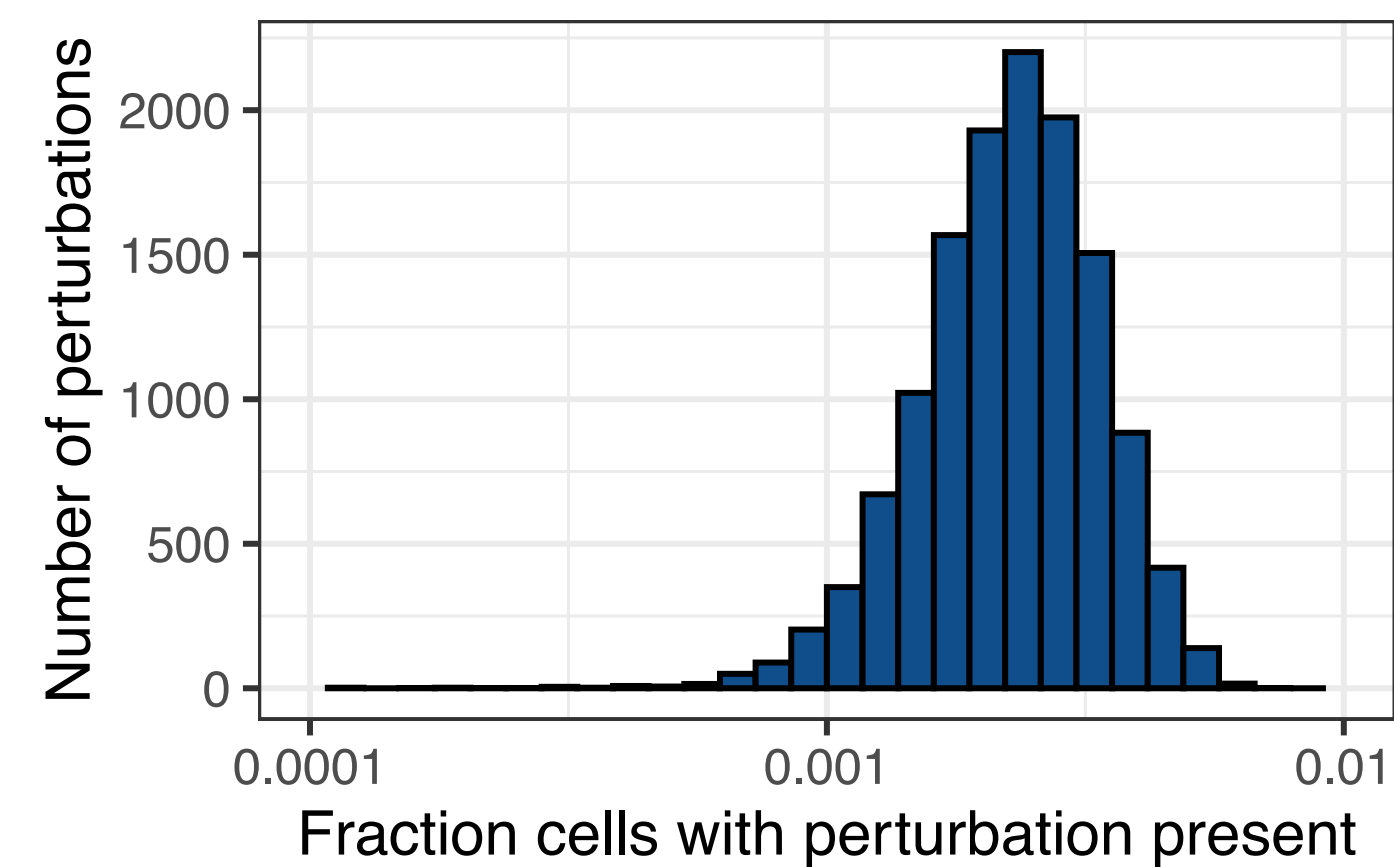
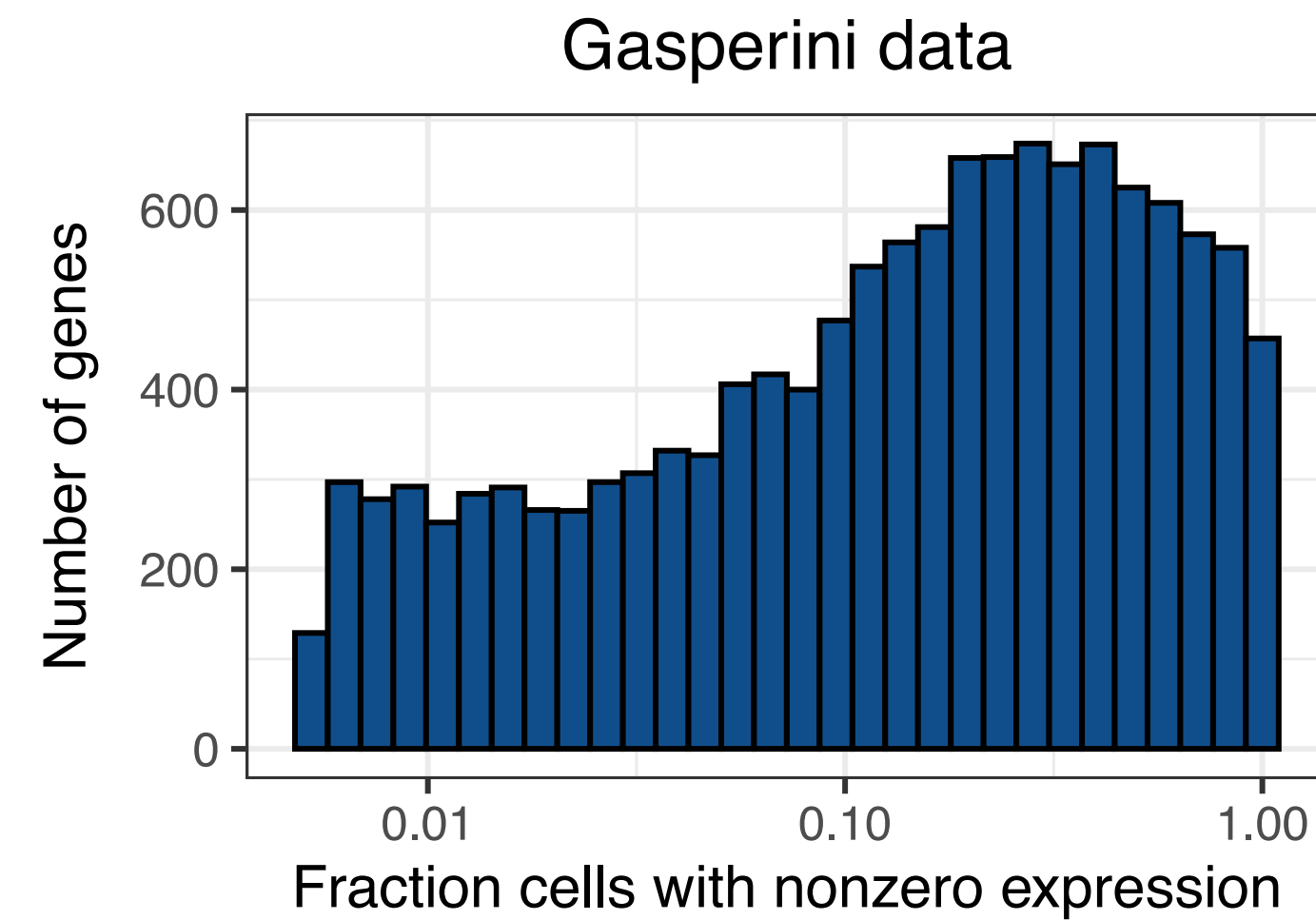
$$H_0 : X_i \perp\!\!\!\perp Y_i \mid Z_i.$$

Challenge: The data are highly sparse

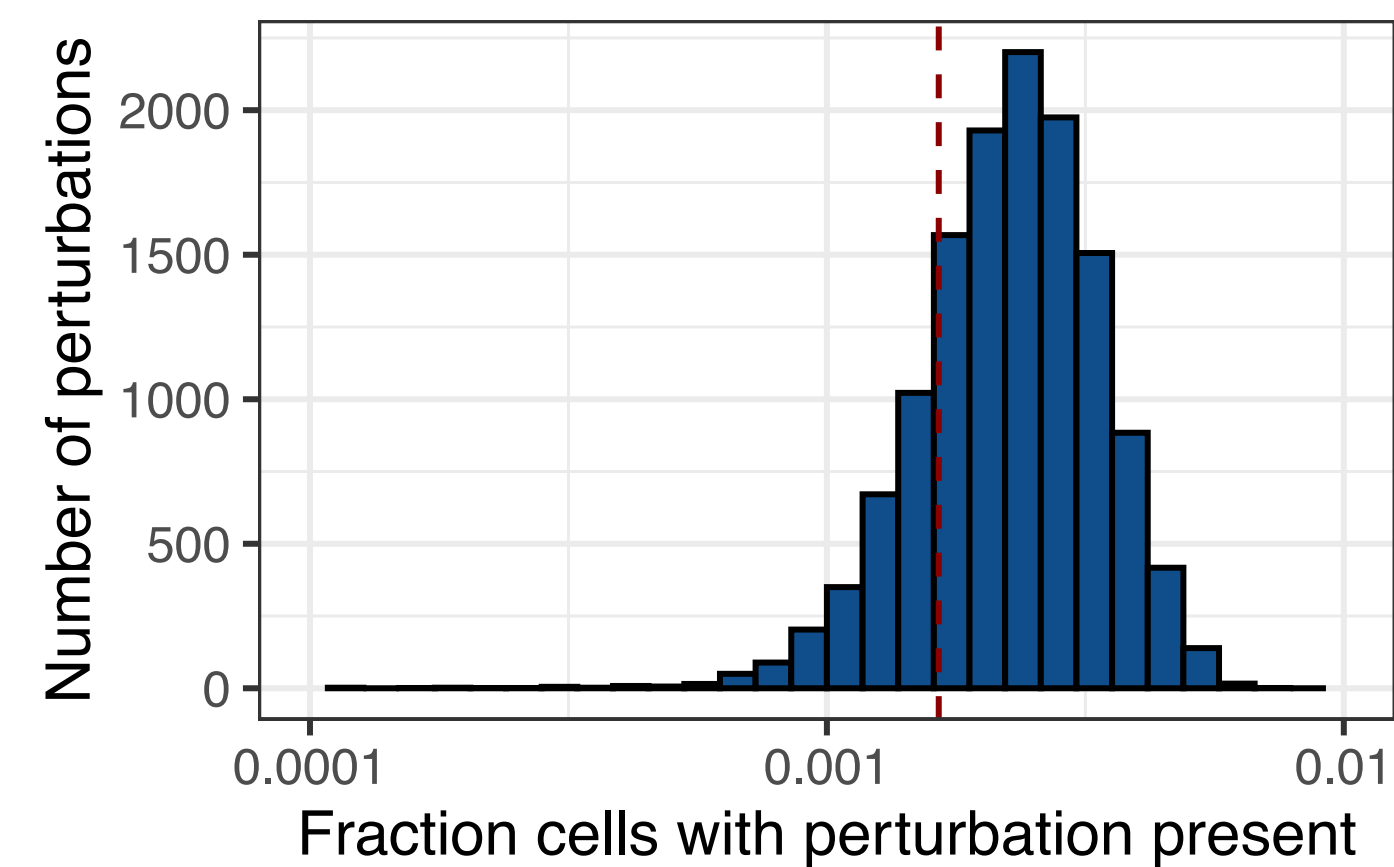
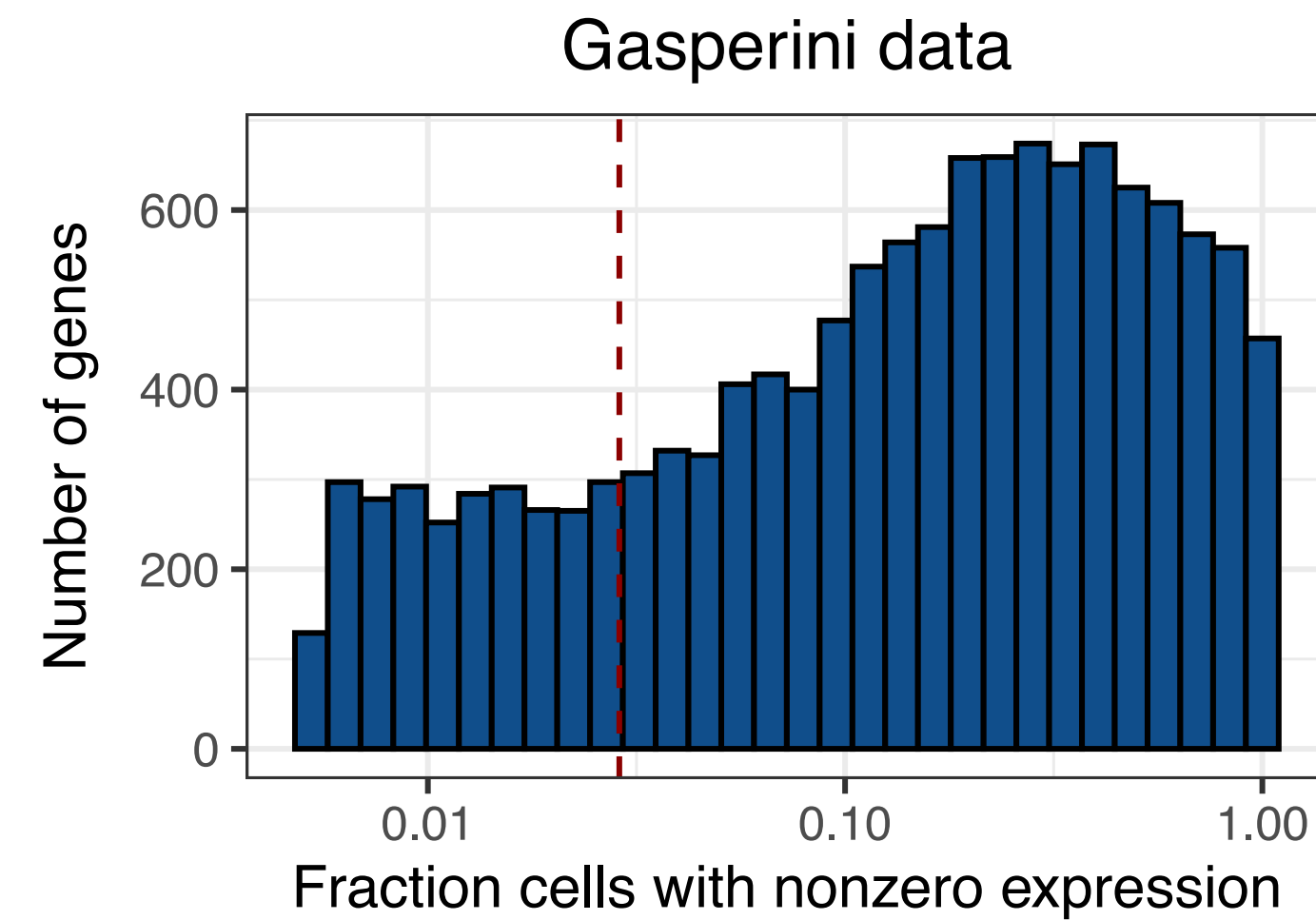
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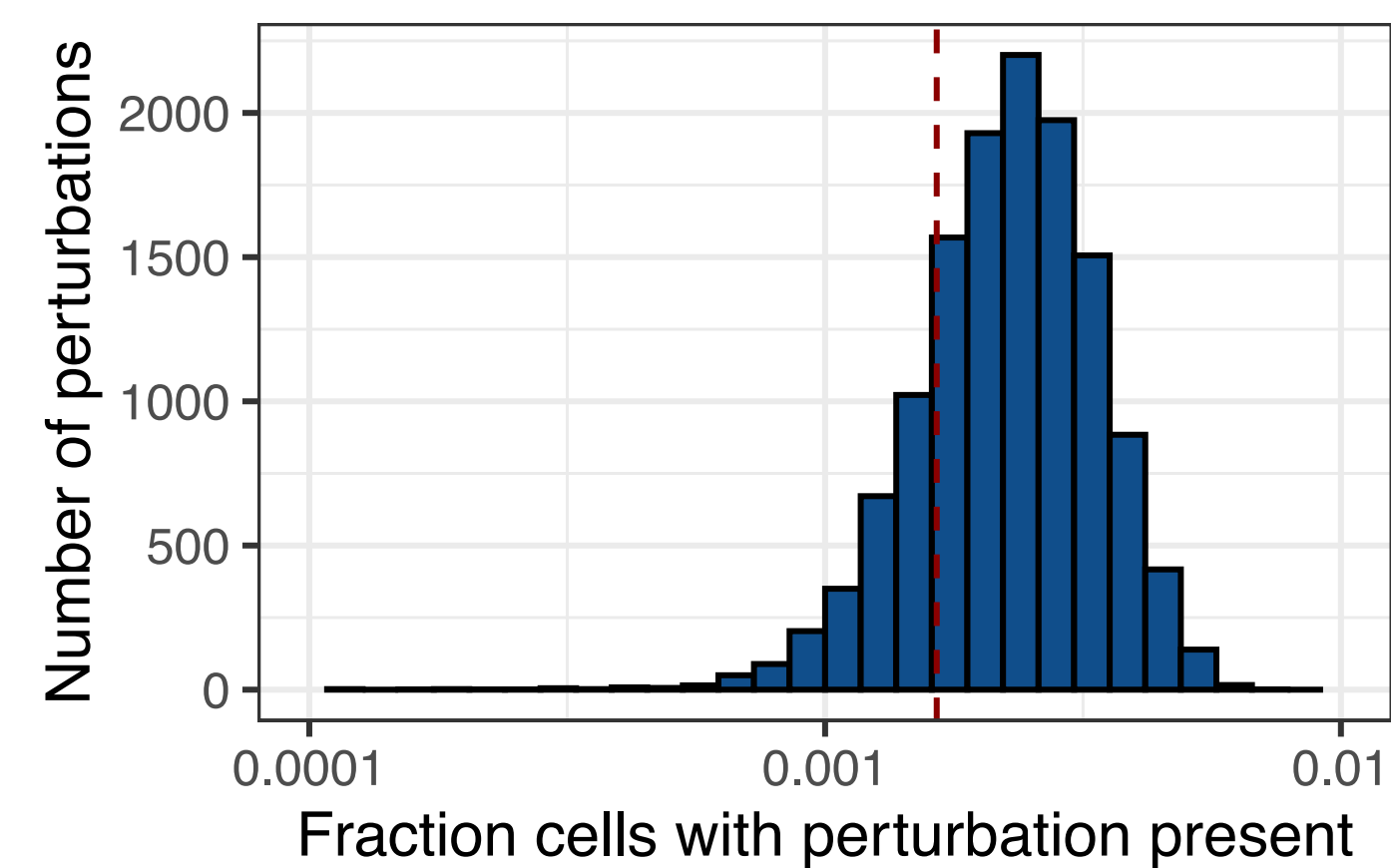
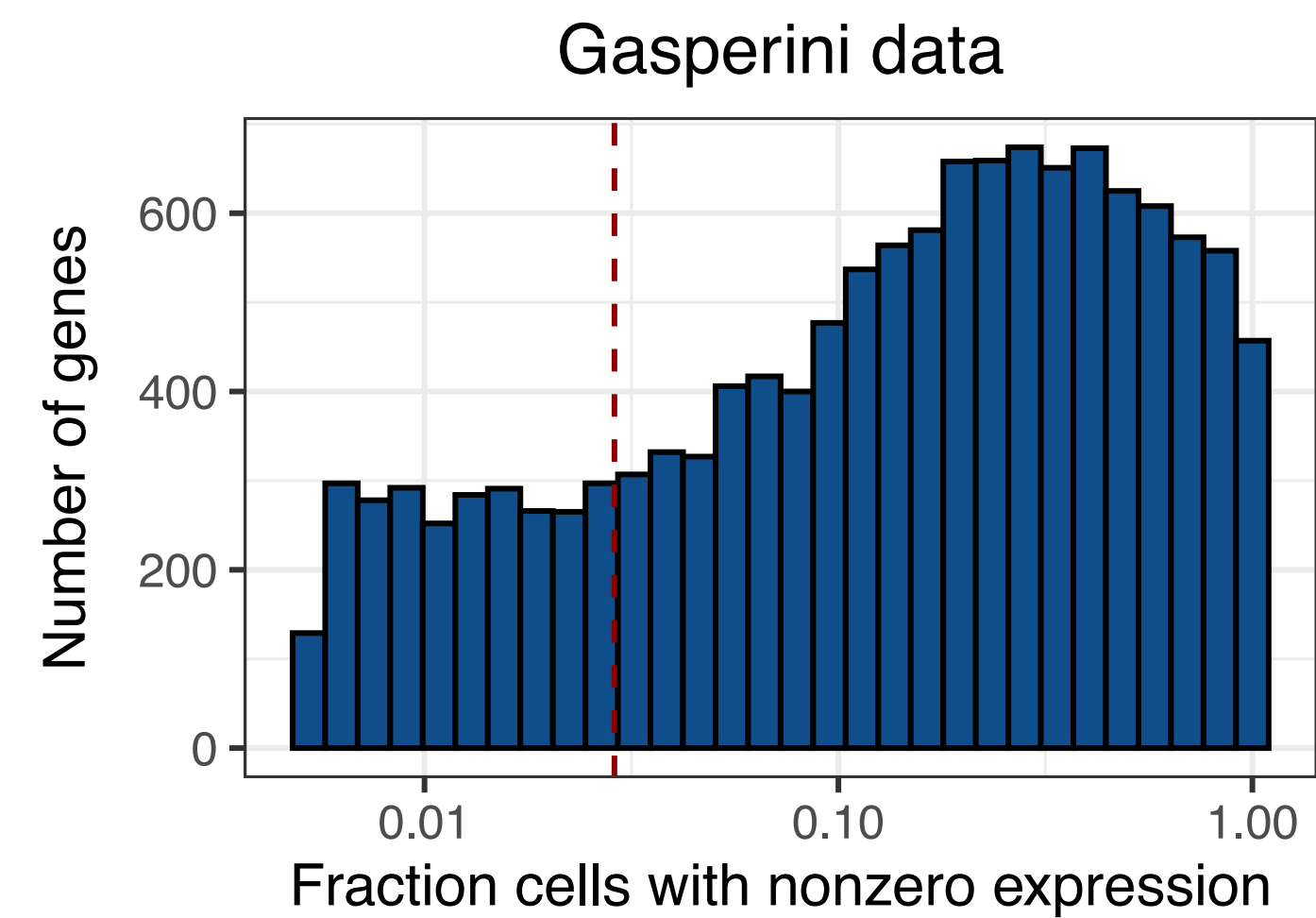
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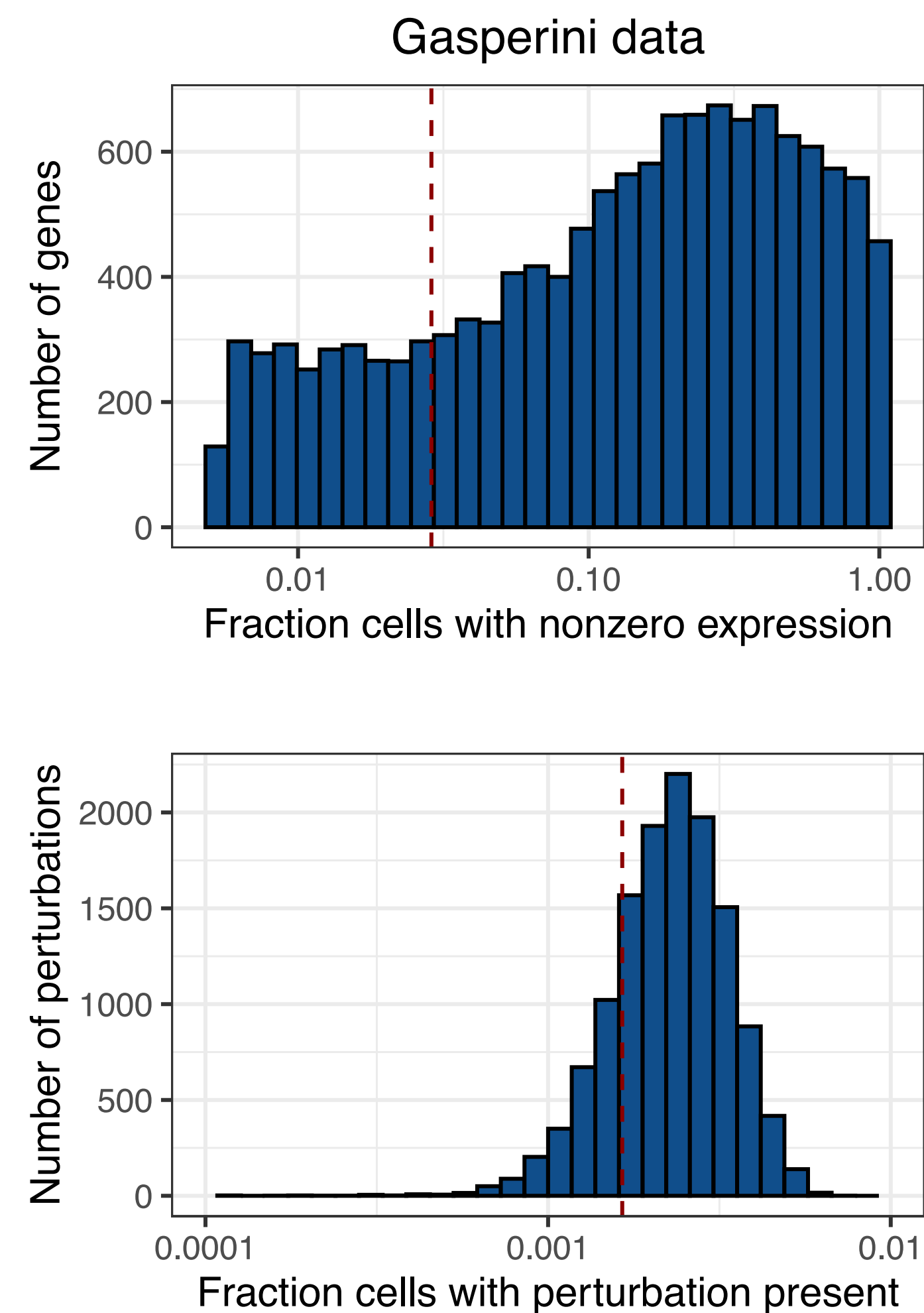


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Cell perturbed	Gene Expression					
	0	1	2	3	4	8
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Consequence: Cannot rely on the CLT, especially in the tails.

Model for the biological data

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Working model:
$$\begin{cases} X_i \mid Z_i \stackrel{\text{ind}}{\sim} \text{Ber}(\pi_i); & \text{logit}(\pi_i) = Z_i^T \delta \\ Y_i \mid X_i, Z_i \stackrel{\text{ind}}{\sim} \text{NegBin}(\mu_i, \theta); & \log \mu_i = X_i \beta + Z_i^T \gamma \end{cases}.$$

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3. Log fold-change β is the signal quantity.

Testing strategies

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$$T_n \equiv \frac{1}{n} \sum_{i=1}^n (X_i - \hat{\mathbb{E}}[X_i | Z_i])(Y_i - \hat{\mathbb{E}}[Y_i | Z_i])$$

¹Shah and Peters (2020)

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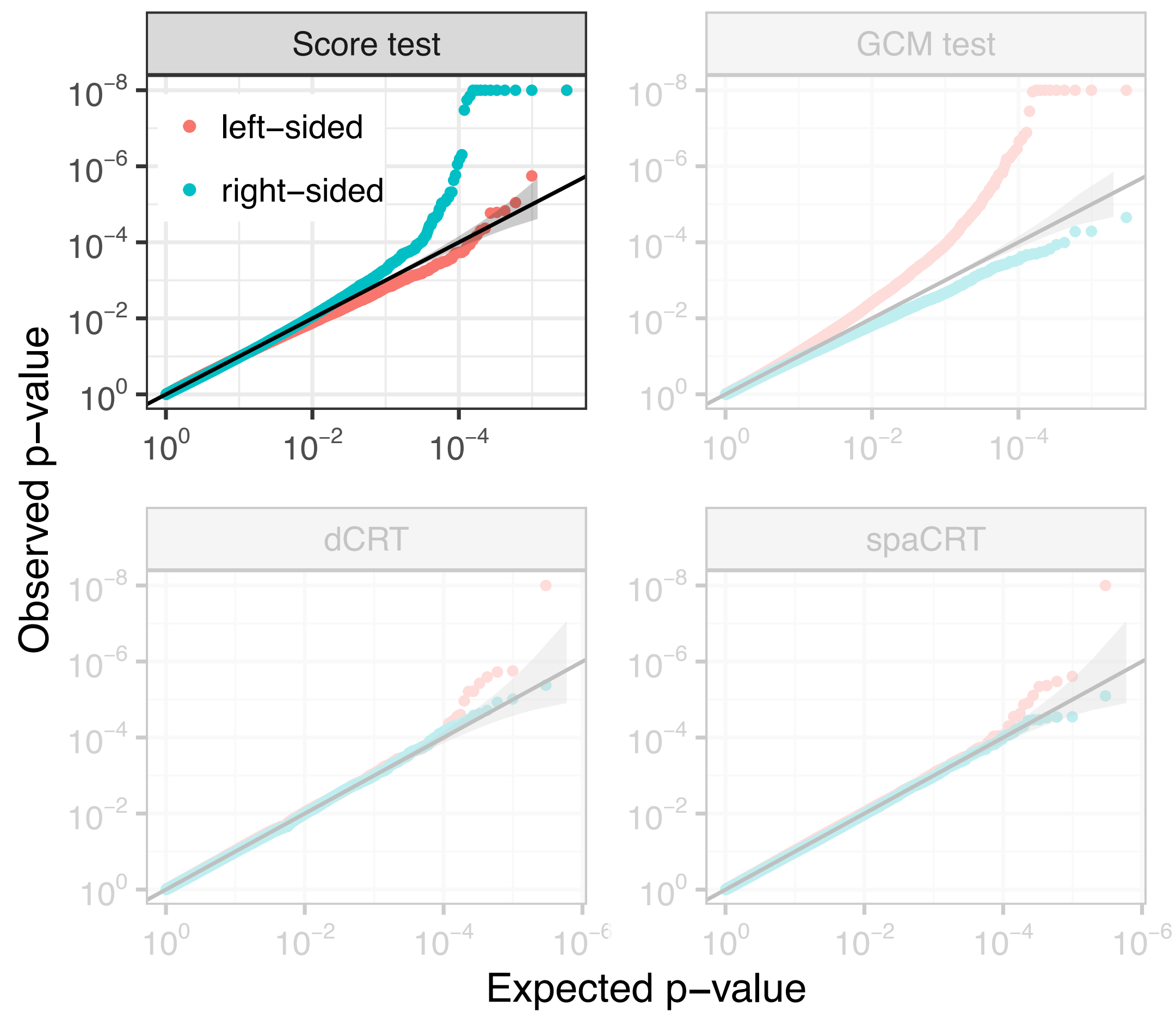
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3. Distilled conditional randomization test (dCRT),² based on T_n and resampling from fitted distribution $X_i | Z_i$

¹Shah and Peters (2020)

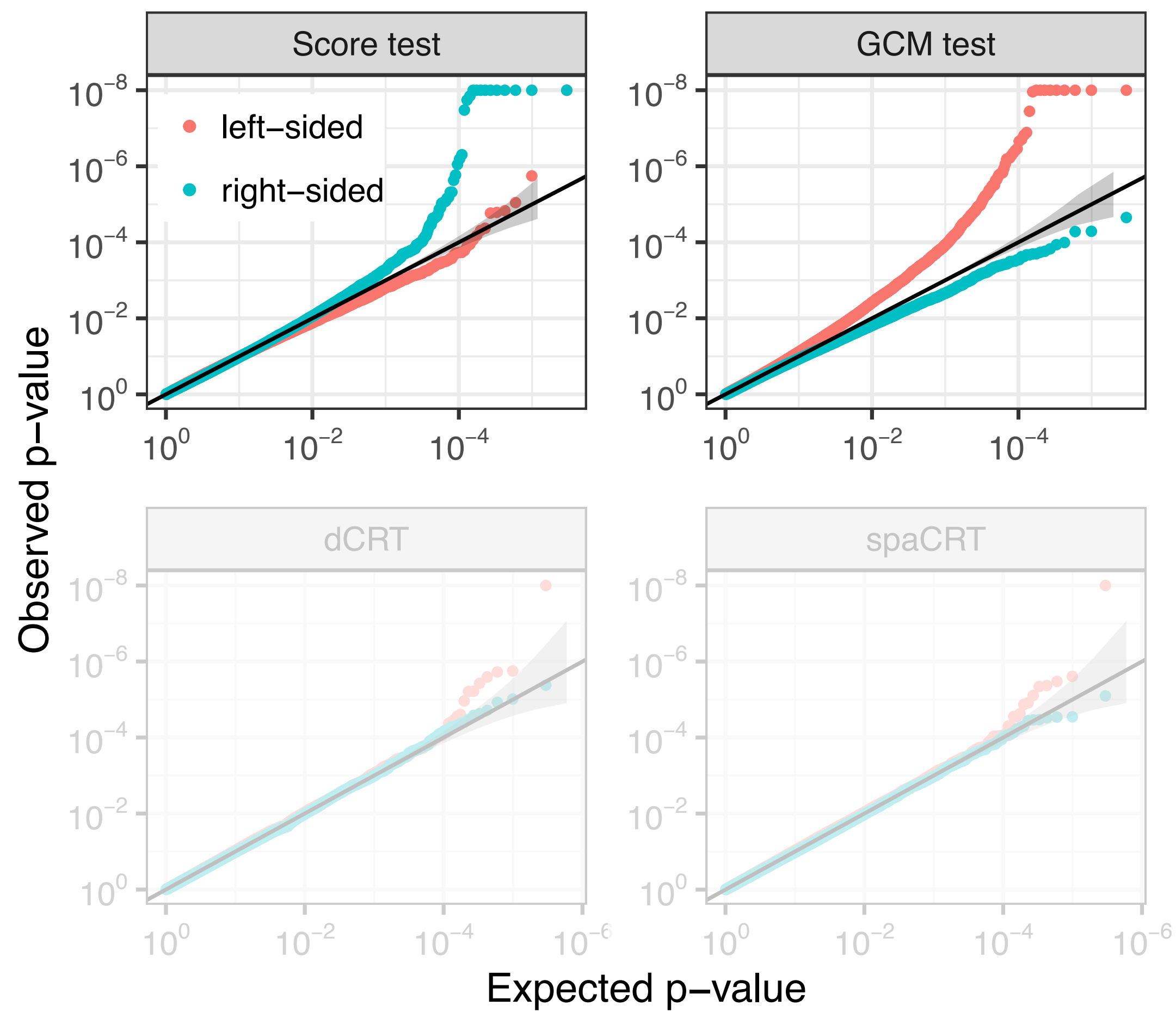
²Liu et al. (2022), Barry et al. (2021)

Comparing Type-I error using negative controls

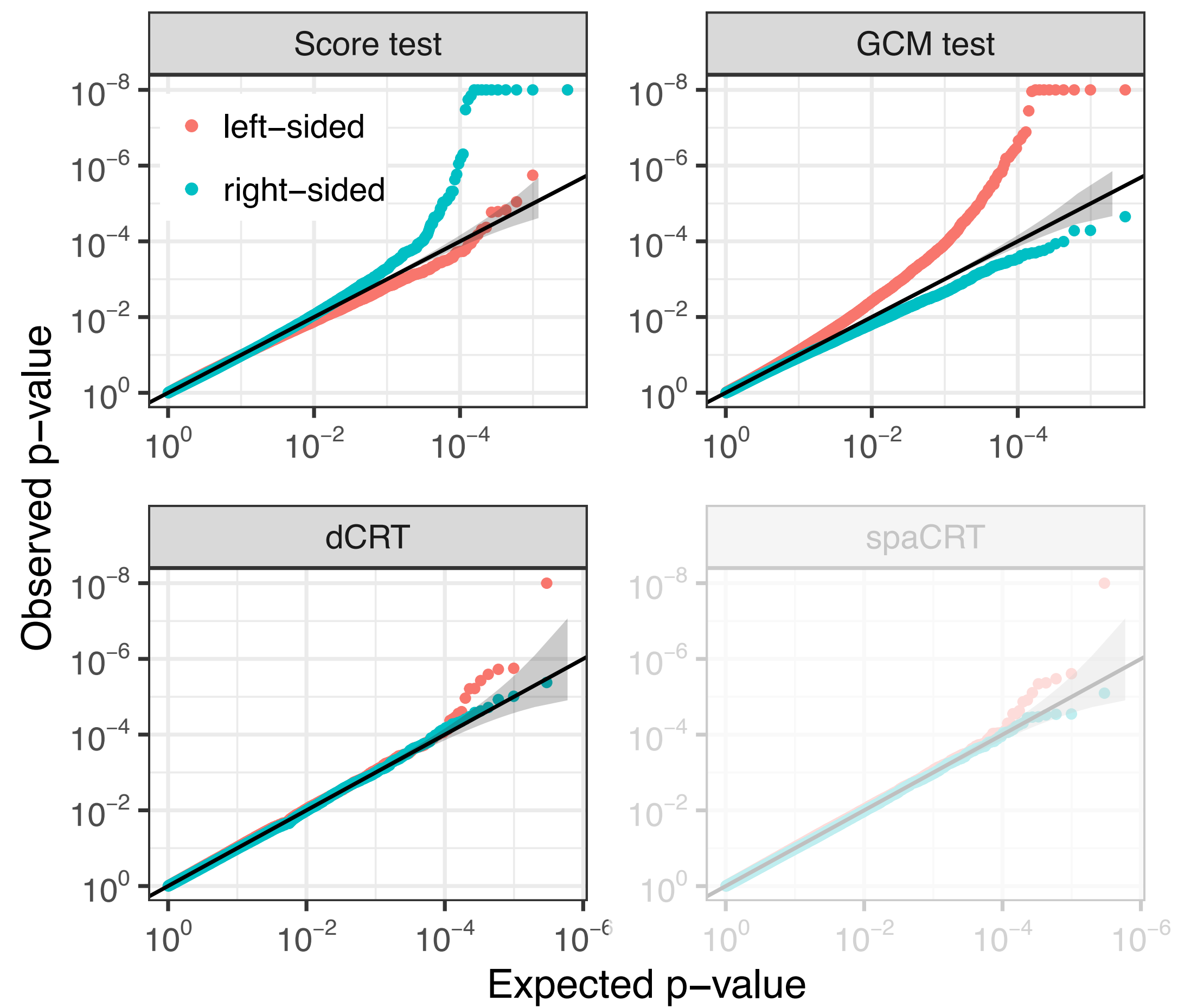
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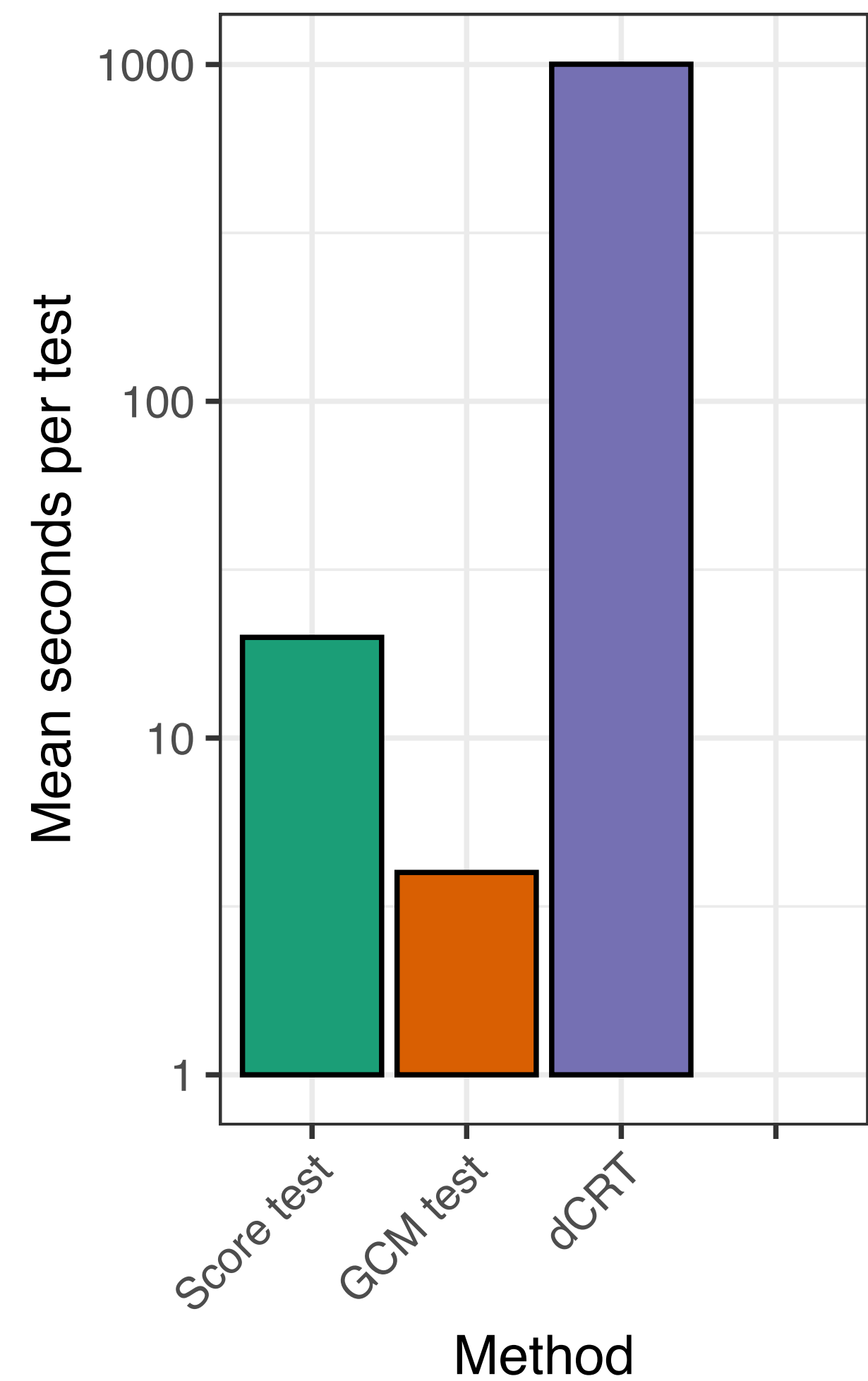
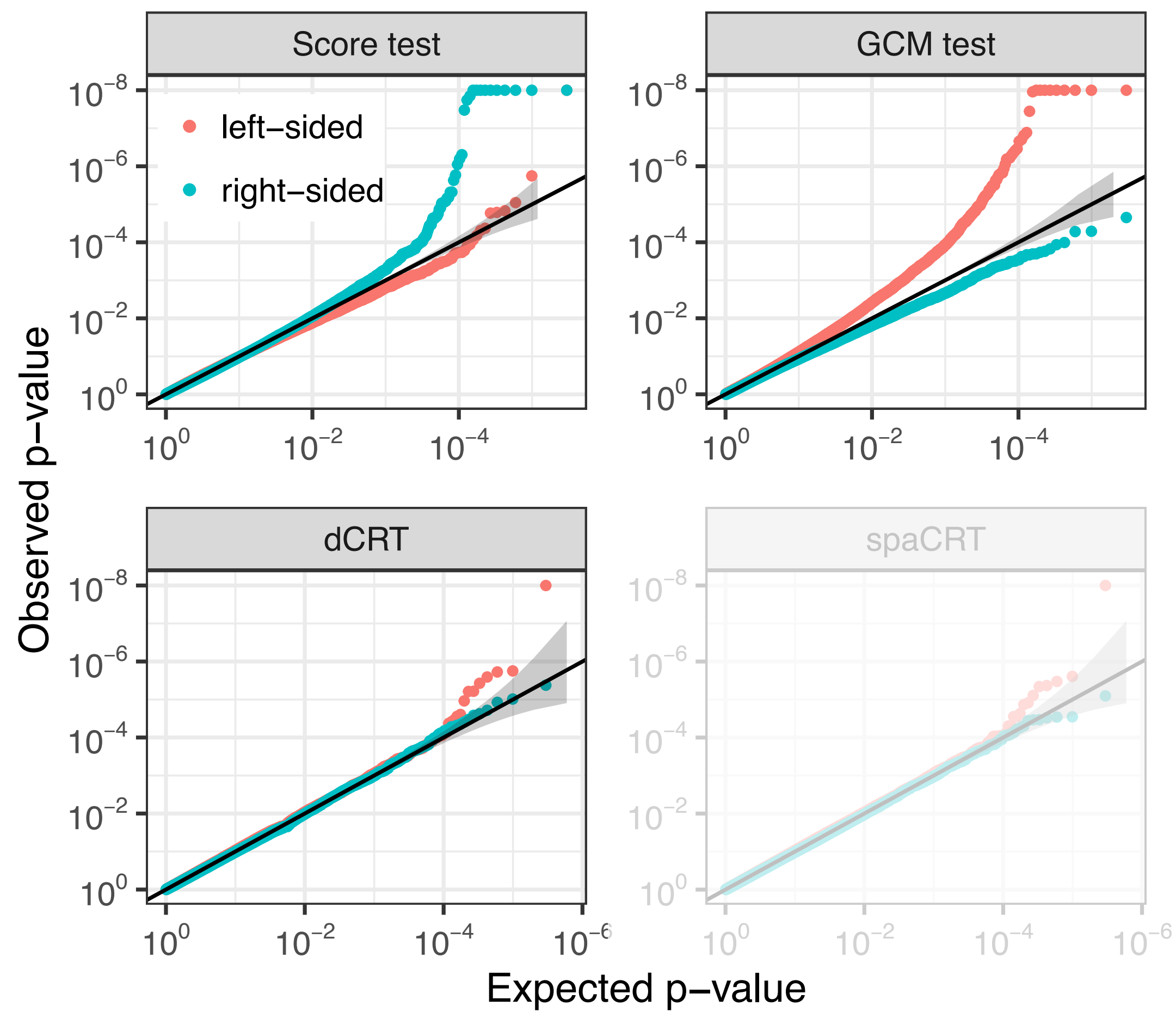
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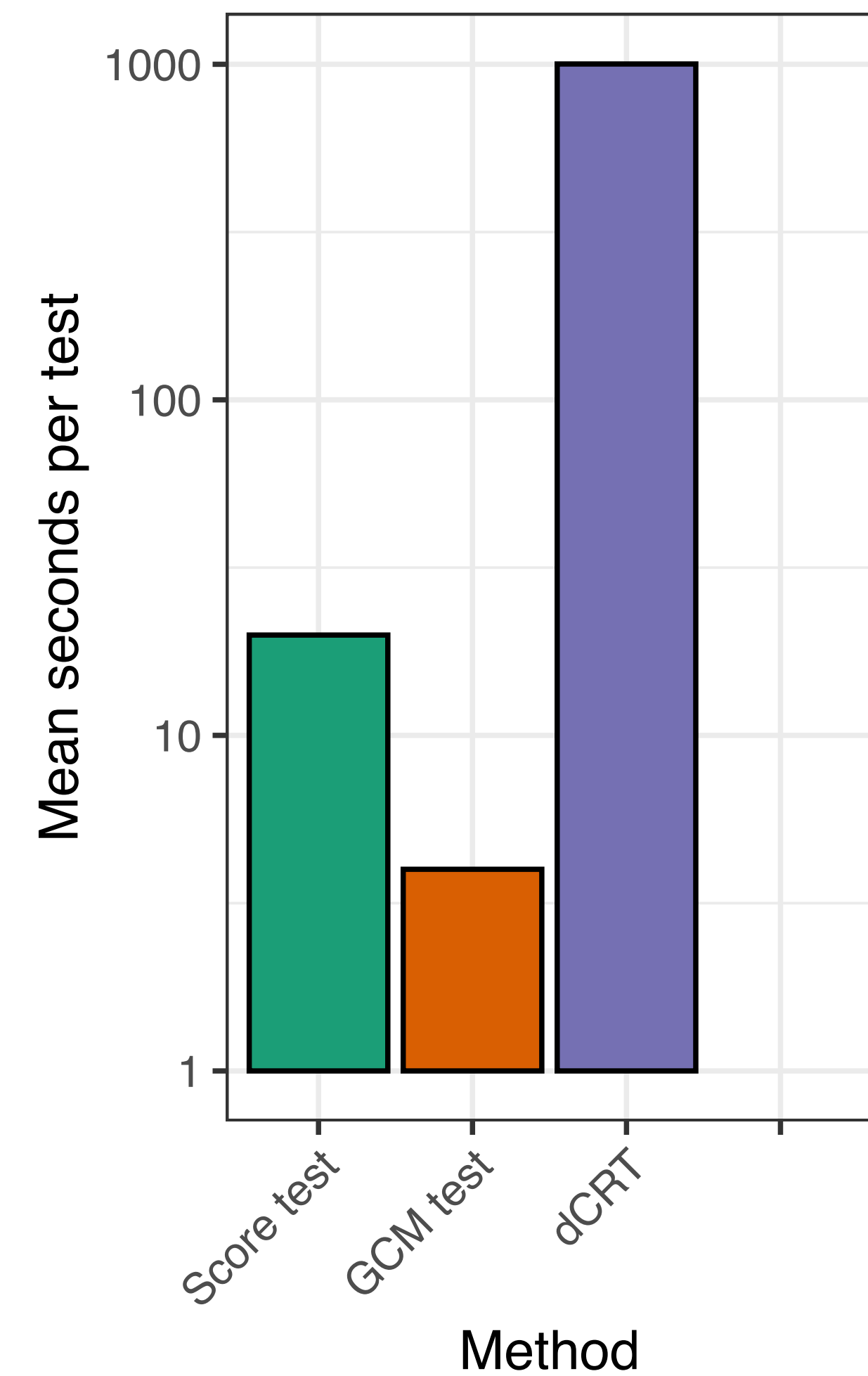
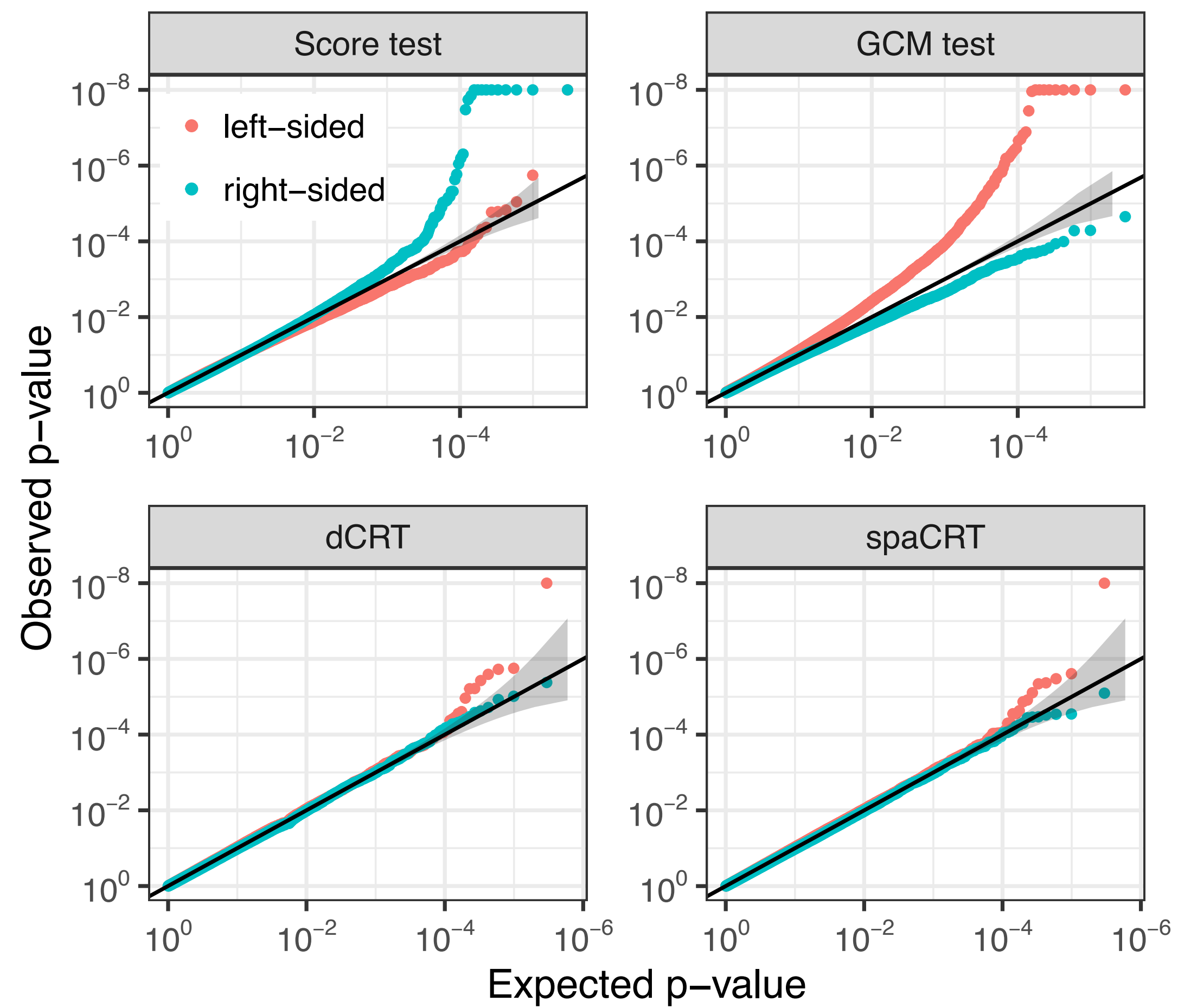
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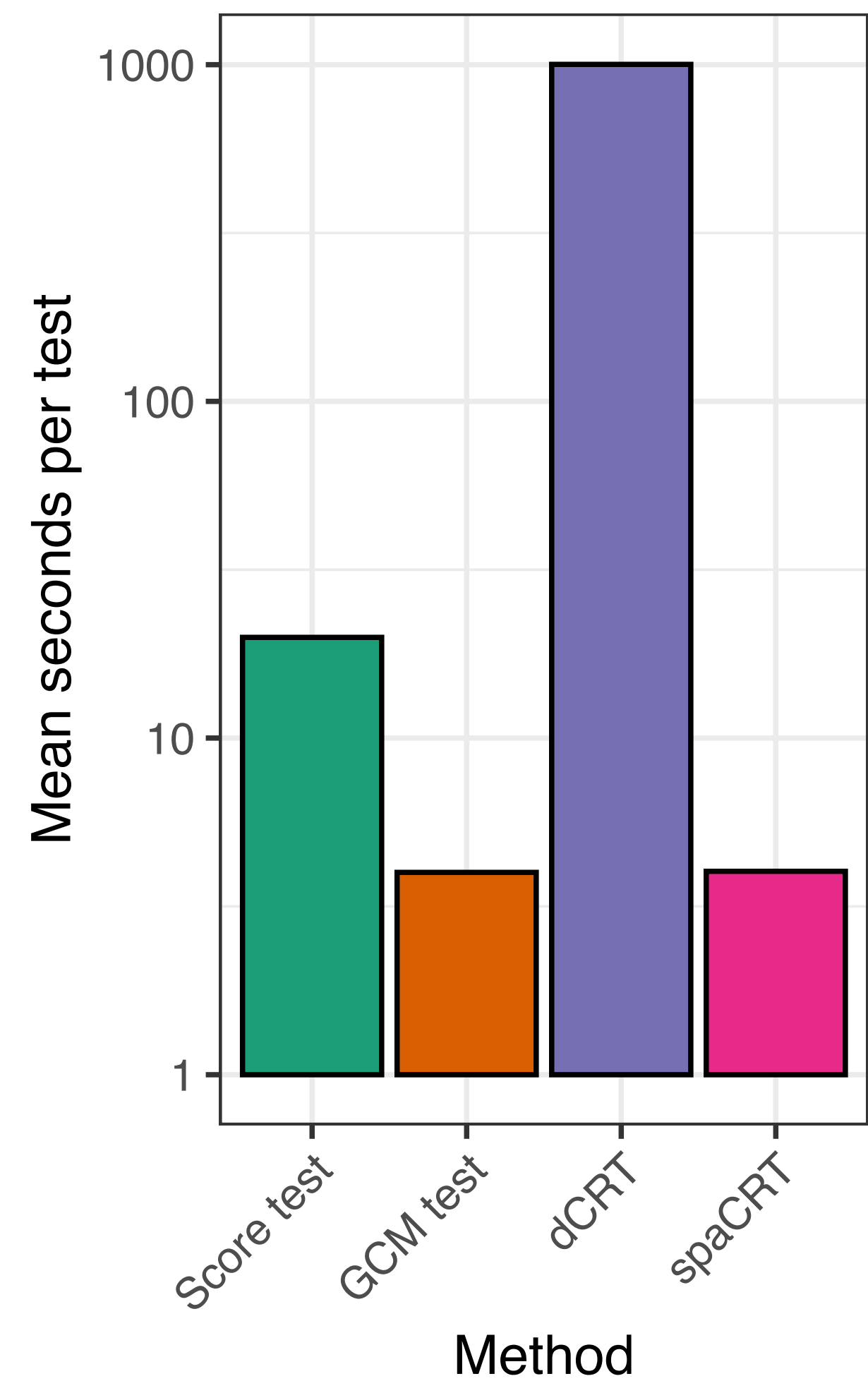
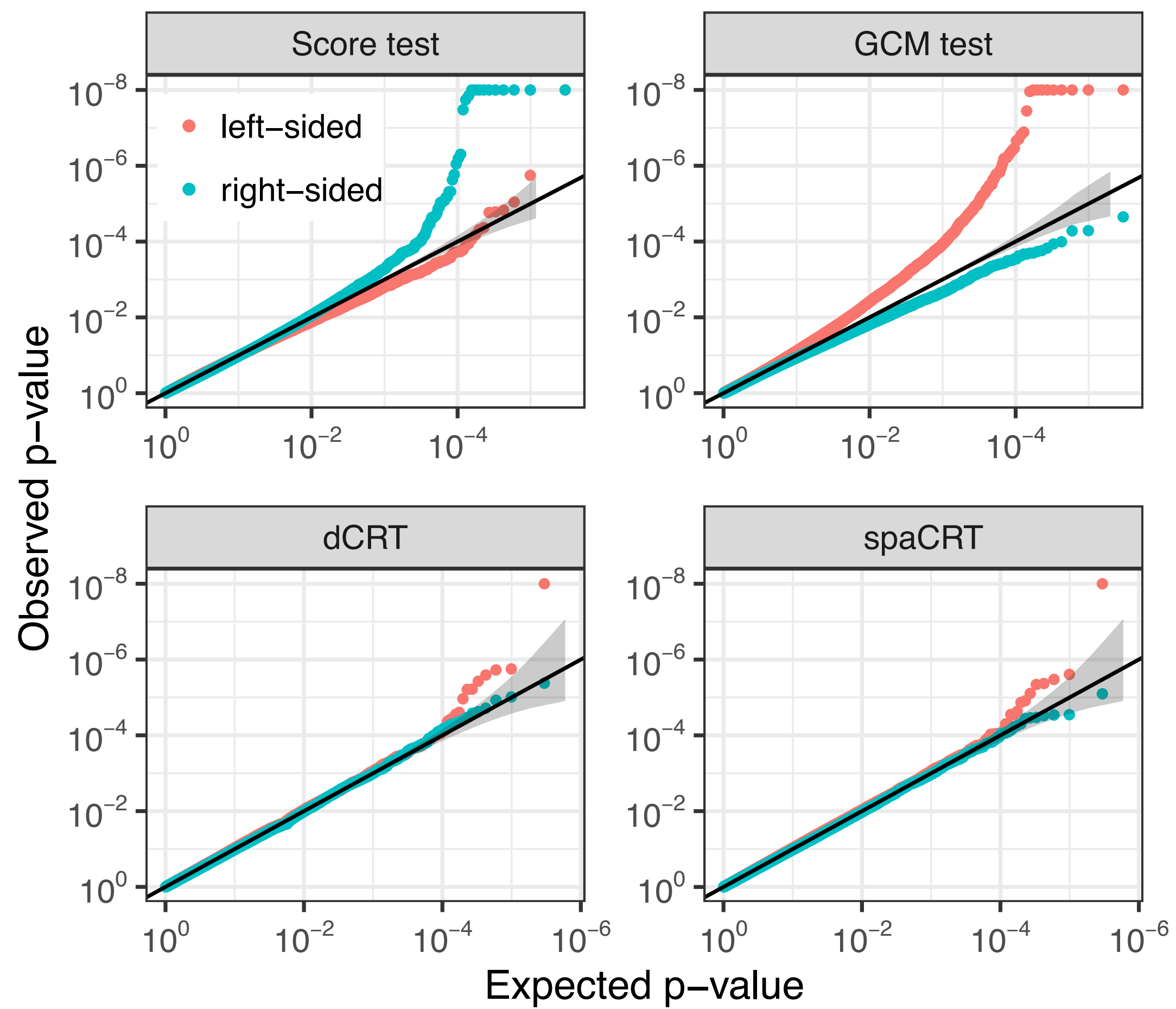
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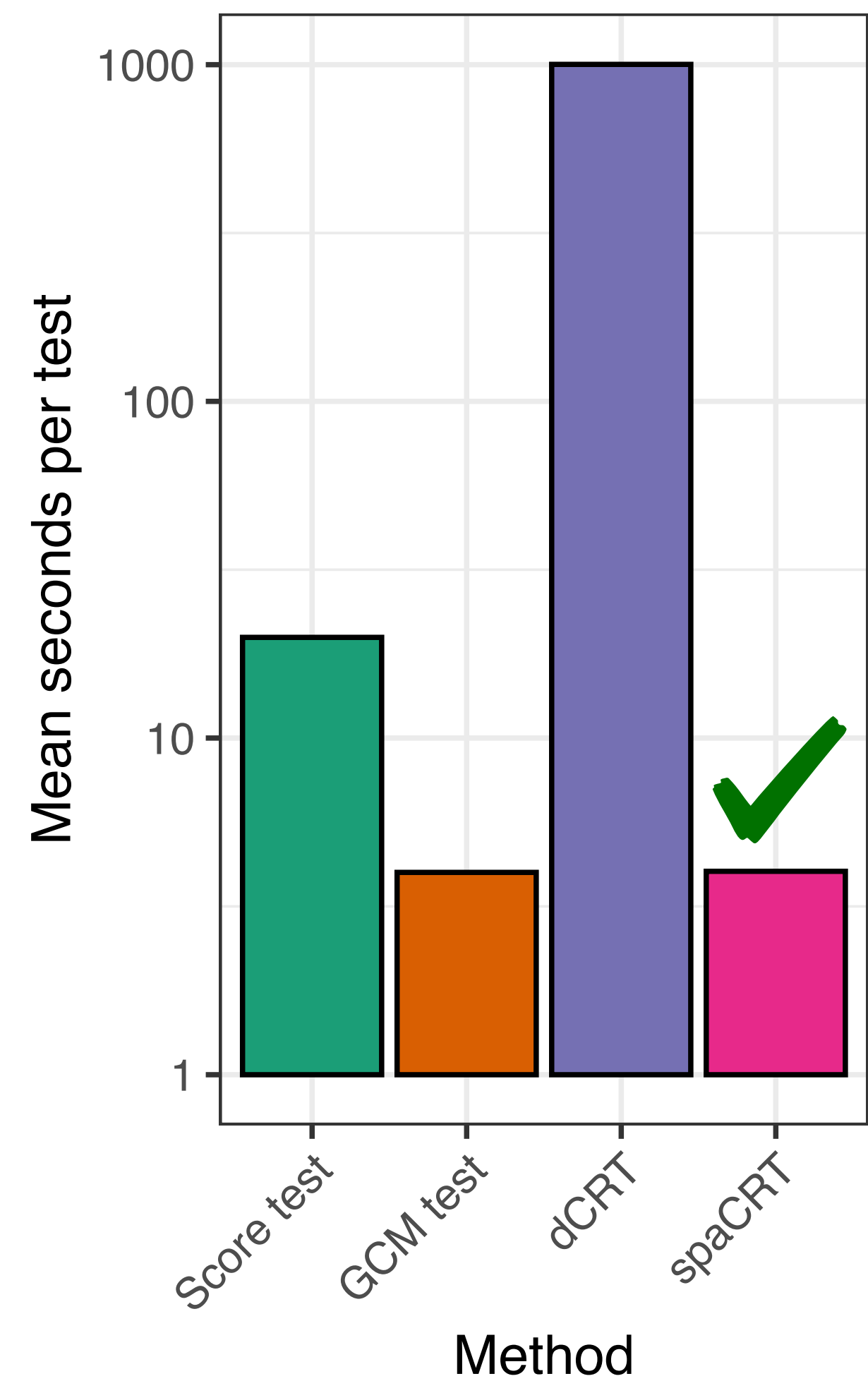
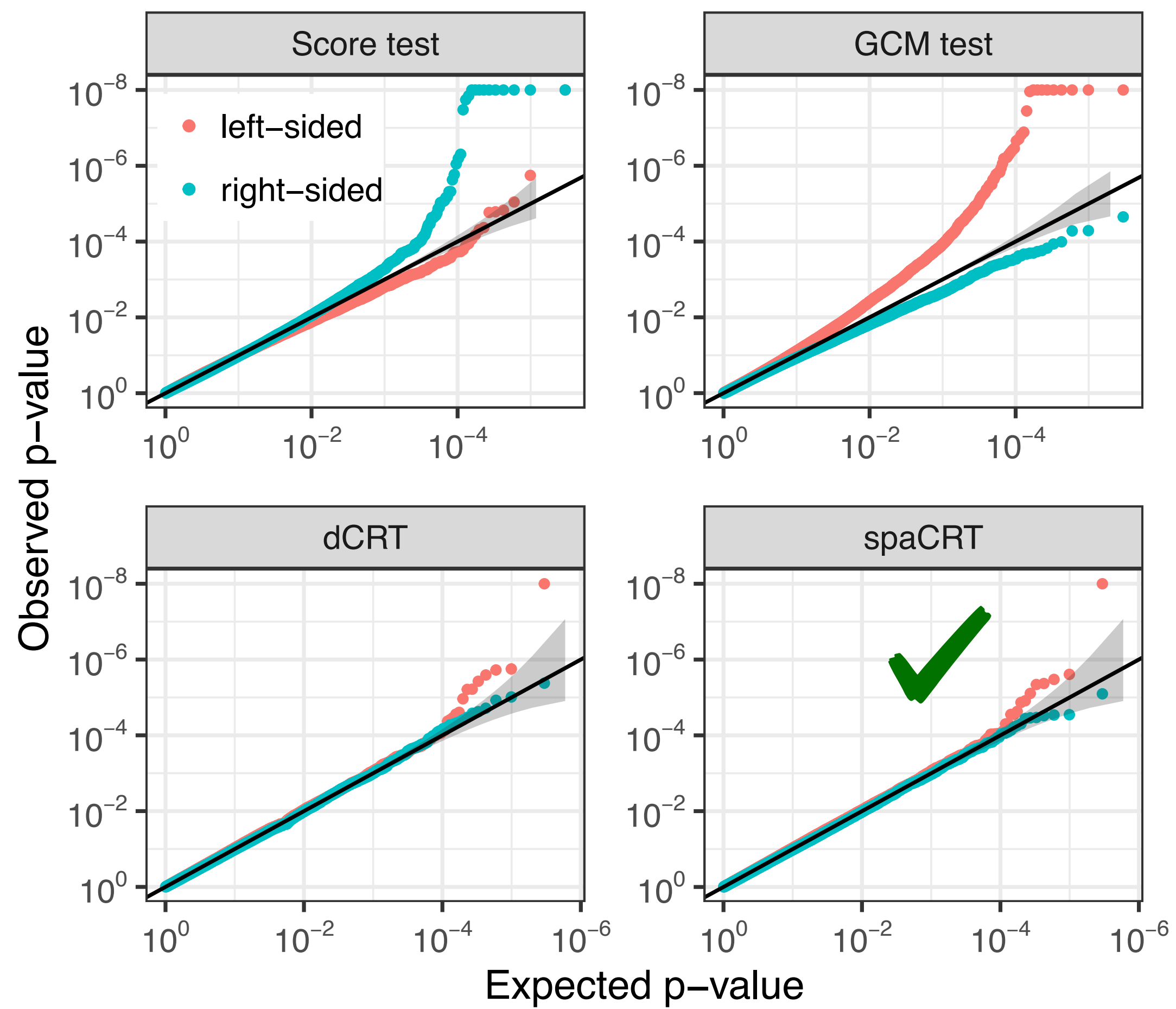
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4. Output dCRT p-value $p_n^{\text{dCRT}} = \frac{1}{B+1} \left(1 + \sum_{b=1}^B 1(\tilde{T}_n^{(b)} \geq T_n) \right)$.

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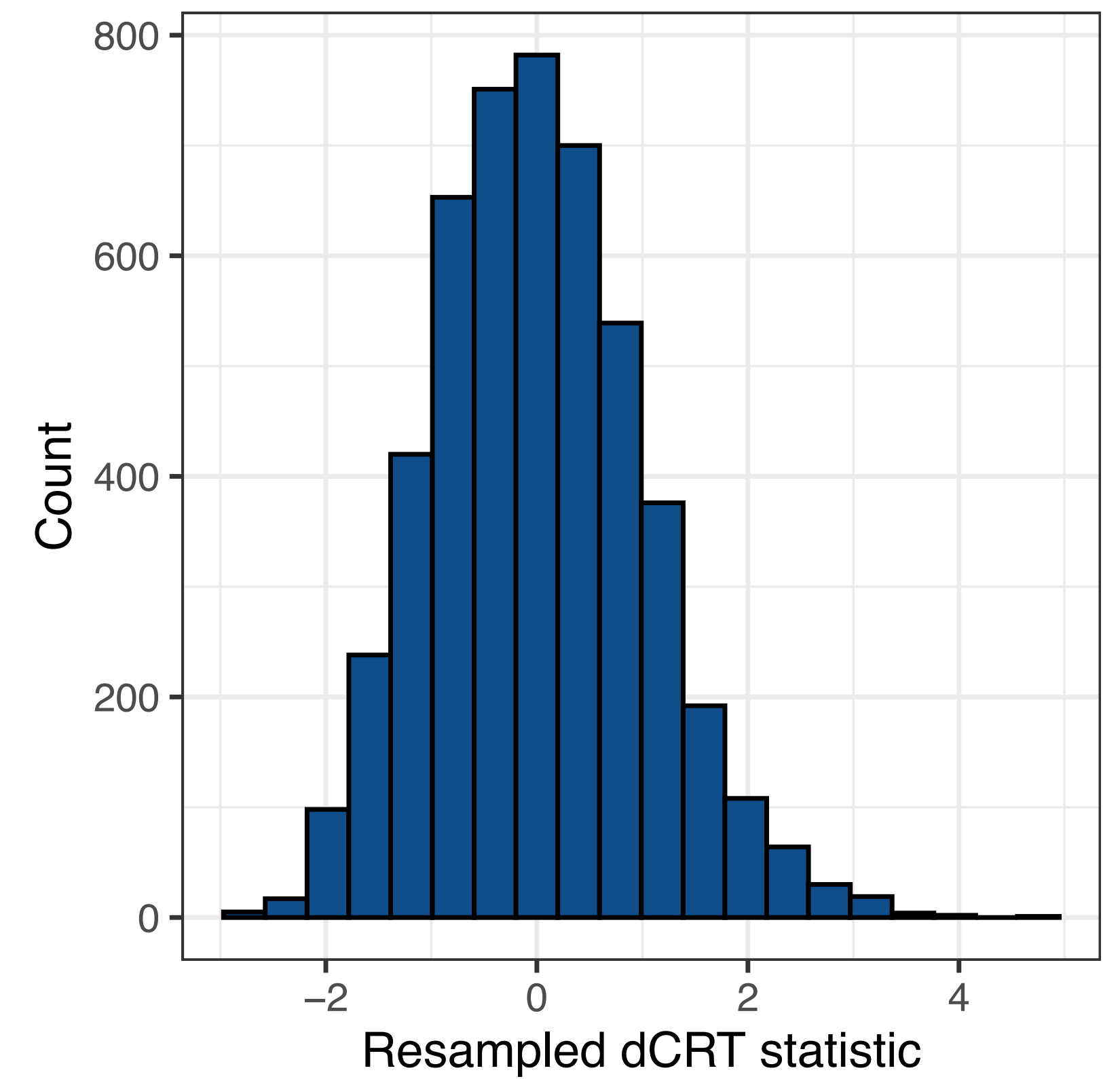
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No hope for resampling-based method if advanced parallelization and heuristic approximation is not available!

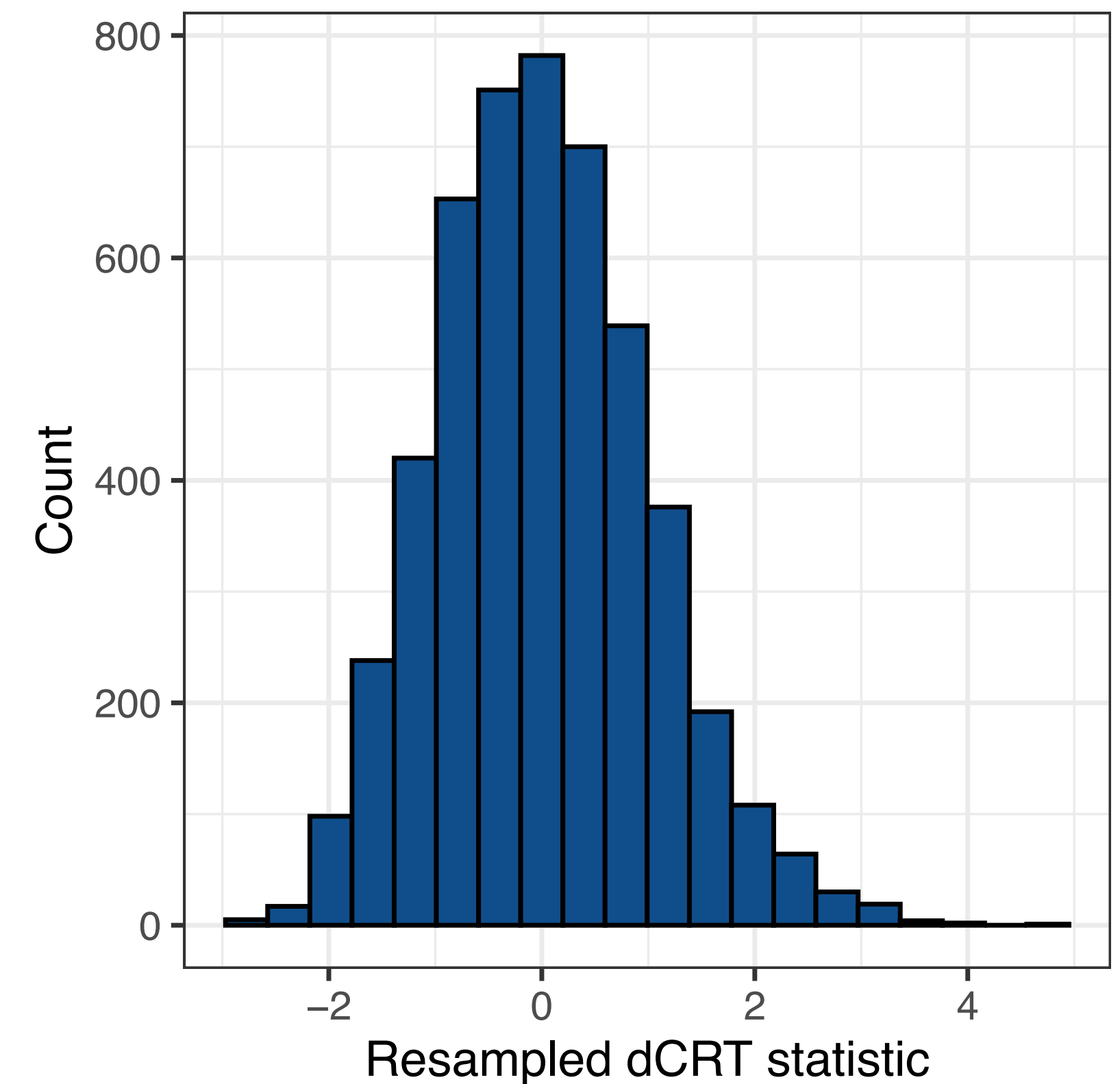
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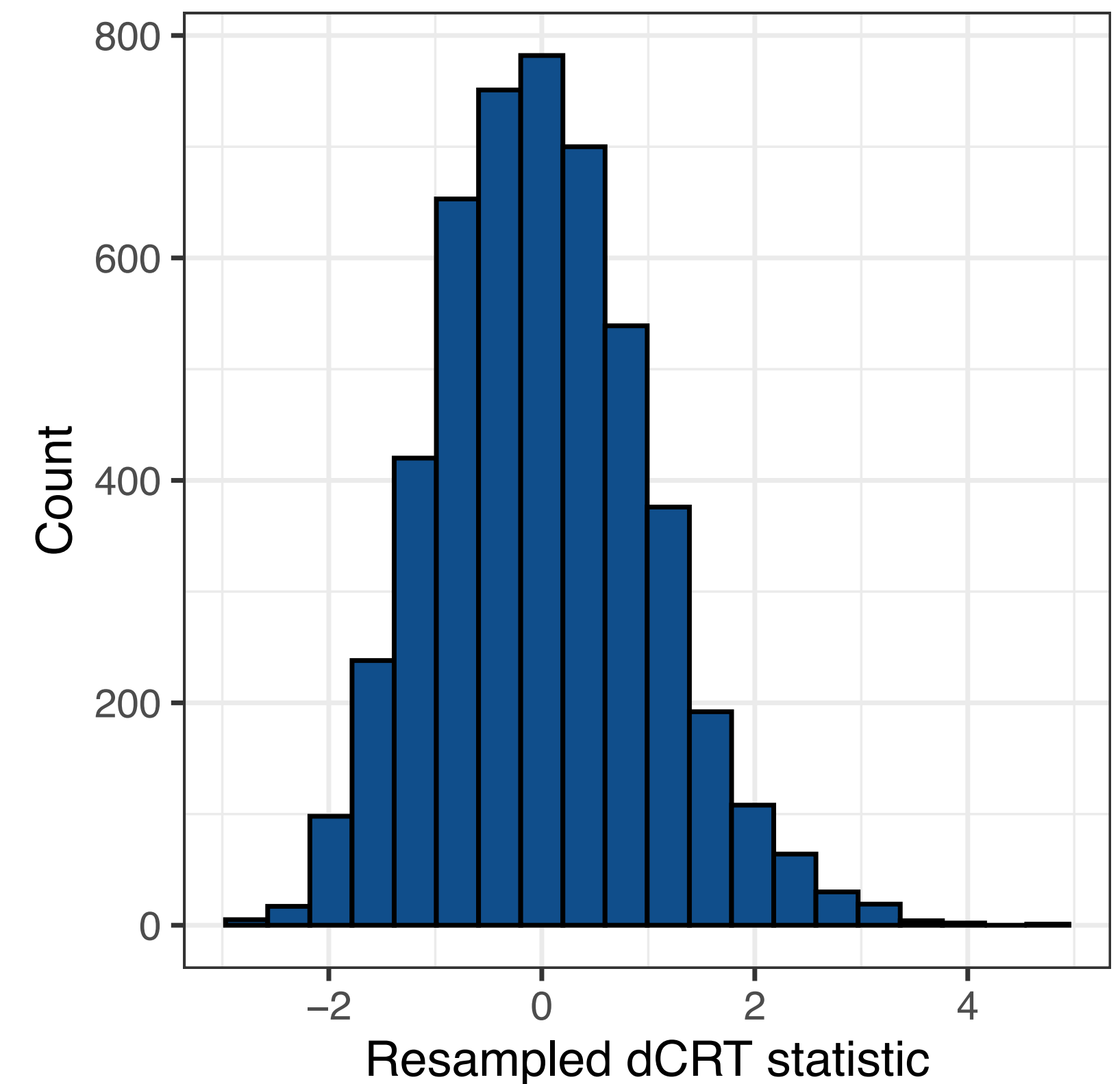
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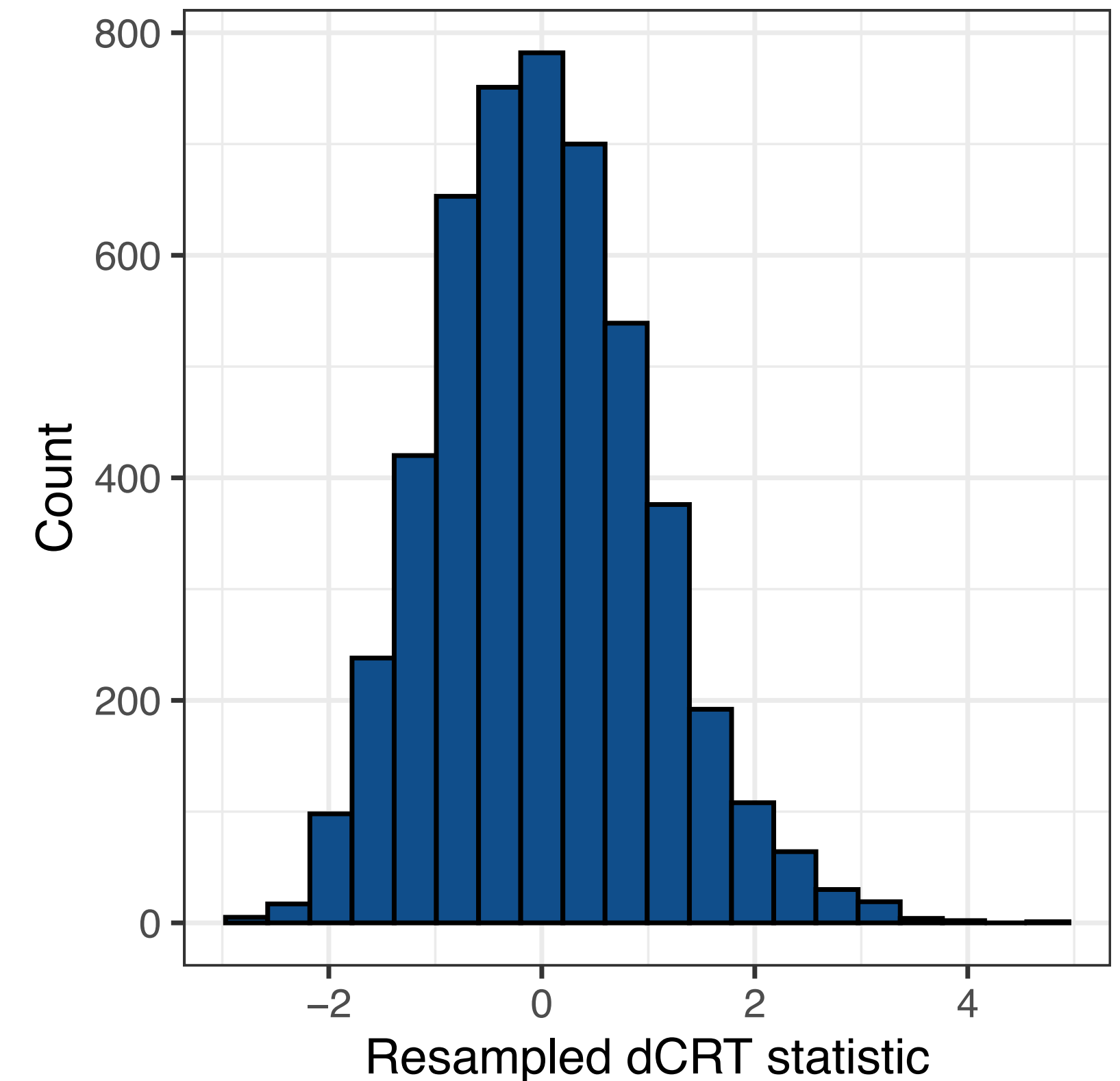
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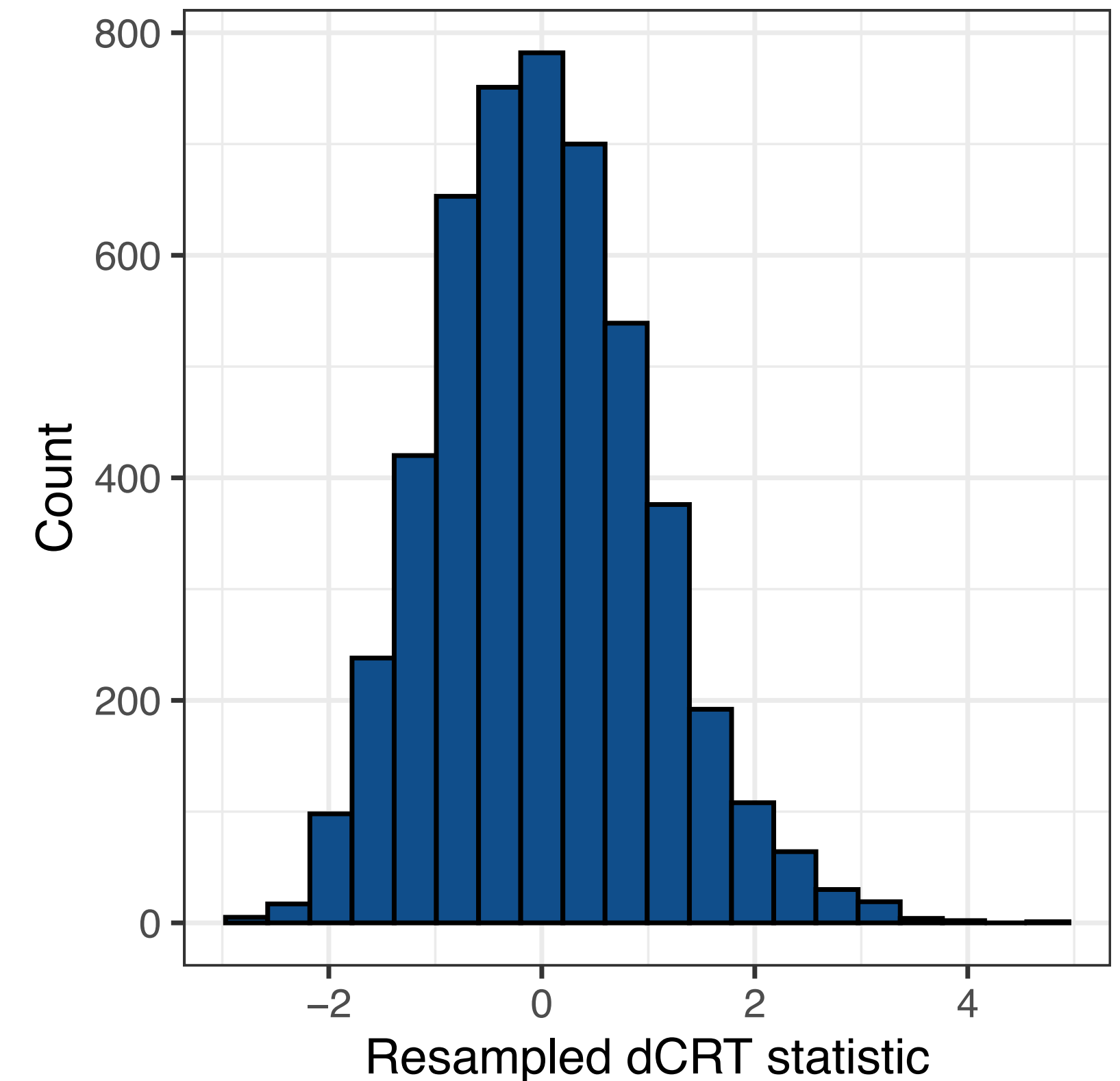
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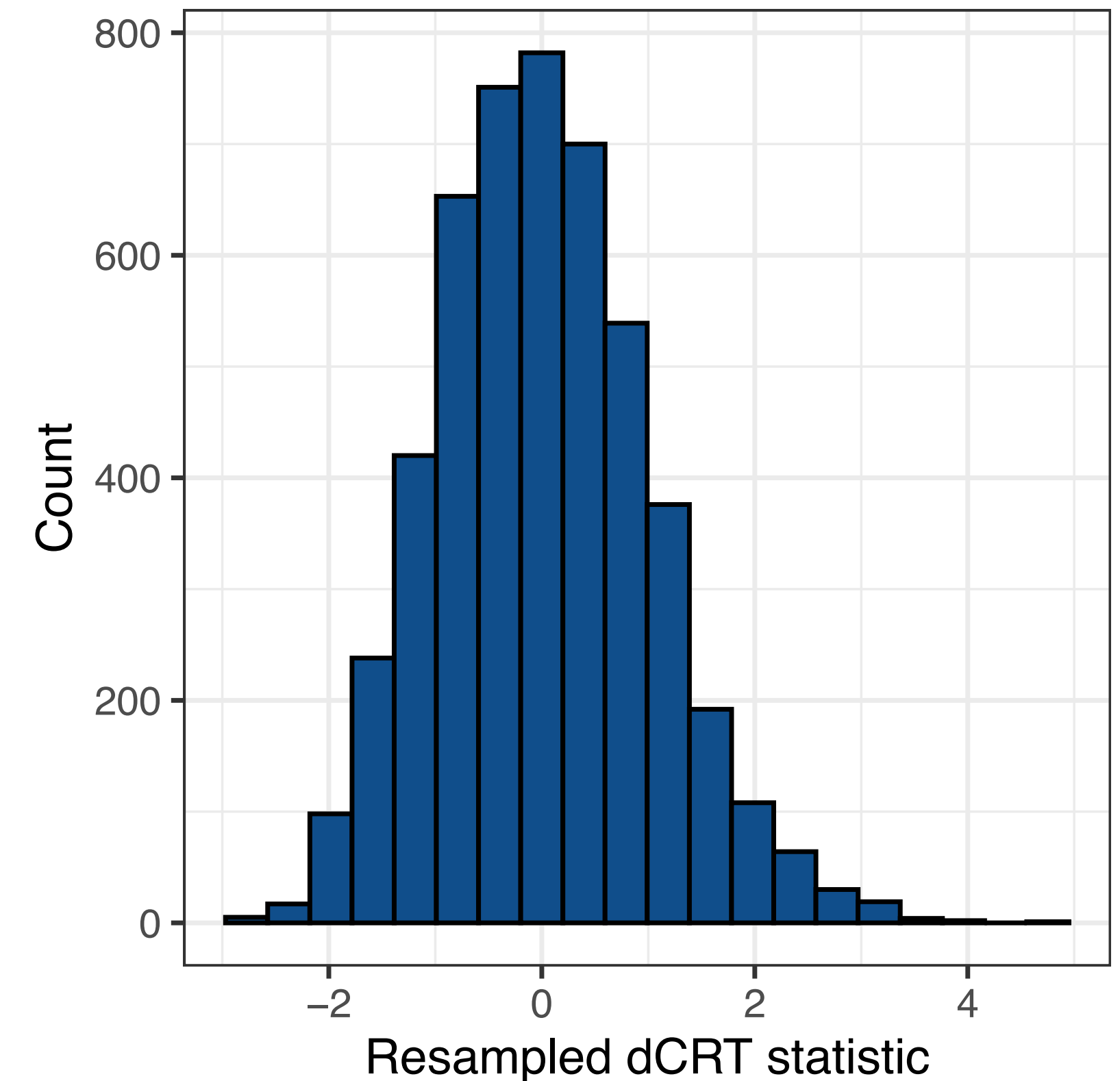


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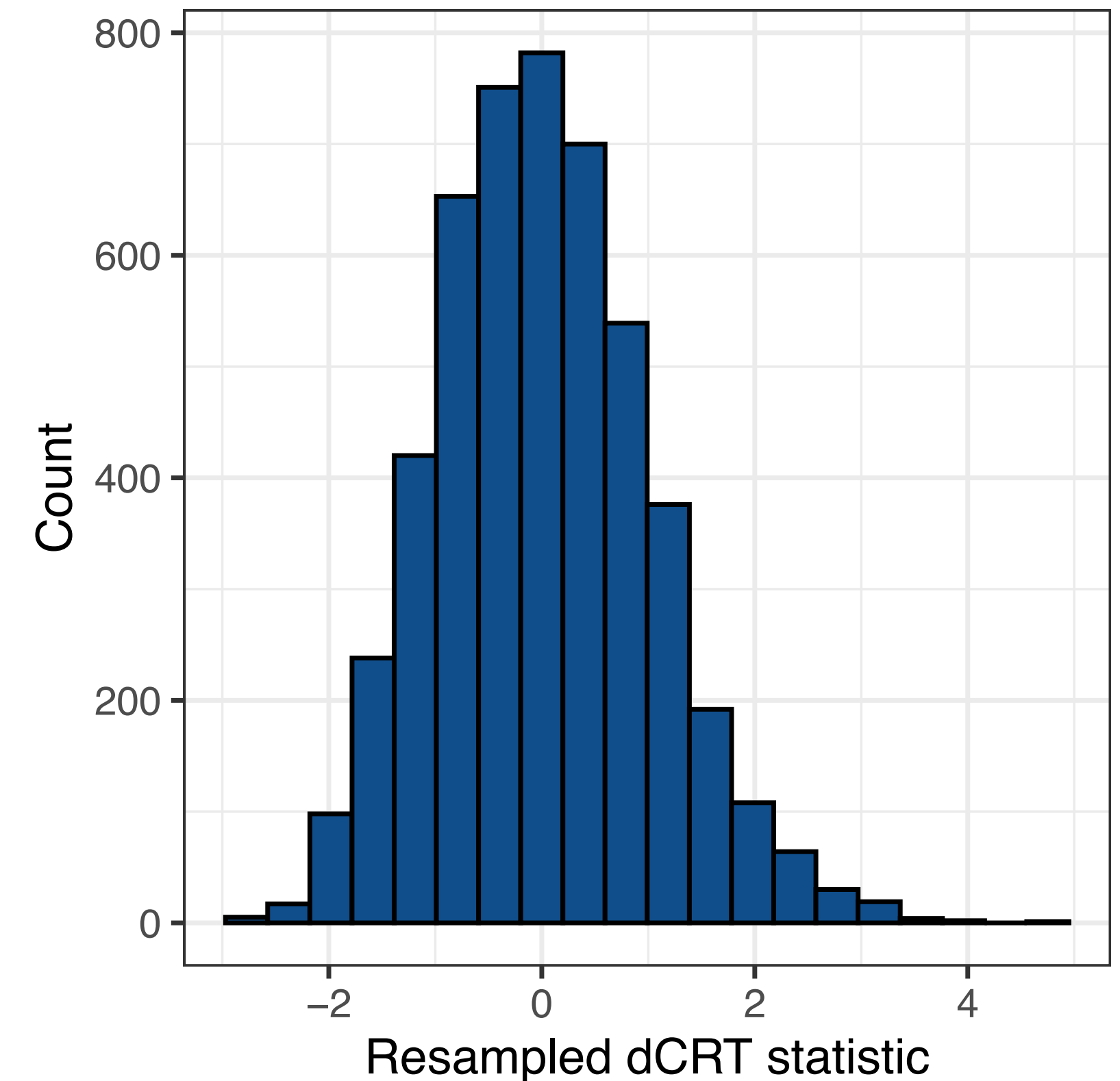


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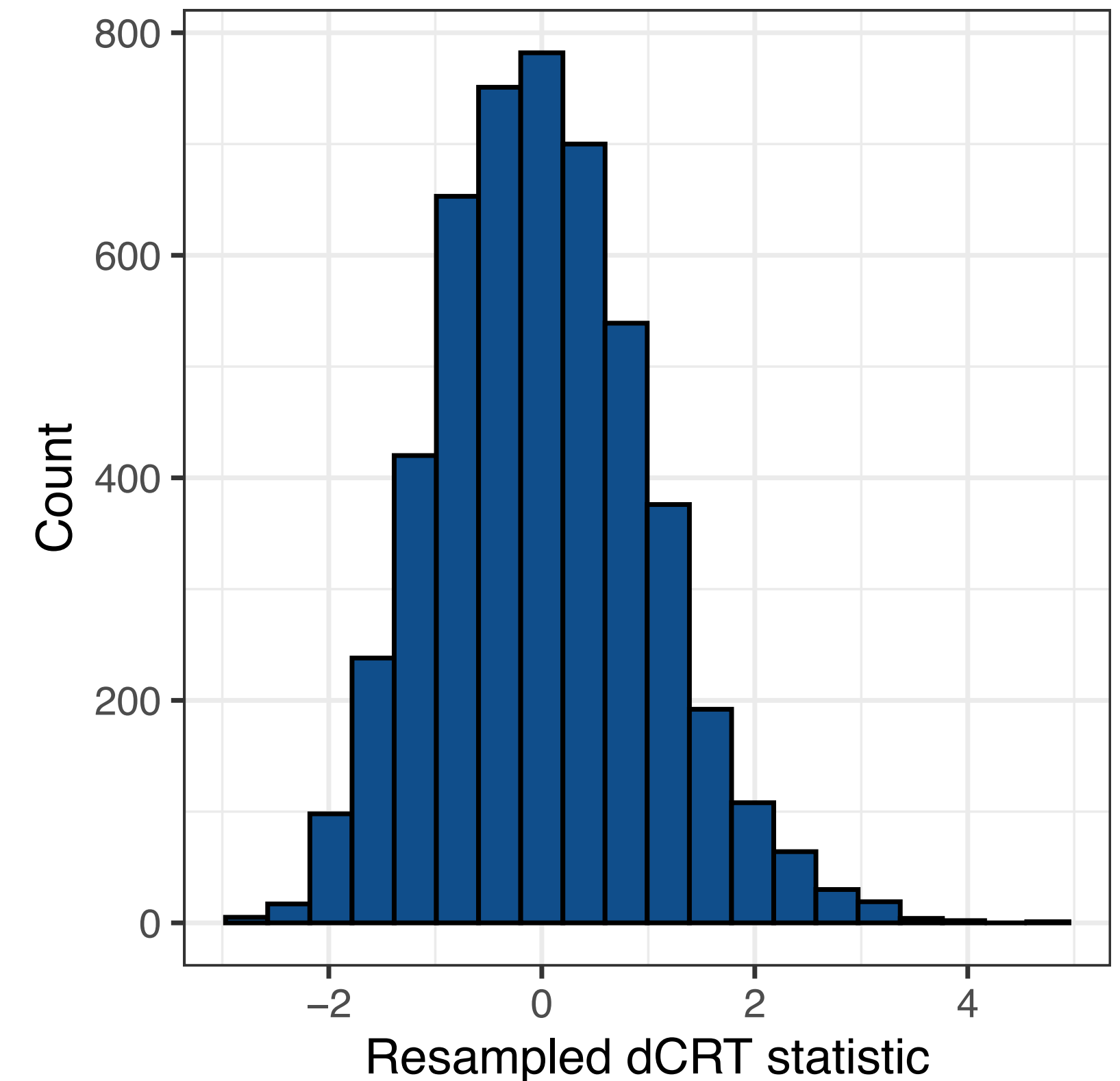
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Therefore, we wish to approximate

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where W_{in} are independent and mean-zero conditionally on data.

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Existing applications of SPA to resampling distributions have ignored these.

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4. Return $p_n^{\text{spaCRT}} = 1 - \Phi(r_n) + \phi(r_n) \left(\frac{1}{\lambda_n} - \frac{1}{r_n} \right).$

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Suppose $\hat{\pi}$ and $\hat{\mu}$ consistent for $\pi(Z_i) = \mathbb{P}[X_i = 1 \mid Z_i]$ and $\mu(Z_i) = \mathbb{E}[Y_i \mid Z_i]$, and $\sup_n \mathbb{E}[Y_{in}^4] < \infty$. Then,

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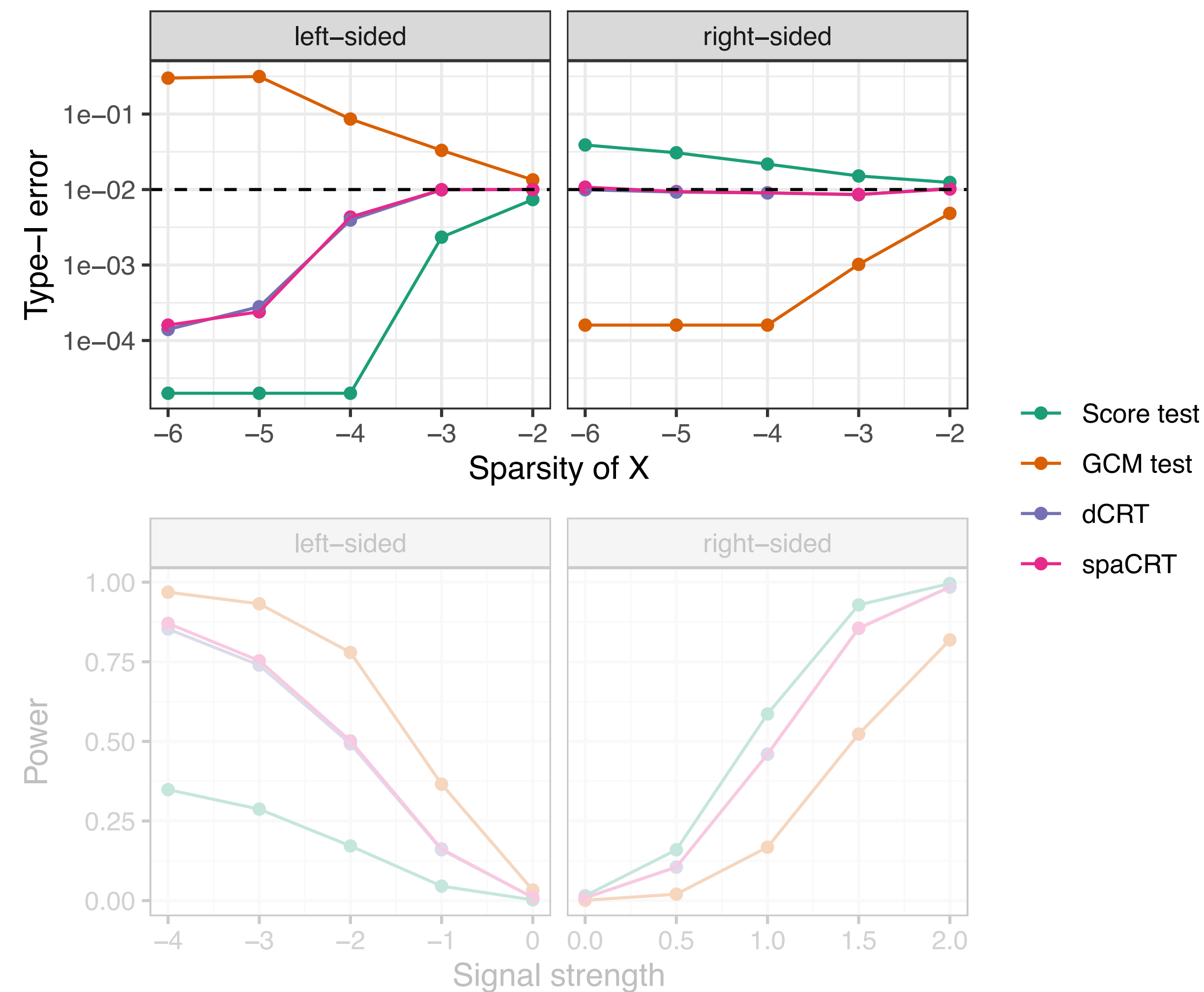
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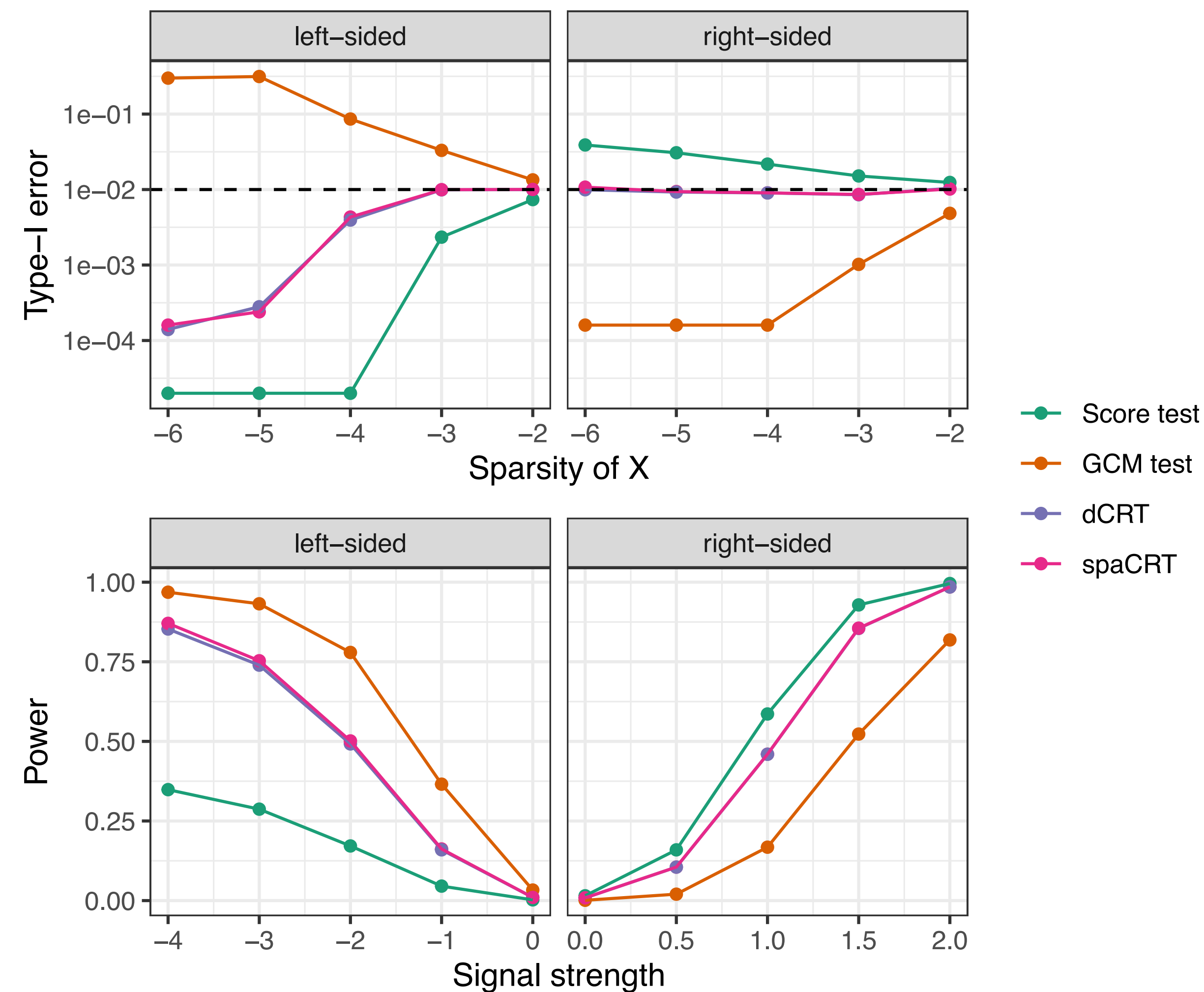
¹Niu et al. (2022)

Numerical simulation results

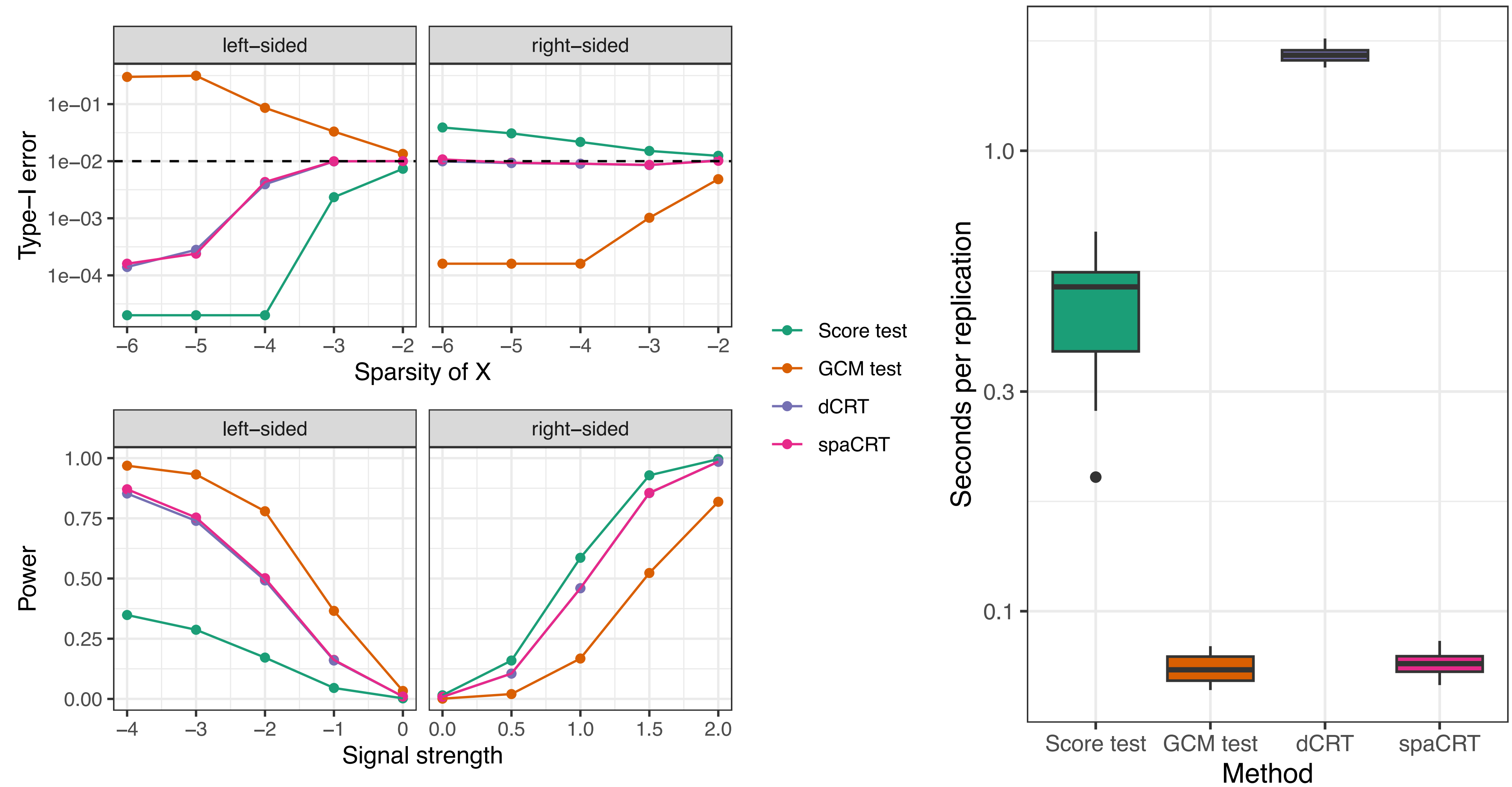
Numerical simulation results



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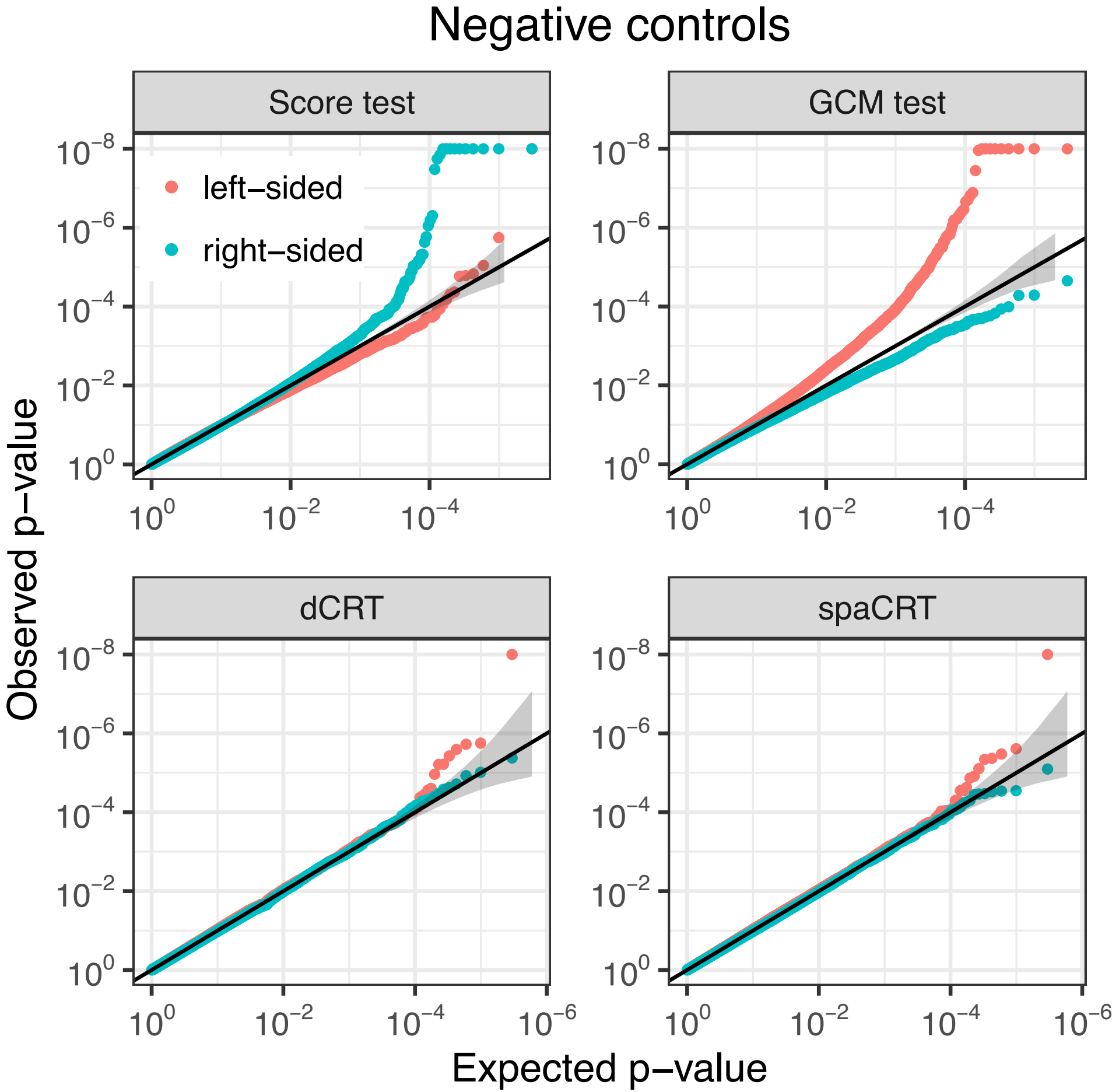


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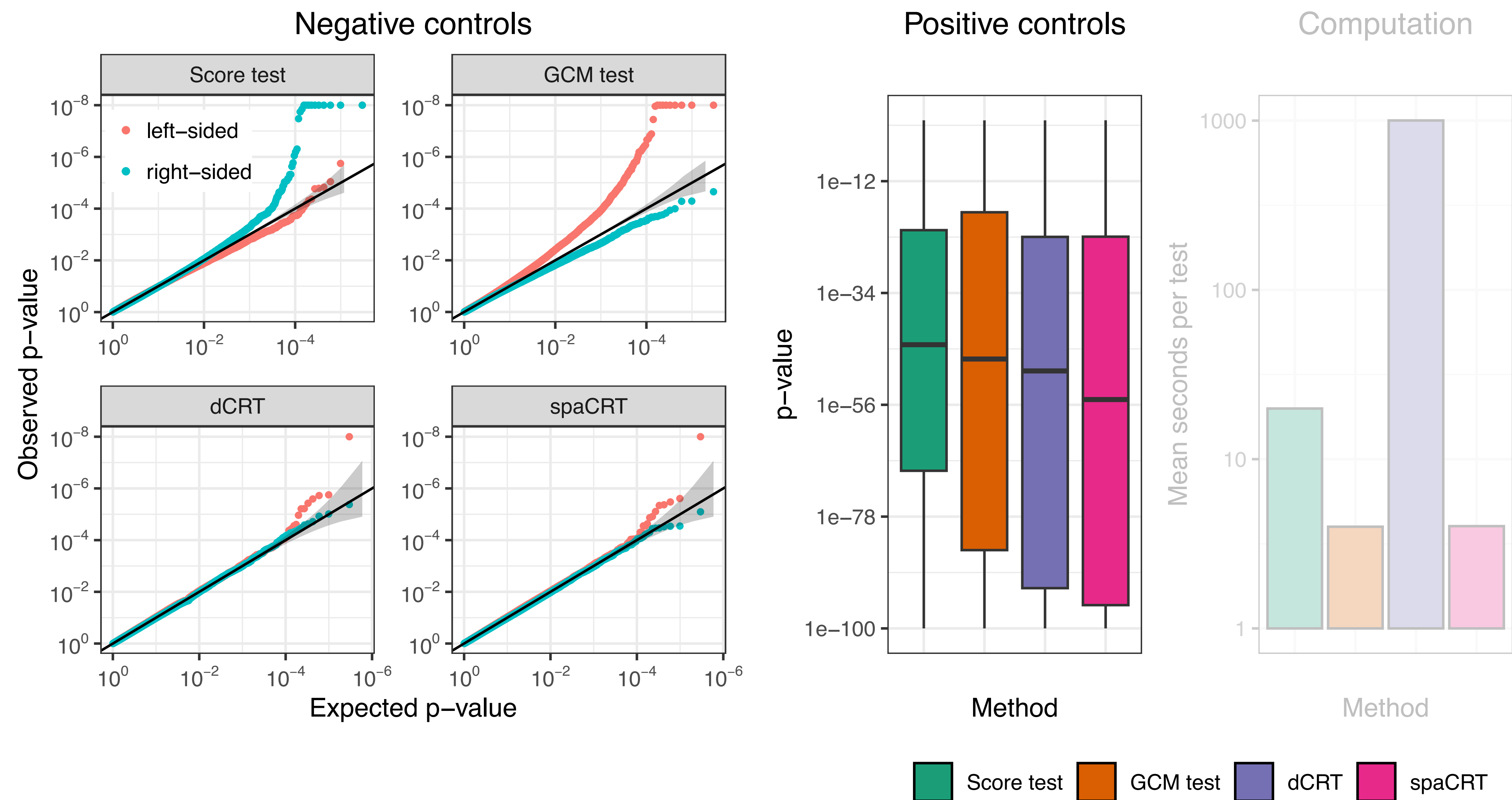


Results on Gasperini data analysis

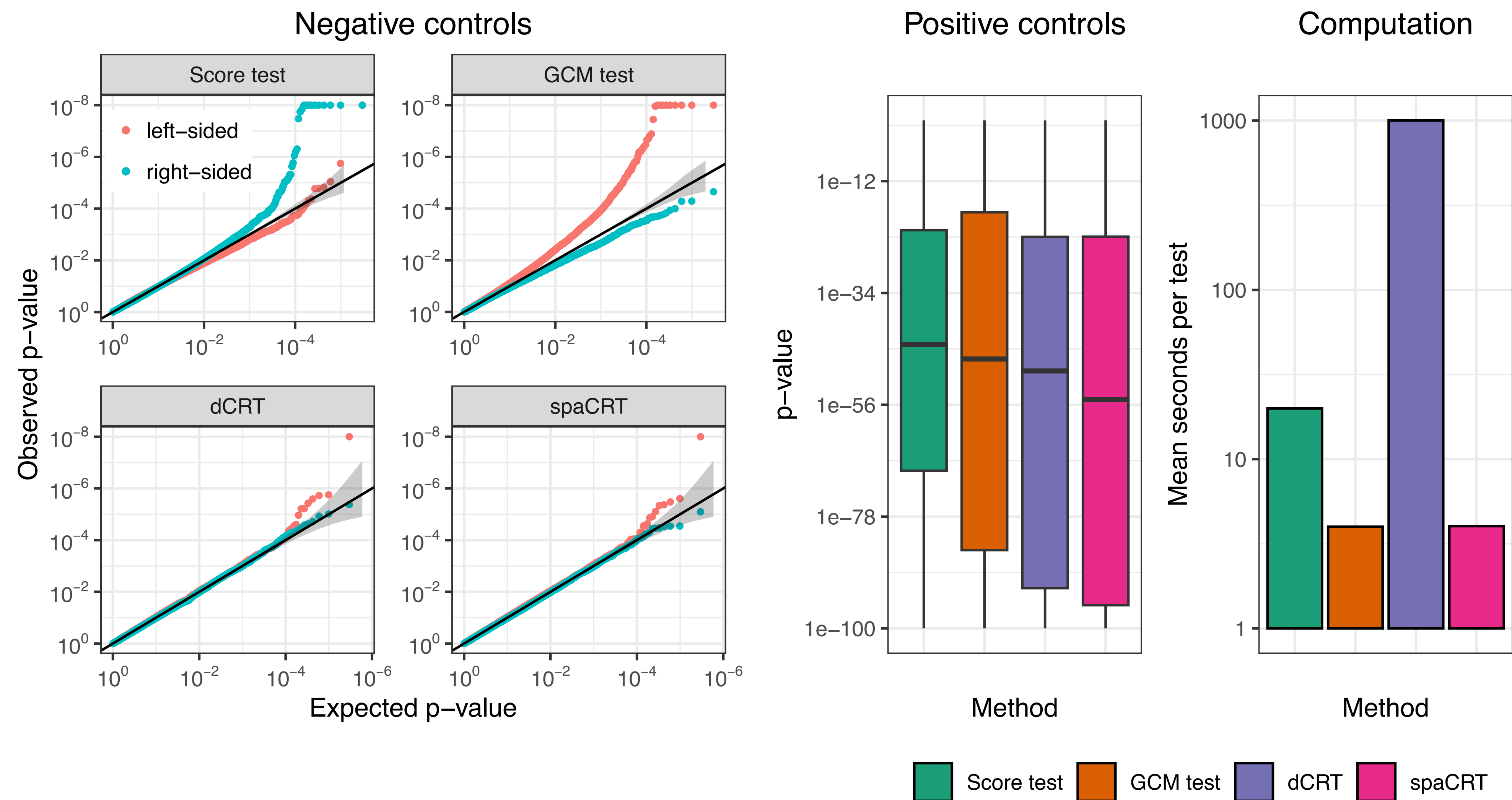
Results on Gasperini data analysis



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Discussion

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Takeaways

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CLT-based		

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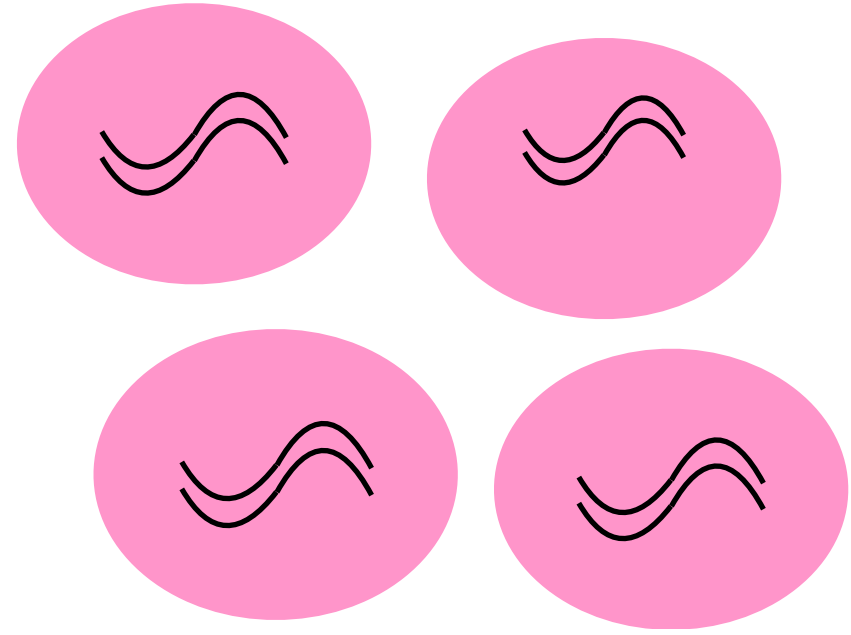
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- Software development.

References

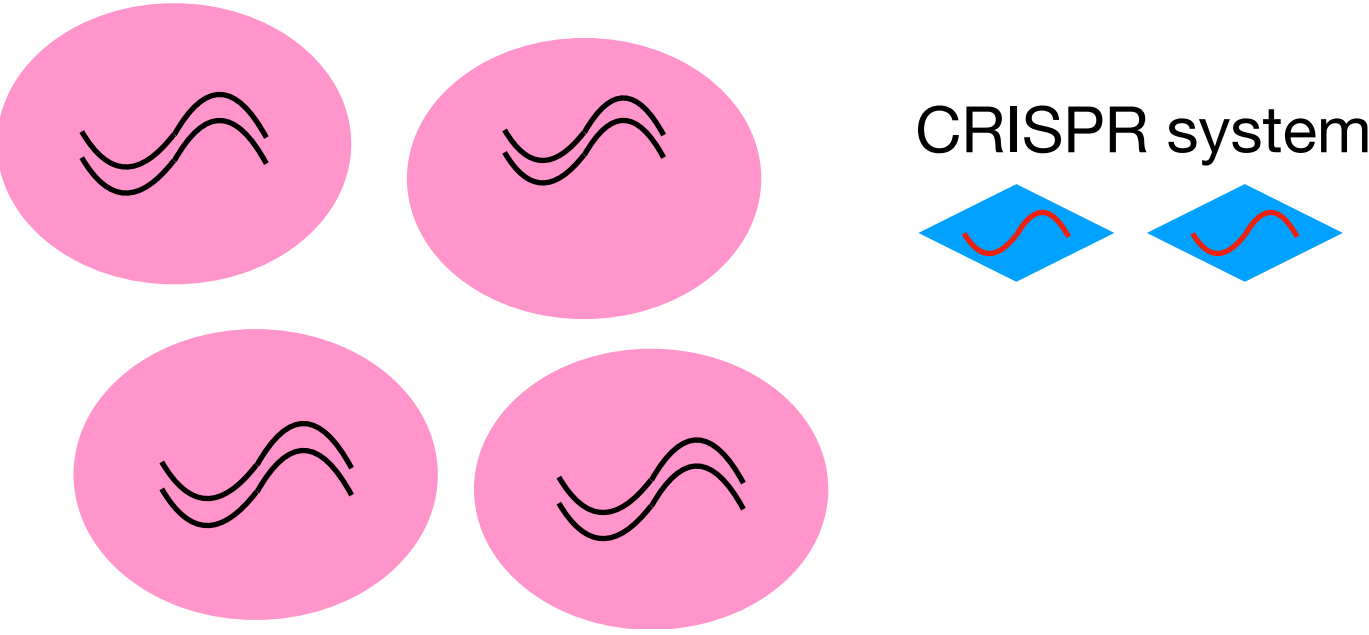
1. The saddlepoint approximation for averages of conditionally independent random variables. arXiv, 2024.
2. Computationally efficient and statistically accurate conditional independence testing with spaCRT. arXiv, 2024.

Background: gene knockout

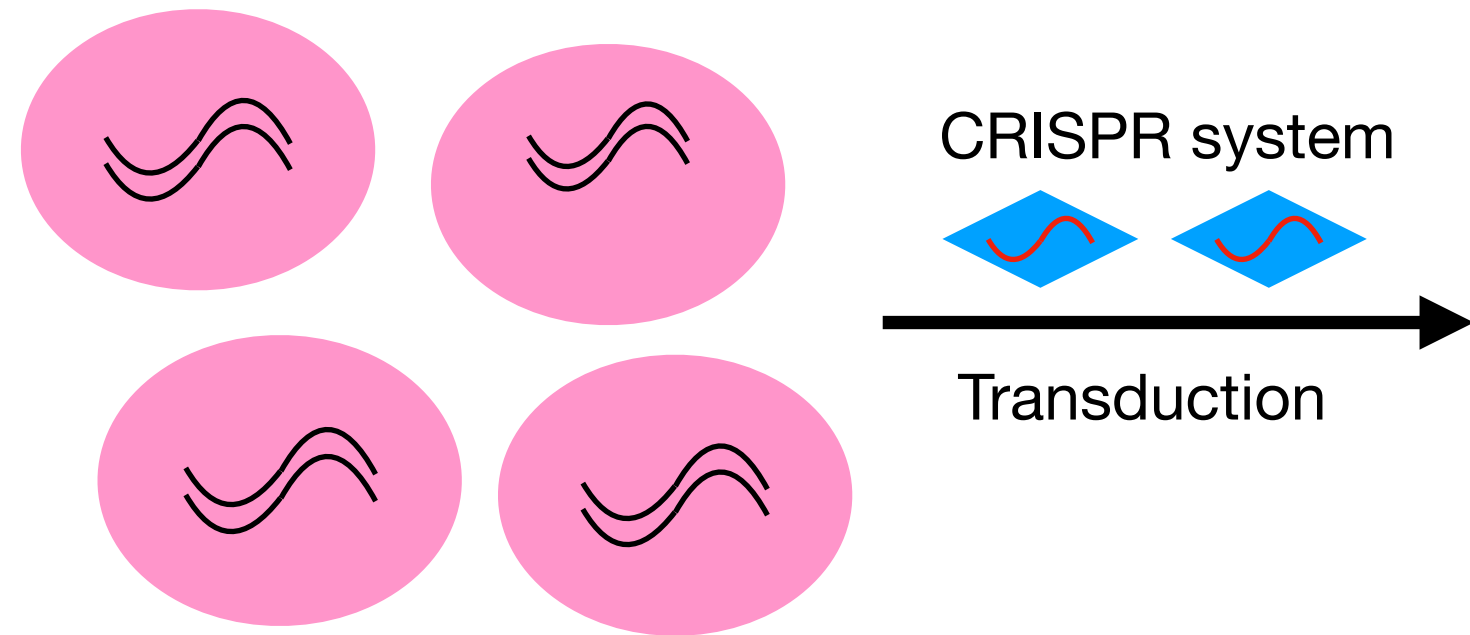
Background: gene knockout



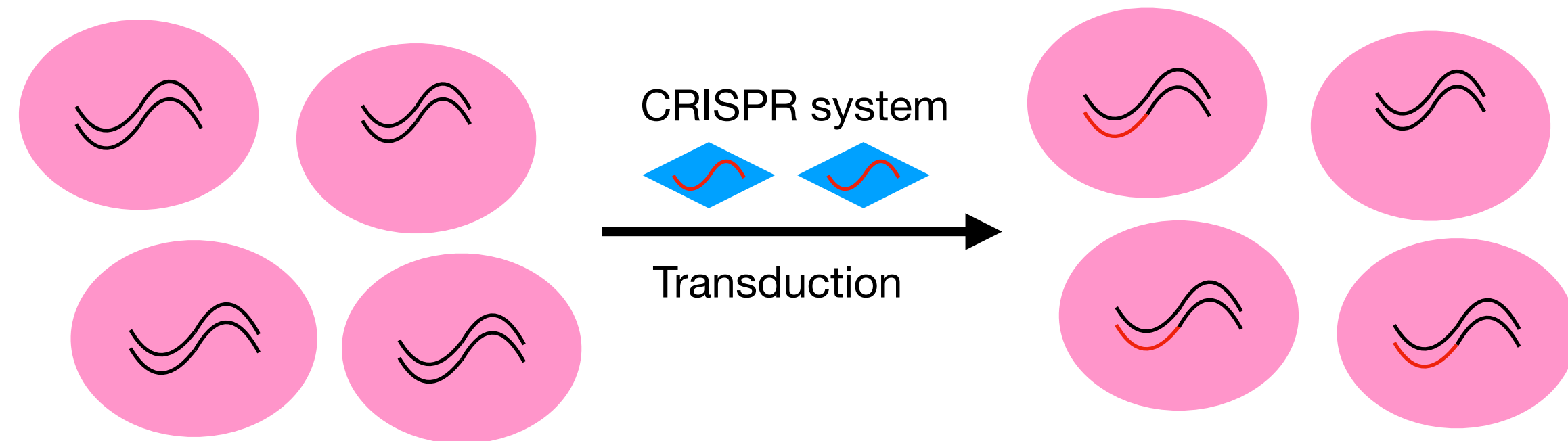
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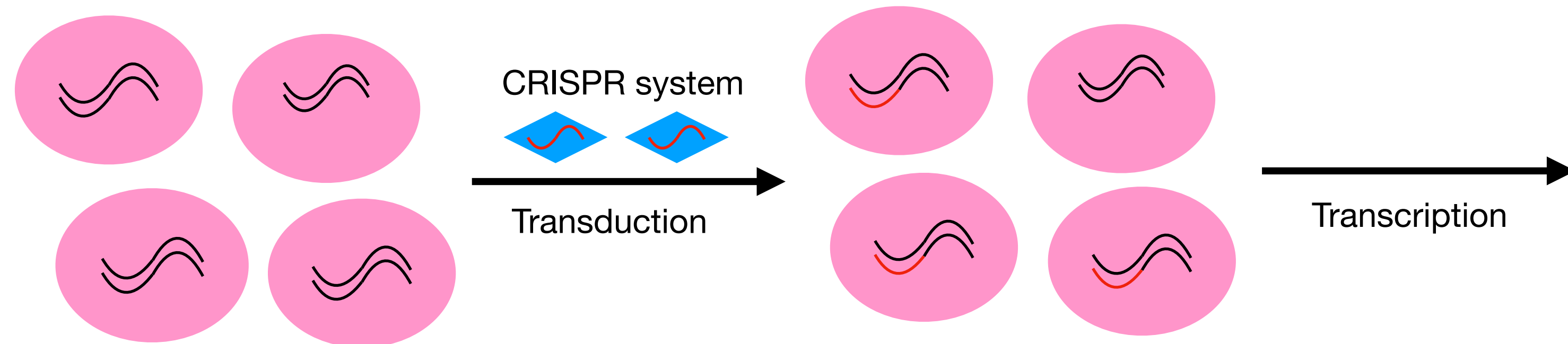
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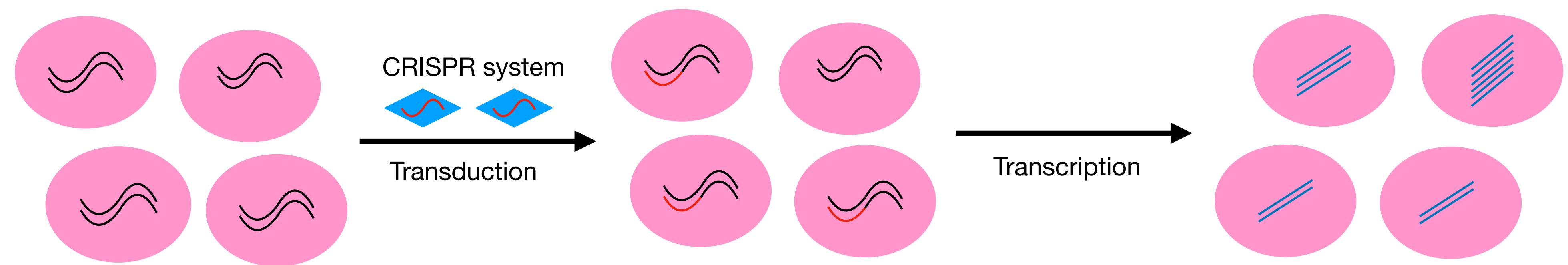
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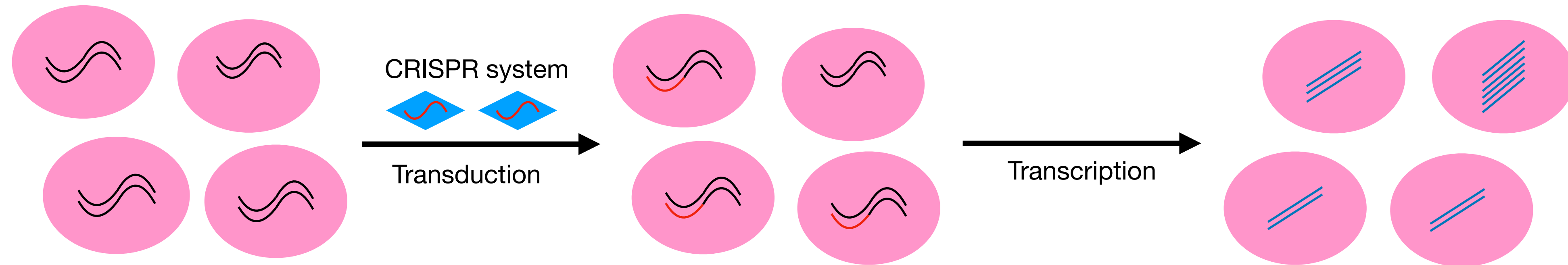
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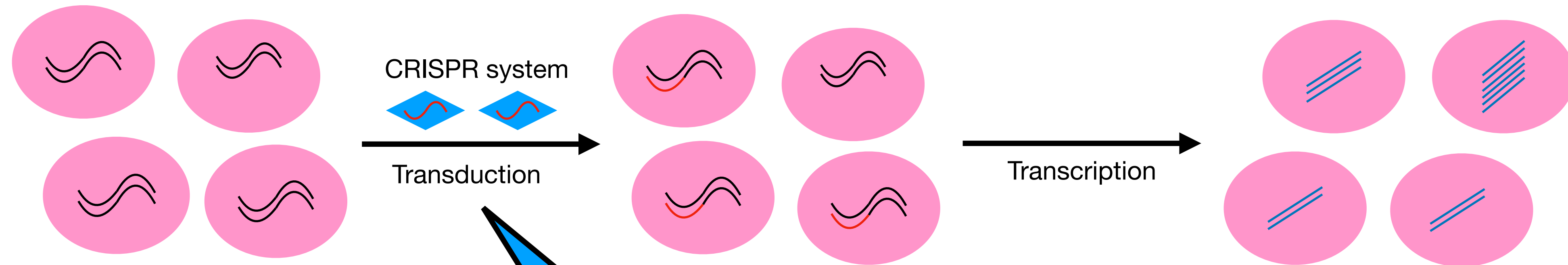
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Gene knockout

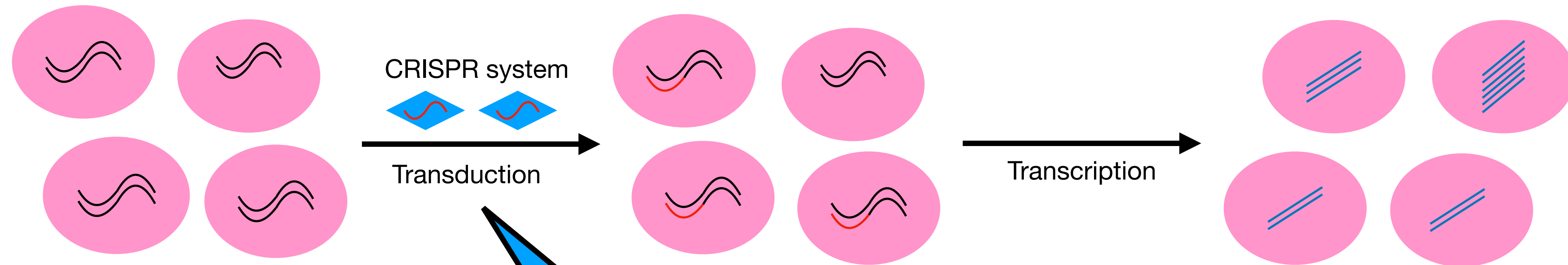


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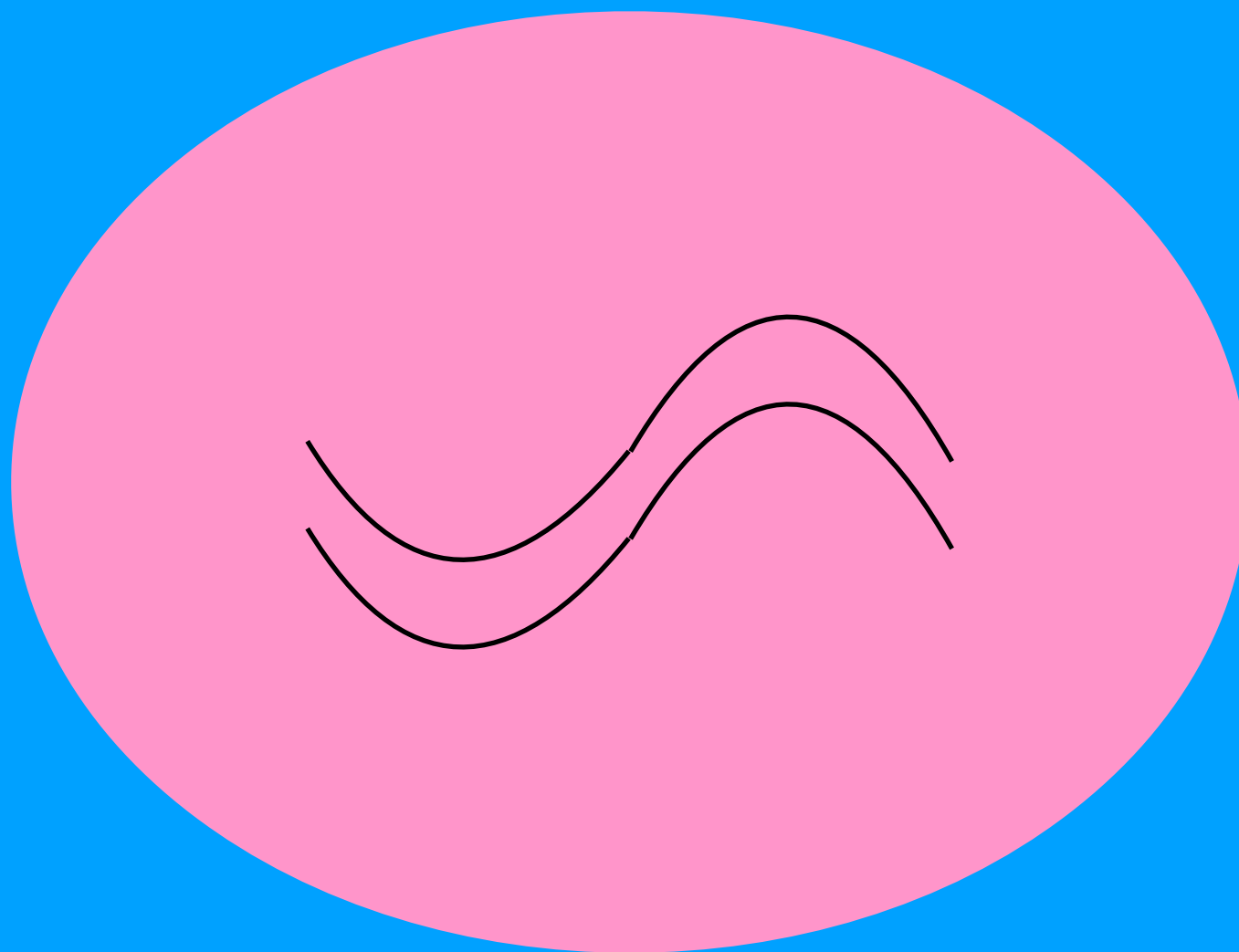


Transduction step

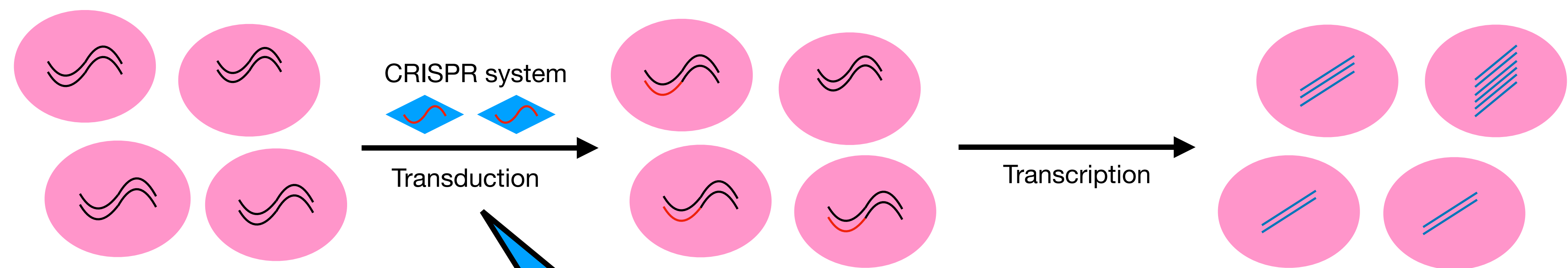
Gene knockout



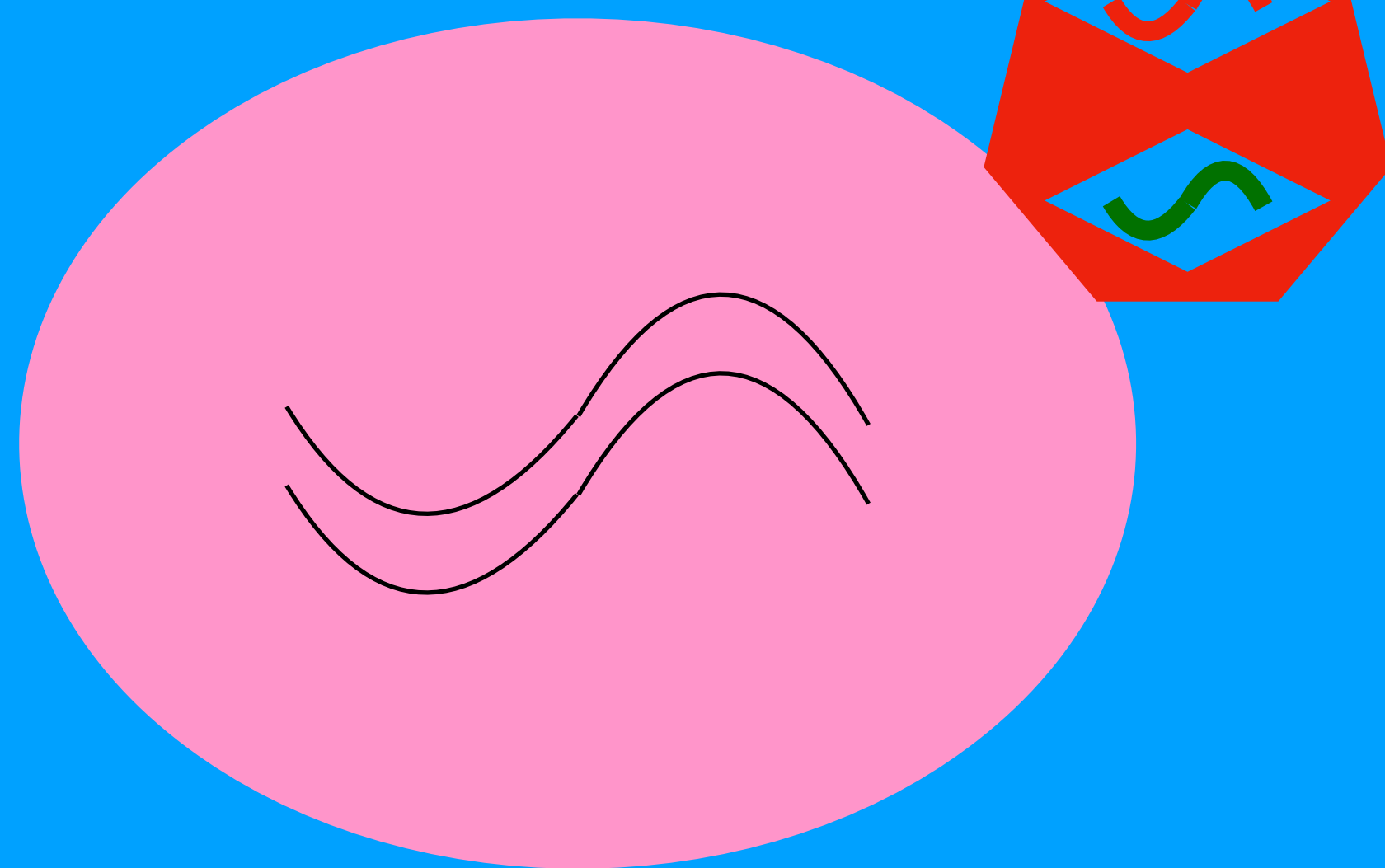
Transduction step



Gene knockout

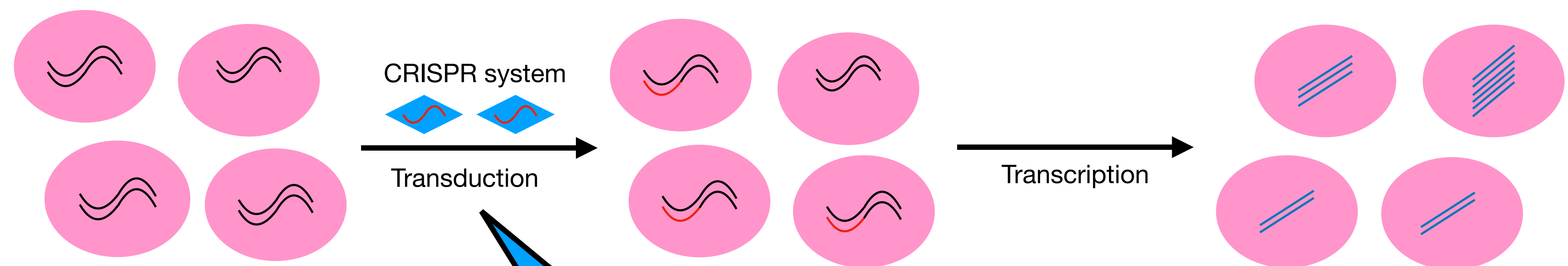


Transduction step



DNA
gRNA information
Cas-9 information

Gene knockout



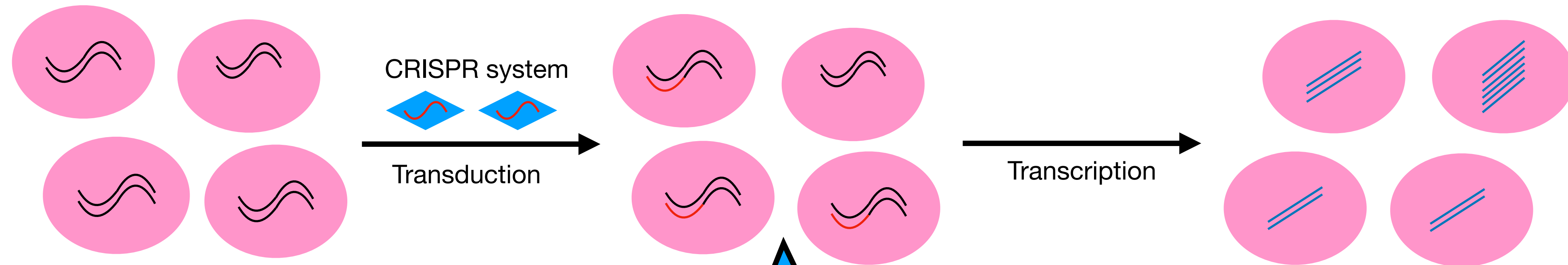
Transduction step

The diagram shows two pink circles representing cells. The left circle contains a black wavy line representing the DNA template. The right circle contains the same black wavy line, but with a red wavy line (gRNA) and a green wavy line (Cas-9) integrated into it. Above the circles is a red hexagonal structure representing the CRISPR array, with a red wavy line (gRNA) and a green wavy line (Cas-9) integrated into it. A legend on the right indicates that the red wavy line represents 'gRNA information' and the green wavy line represents 'Cas-9 information'.

DNA

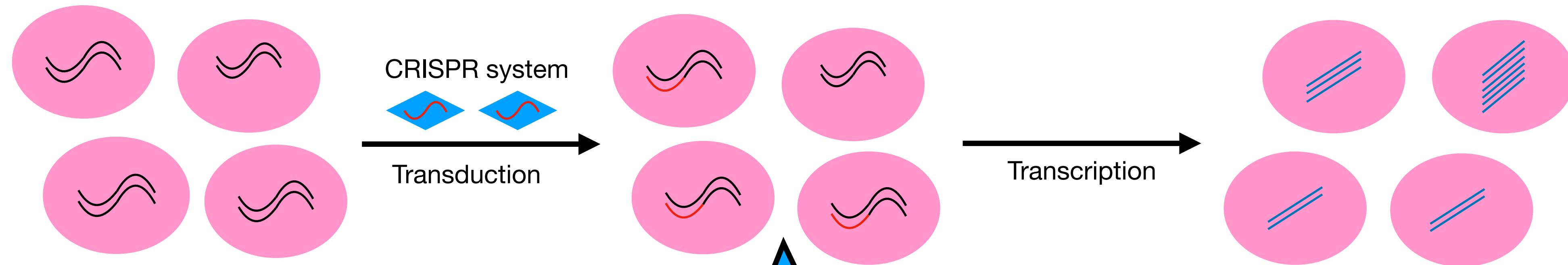
- gRNA information
- Cas-9 information

Gene knockout



Editing step

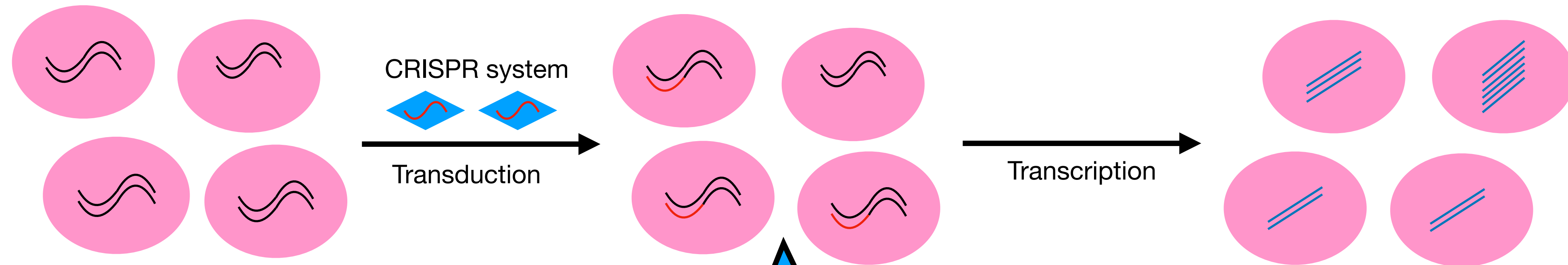
Gene knockout



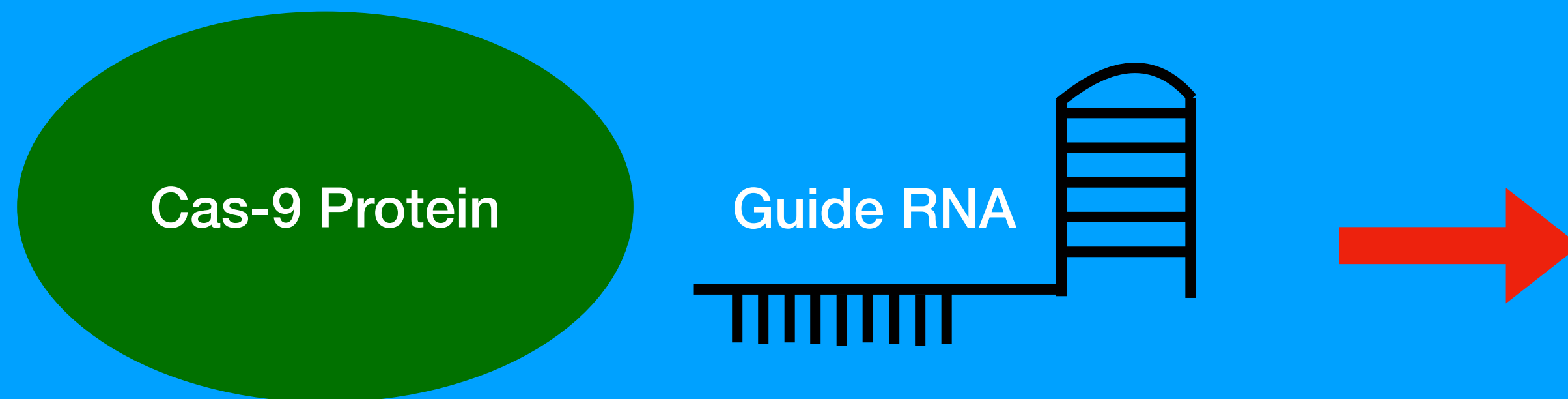
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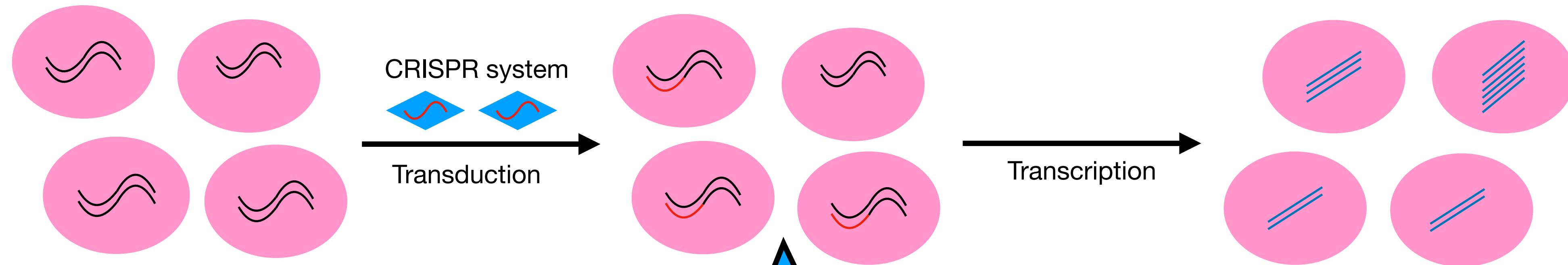
Gene knockout



Editing step



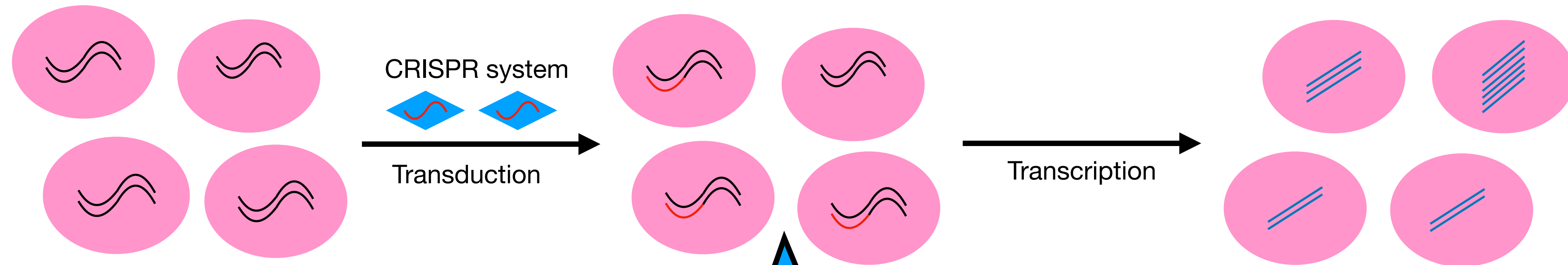
Gene knockout



Editing step

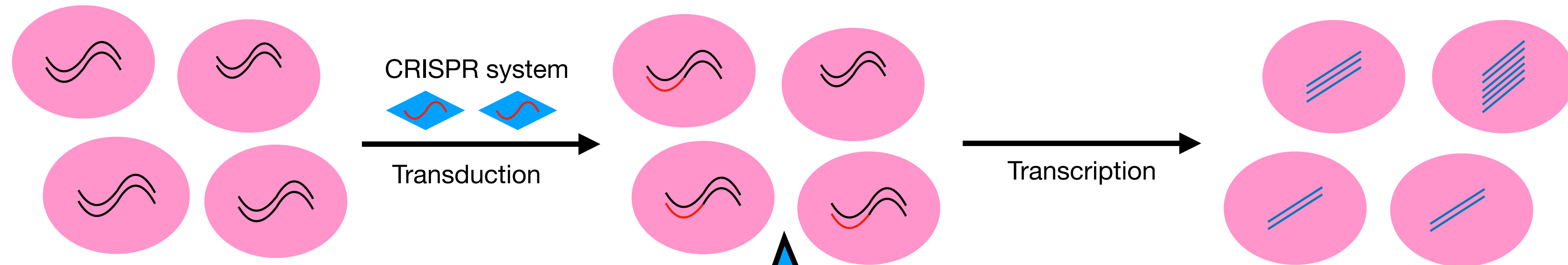


Gene knockout

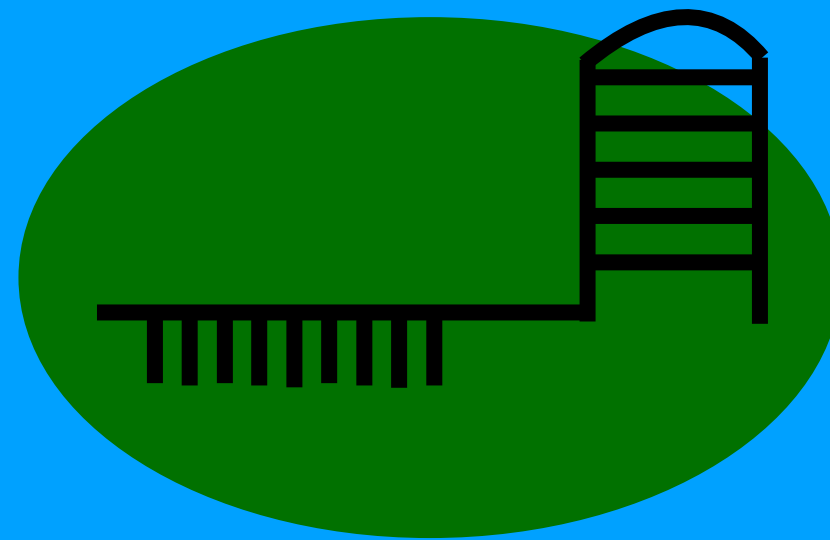


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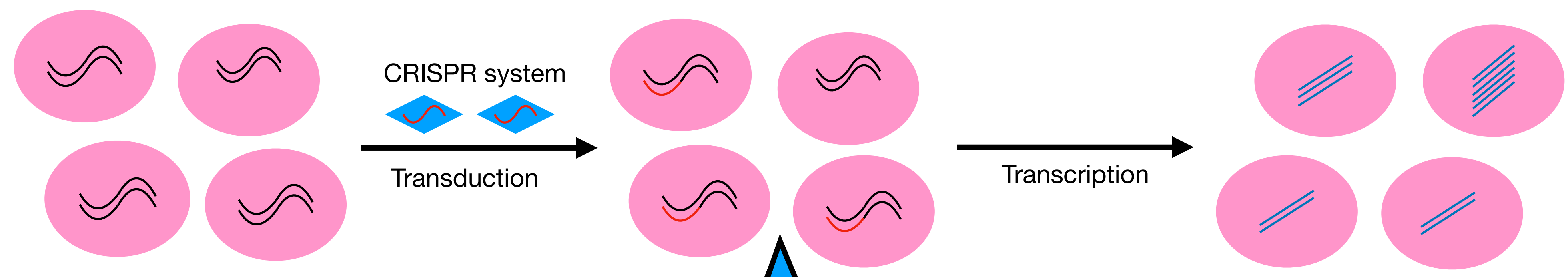
Gene knockout



Editing step



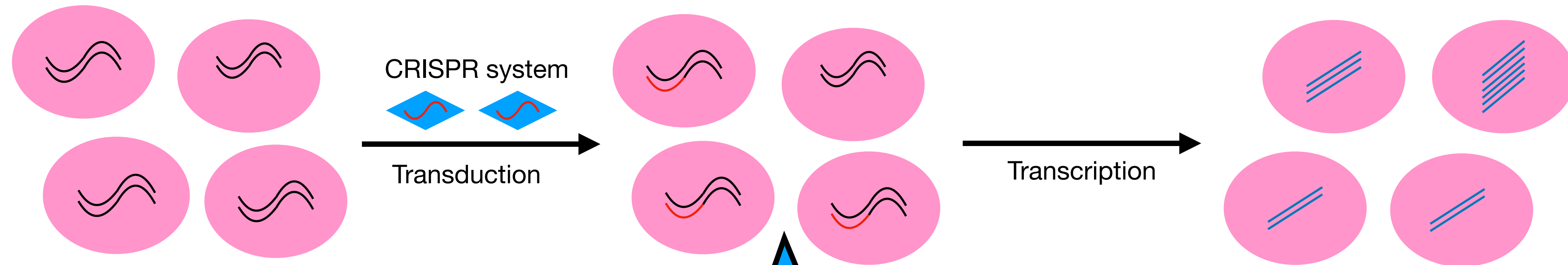
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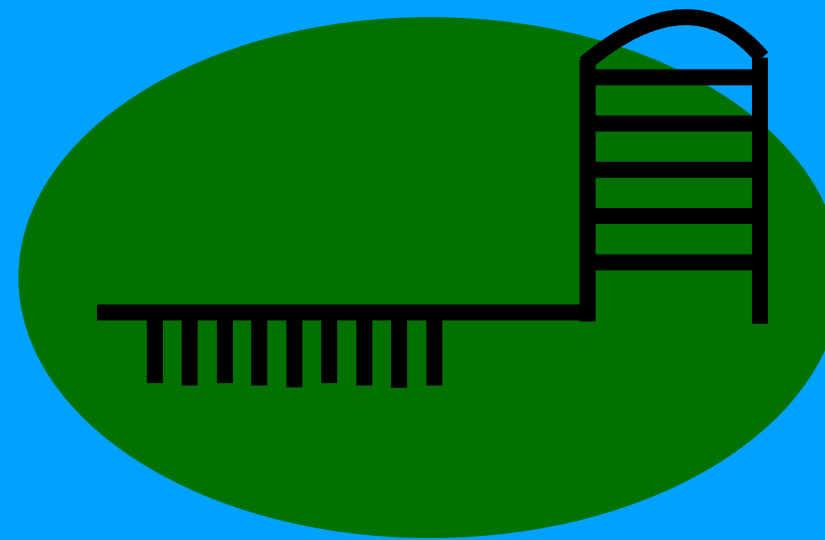
Editing step

This section provides a detailed view of the editing step. It features a green oval representing a cell nucleus. Inside the nucleus, a black ladder-like structure represents the DNA template. A blue ladder-like structure, representing the guide RNA, is shown binding to the DNA template. Below the nucleus, a long black ladder-like structure represents the full length of the DNA template.

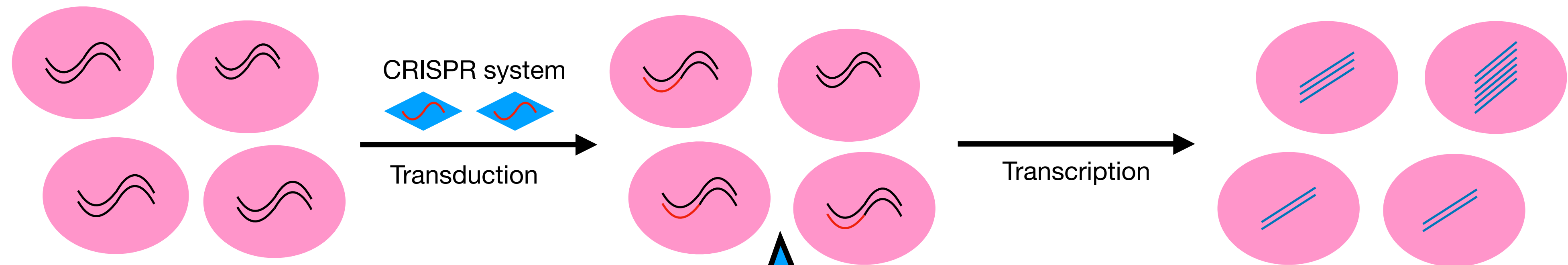
Gene knockout



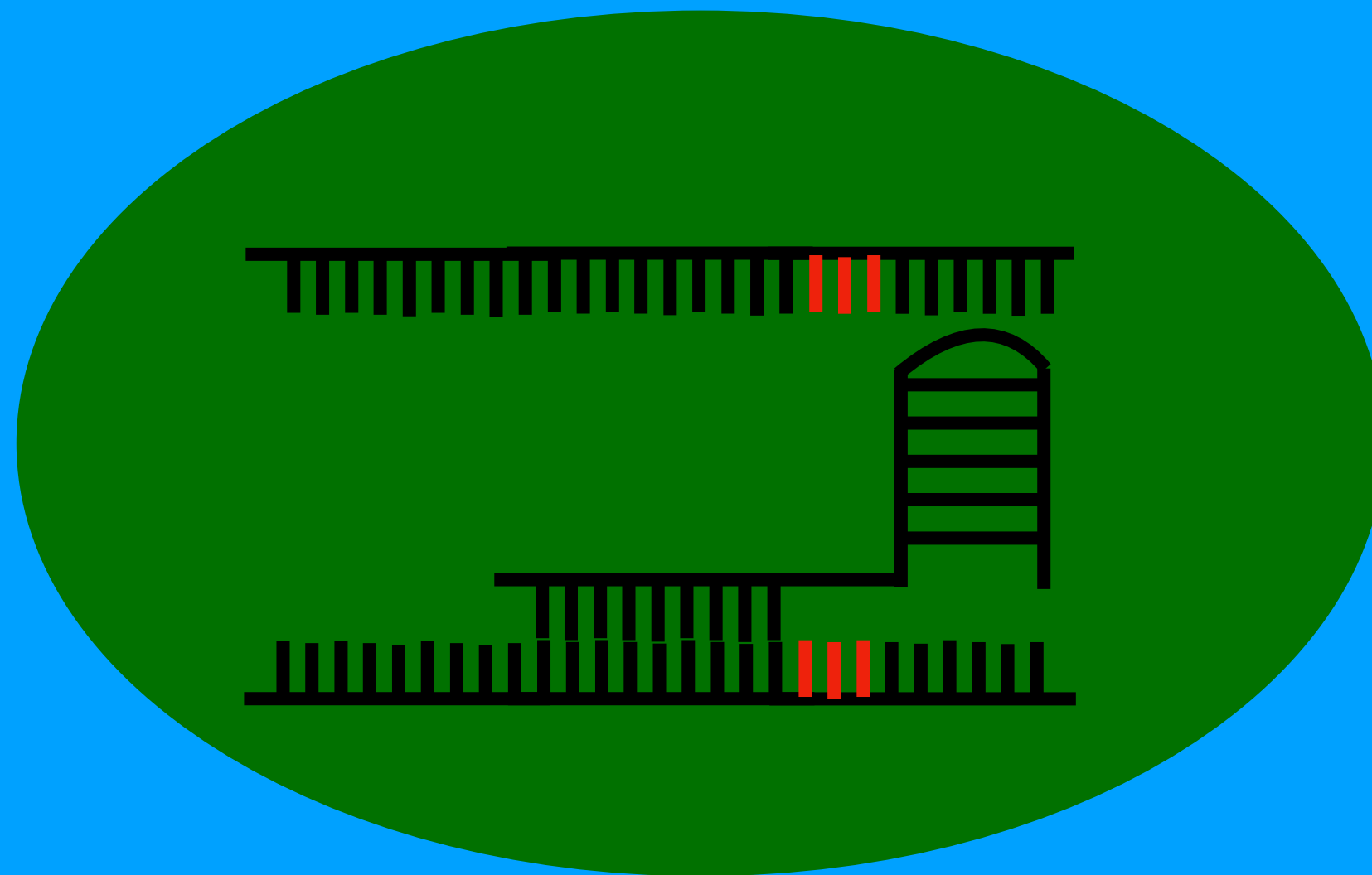
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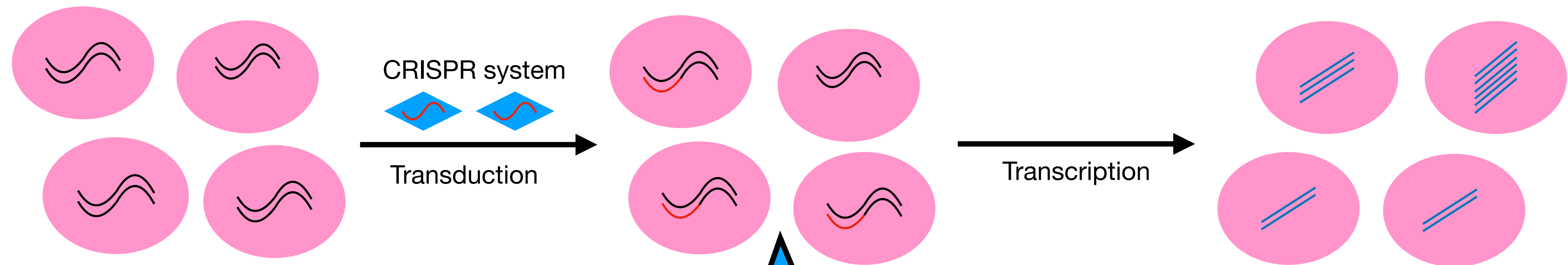
Gene knockout



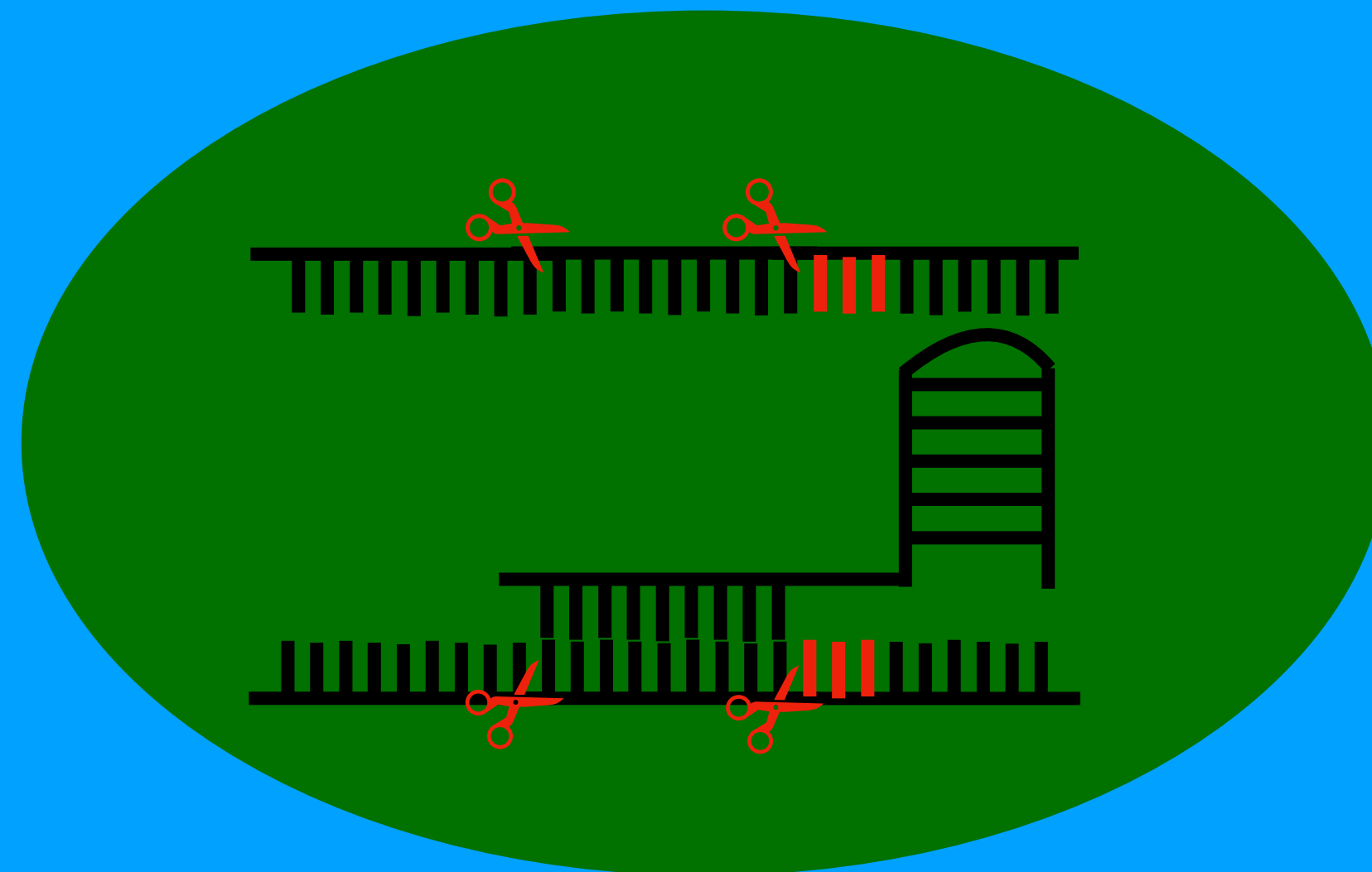
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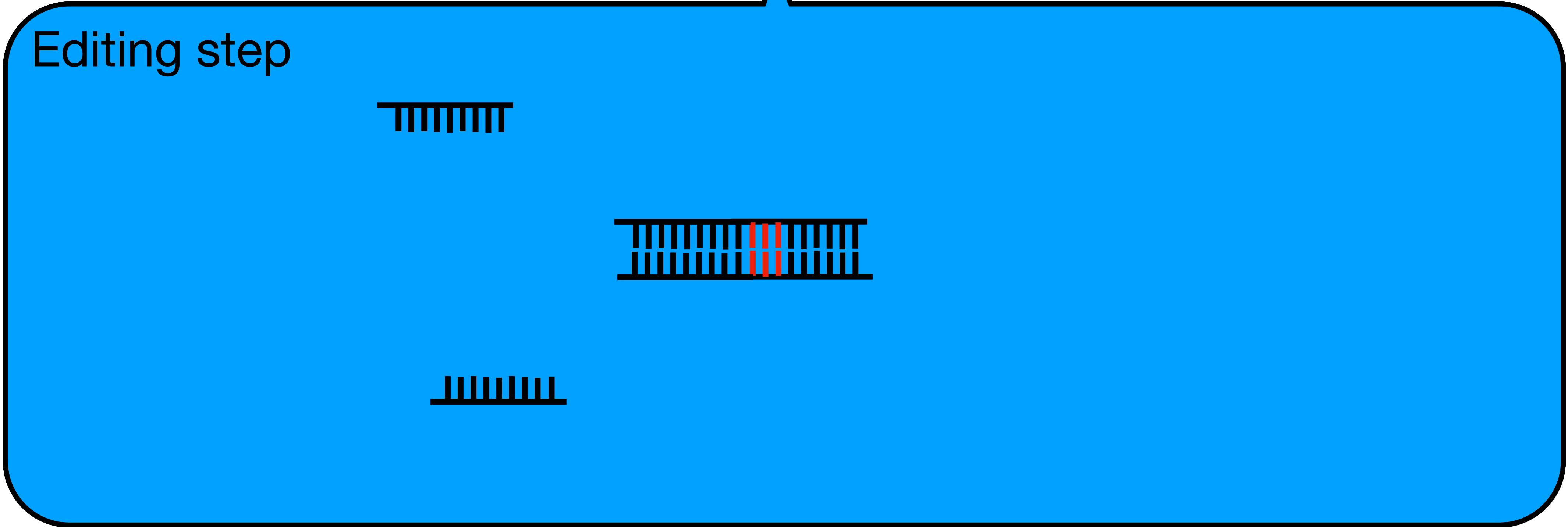
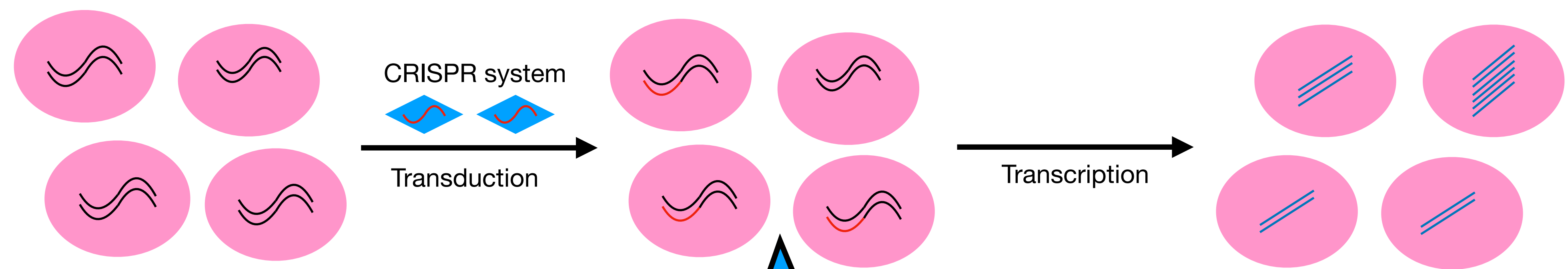
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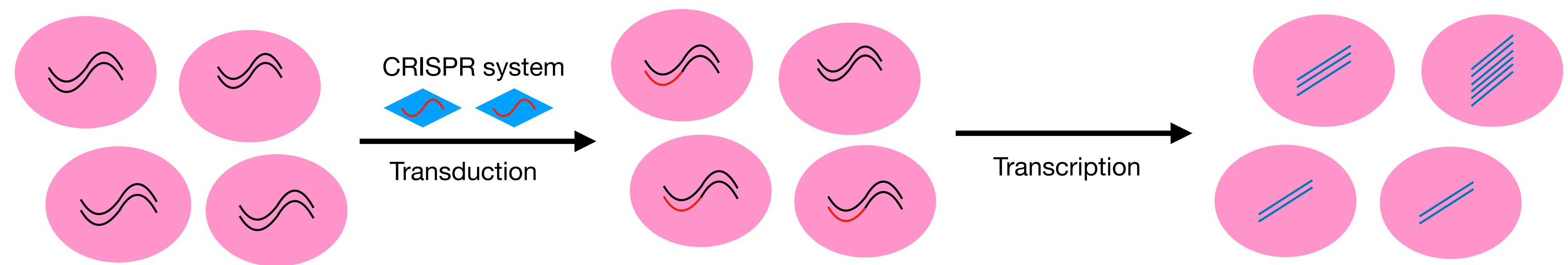
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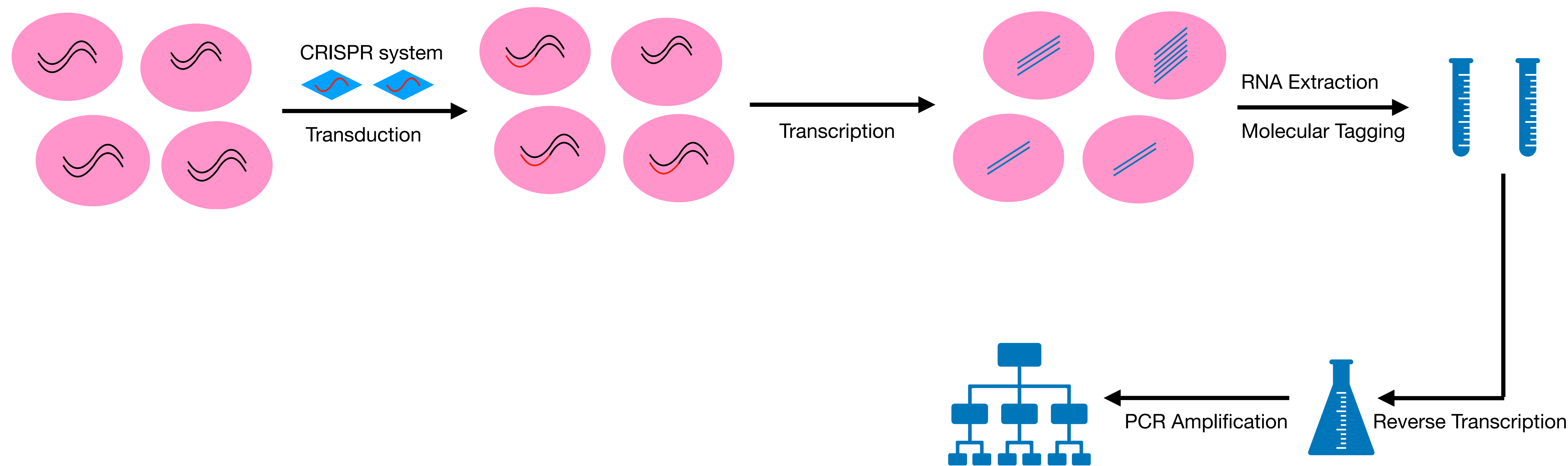
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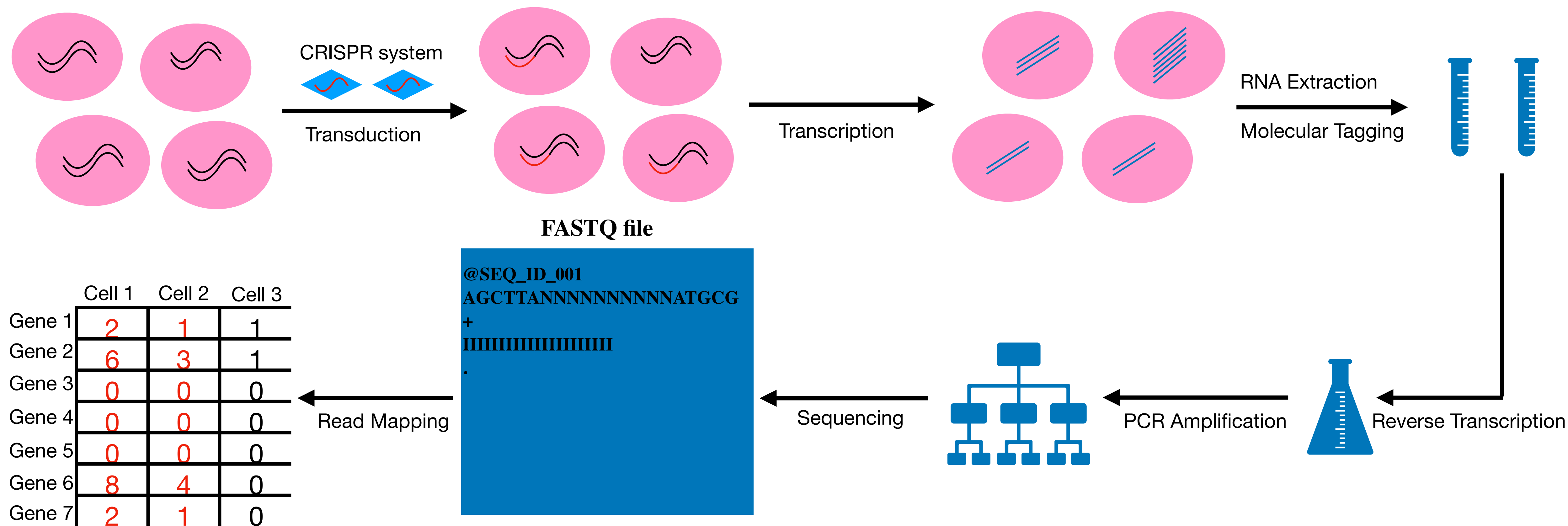
Background: RNA-sequencing



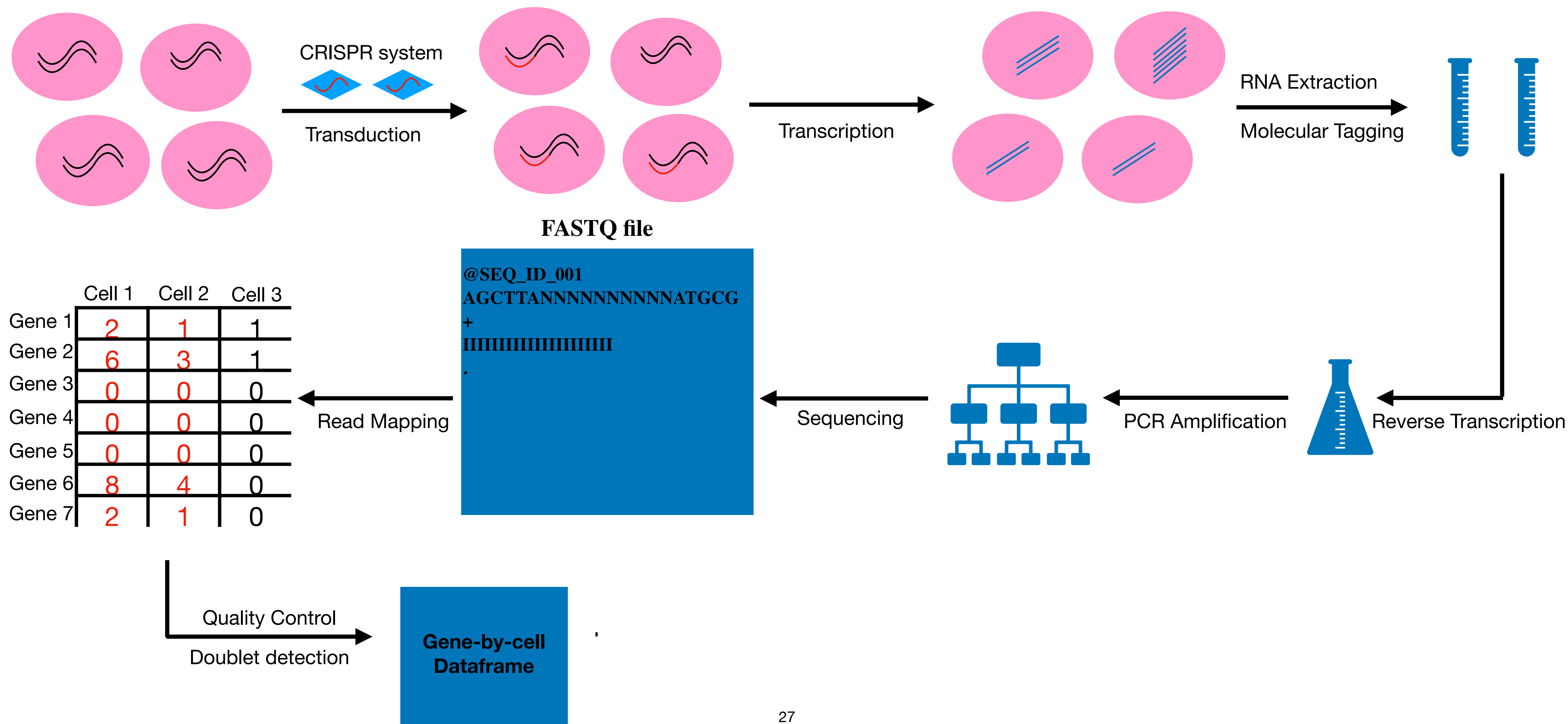
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